



A survey of large language models for legal tasks: Progress, prospects and challenges

Congqing He^{a,b,1} , Haichuan Hu^{c,1} , Yanli Li^d , Hao Zhang^{b,*} , Qunjun Zhang^e

^a School of Information Engineering, Huzhou University, Huzhou, 313000, China

^b School of Computer Sciences, Universiti Sains Malaysia, Penang, 11800, Malaysia

^c Alibaba Cloud, Hangzhou, Zhejiang, 310058, China

^d School of Electrical and Computer Engineering, The University of Sydney, Sydney, NSW 2006, Australia

^e School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, 210000, China

ARTICLE INFO

Keywords:

Large language models
Legal tasks
Legal reasoning
Legal datasets
Legal benchmarks

ABSTRACT

Recent advances in large language models (LLMs) have unlocked new opportunities for machine learning and deep learning applications in the legal domain. LLMs demonstrate remarkable capabilities in comprehending complex legal language, analyzing lengthy documents, and generating contextually relevant legal text. This survey provides a task-oriented overview of the application of LLMs in the legal domain, focusing on the categorization of tasks and a review of associated methods, datasets and benchmarks. We first review both general and legal-domain LLMs, summarize their foundational architectures and adaptation methods, and briefly analyze their suitability for various legal tasks. Subsequently, we categorize LLM-based legal tasks along the typical legal workflow, including legal information retrieval, legal document analysis, judicial decision prediction, legal question answering, legal document generation, legal agent-based modeling, and legal education and training. Within each task, we primarily analyze specific methodologies such as retrieval-augmented generation, prompting strategies, and reasoning. Furthermore, a comprehensive collection of legal datasets, benchmarks, and model resources is presented as a practical reference for researchers and practitioners. Finally, we outline open challenges and future directions, addressing issues such as bias, interpretability, data privacy, and regulatory compliance in legal LLMs. This survey provides a structured and comprehensive overview to facilitate the adoption and further development of LLMs in the legal domain. The collection is available at <https://github.com/hecongqing/Awesome-LLM4Law>.

1. Introduction

The legal domain is characterized by complex rules, jurisdiction-specific distinctions, and an evolving body of statutes and case law. The growing integration of digital technologies has paved the way for artificial intelligence (AI) to play an increasingly influential role in legal practice [1–4]. Large language models (LLMs), such as OpenAI's GPT-3 and GPT-4 [5,6], have demonstrated a remarkable ability to process vast volumes of text, interpret complex contexts, and generate human-like language [7,8]. The use of LLMs in legal tasks presents substantial opportunities to transform legal workflows by streamlining document review and drafting, enhancing the accuracy of legal research, and expanding access to legal information for both professionals and the general public.

LLMs pretrained on large-scale general text corpora (e.g., T5 [9], GPT-4 [6]) demonstrate strong language understanding capabilities. For example, GPT-4 has shown impressive performance in complex legal reasoning tasks, even surpassing average human performance on the bar exam [10]. However, general LLMs typically lack legal expertise, including specialized terminology and jurisdictional details. To bridge this gap, researchers have developed legal LLMs by further pretraining on legal corpora or by training models from scratch using legal texts [11,12]. Models such as Lawyer LLaMA [13], InternLM-Law [14], and LawGPT [12], fine-tuned for specific jurisdictions and domains, can effectively process legal documents and handle legal tasks.

In recent years, the legal domain has increasingly adopted LLMs across various legal tasks, reshaping legal practice by streamlining

* Corresponding author.

Email addresses: hecongqing@student.usm.my (C. He), huhaiquan.hhc@alibaba-inc.com (H. Hu), yanli.li@sydney.edu.au (Y. Li), zhanghao666@student.usm.my (H. Zhang), qunjunzhang@njnu.edu.cn (Q. Zhang).

¹ Congqing He and Haichuan Hu are co-first authors.

<https://doi.org/10.1016/j.cosrev.2026.100906>

Received 1 June 2025; Received in revised form 24 November 2025; Accepted 13 January 2026

Available online 21 January 2026

1574-0137/© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

workflows, enhancing expertise, and offering new insights into judicial processes [15]. For example, LLMs efficiently transform complex legal texts into clear summaries in document summarization and contract analysis, thereby improving information processing [16,17]. Legal question-answering has greatly advanced with LLMs [18–20]. LLMs can efficiently retrieve and synthesize information from statutes, case law, and regulations, providing valuable insights for decision-making and expanding public access to justice. Moreover, LLMs show potential in judgment prediction and generating explanations for judicial decisions [21,22]. Notably, LLMs significantly surpass previous AI in law through enhanced complex reasoning capabilities, allowing them not only to retrieve information but also to simulate human-like reasoning processes [23,24]. Leveraging these capabilities, LLMs facilitate multi-agent legal simulations and interactive educational platforms, effectively modeling courtroom proceedings and legal debates [25]. Therefore, LLMs are emerging as essential tools for enhancing the efficiency, accuracy, and accessibility of legal services.

Despite notable progress, the application of LLMs in legal tasks still presents several persistent challenges. Confidentiality and data privacy are critical due to the sensitivity of legal information. Model hallucinations, biases, and variations across legal systems can limit reliability. Moreover, explainability and alignment with legal reasoning are essential for transparency and accountability. Practical constraints, including computational cost, inference latency, and the requirement for up-to-date legal knowledge, also limit real-world application. Prior surveys have highlighted these issues. For example, Qin and Sun [26] emphasize bias, interpretability, and ethical compliance, while Chen et al. [27] focus on regulatory alignment, fairness, and robustness. Addressing these challenges is vital for ensuring that LLMs provide trustworthy, transparent, and equitable benefits to both legal practitioners and the public.

Recently, several surveys have explored the applications of LLMs in the legal domain. Lai et al. [28] present a review of legal LLMs, covering foundational AI and law concepts, differences between traditional

and AI-assisted judicial decision-making, fine-tuning and evaluation methods, and future research directions. Qin and Sun [26] investigate the relationship between LLMs and the legal system, focusing on tasks such as legal text comprehension, case retrieval, and domain-specific challenges. Chen et al. [27] compare LLM applications in finance, healthcare, and law domains, emphasizing the importance of regulatory compliance and ethical considerations in each field. Table 1 summarizes the differences between these existing surveys and ours.

However, existing surveys have primarily focused on technical methodologies, often neglecting the overview of LLM-based methods for legal tasks and rarely providing or systematically evaluating legal datasets and benchmarks. In contrast, our survey addresses these gaps by systematically organizing LLM-based methods across core legal tasks, consolidating publicly available legal datasets, benchmarks, and code resources, and synthesizing cross-cutting ethical, regulatory, and other challenges. Fig. 1 outlines the paper's structure, organizing models, LLM-based legal tasks, datasets, code and benchmarks, as well as challenges and future directions.

This paper provides a comprehensive survey of LLMs in the legal domain, covering recent progress in model development, key legal tasks, available datasets and benchmarks, and the main open challenges. Specifically, the survey addresses four main research questions:

- RQ1:** Which general LLMs and legal-domain LLMs are currently used in the legal domain?
- RQ2:** What LLM-based methods have been developed for legal tasks, and how are they applied technically?
- RQ3:** What datasets and benchmarks are available for developing LLMs in the legal domain?
- RQ4:** What are the main challenges and open questions, and what solutions are emerging?

Contributions: Our main contributions include:

- We provide a task-oriented synthesis of LLM applications in key legal tasks, including legal information retrieval, legal document analysis, legal decision support, legal question answering, and legal document generation. In addition, we review legal agent-based modeling that integrates legal tasks into multi-stage workflows and LLM-based applications in legal education and training.
- We review representative general LLMs and legal-domain LLMs, outlining their model architectures and typical pre-training and fine-tuning strategies. Additionally, legal datasets and evaluation benchmarks covering multiple tasks, languages, and jurisdictions are systematically organized to support future research.

Table 1

Comparison between our survey and related surveys. Circles indicate areas covered but lacking extensive detail.

Survey	Legal LLMs	Datasets, benchmarks and codes	Legal tasks	Challenges
[28]	✓	○	○	✓
[26]	✓	○	✓	○
[27]	○	○	○	✓
Ours	✓	✓	✓	✓

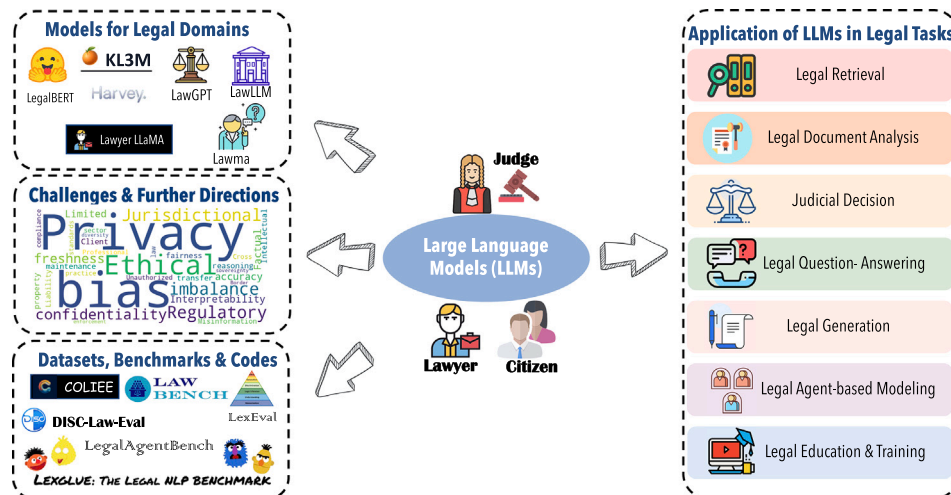


Fig. 1. Overview of the structure of this survey, covering models for legal domains, applications of LLMs in legal tasks, datasets and benchmarks (including code resources), and challenges and future directions.

- An extensive analysis of legal domain issues, addressing data quality and coverage, robustness and bias, explainability and legal reasoning, regulatory and ethical constraints, and deployment risks, along with proposing future research directions.

Organization: The remainder of this paper is structured as follows. [Section 2](#) defines the scope of the survey and details the methodology employed for literature collection and analysis. In [Section 3](#), we review representative general LLMs and legal-domain LLMs, outlining their model architectures and typical pre-training and fine-tuning strategies (addresses [RQ1](#)). [Section 4](#) provides a task-oriented synthesis of how LLMs are currently used across key legal tasks (addresses [RQ2](#)). [Section 5](#) surveys datasets, benchmarks, and code resources for the training and evaluation of legal LLMs across different tasks, languages, and jurisdictions (addresses [RQ3](#)). [Section 6](#) discusses open challenges and future research directions (addresses [RQ4](#)). Overall, this organization follows a progression from models to task-level applications, supporting resources, open challenges, and future directions, aligning with the overview summarized in [Fig. 1](#).

2. Survey methodology

This section outlines the scope of the survey and details the methodology used to collect and analyze the relevant literature.

(“llm”) AND (“legal” OR “law” OR “legislative” OR “judicial” OR “court”) AND (“retrieval” OR “summarization” OR “judgment prediction” OR “question answering” OR “dialogues” OR “agent” OR “drafting” OR “document generation” OR “exam” OR “benchmark” OR “statutory” OR “argument mining”)

Table 2
Illustrative examples of keywords and synonyms employed in the literature search.

Concept	Keywords and synonyms
Legal domain	legal, law, lawyer, court, judiciary, judicial, case, legislation, legal case, legal text, legal document.
LLM	large language model, LLM, LLMs.
Legal tasks	summarization, summary, legal summarization, case summarization; court judgment prediction, statutory prediction, case outcome prediction, judicial prediction; question answering, QA, legal QA; legal agent, legal assistant; contract drafting, contract generation, contract analysis, contract review; legal generation; legal reasoning; benchmark.

The query was adapted to match the syntax requirements of each digital library. To maximize recall and precision, we conducted iterative pilot searches, refining the query to ensure comprehensive keyword coverage. [Table 2](#) presents the key search terms used in each component of the query.

Secondary sources: Supplementary searches were conducted to ensure comprehensive coverage. These included backward reference searches, in which the bibliographies of selected articles were reviewed to identify additional relevant studies, and forward citation searches using Google Scholar to locate recent publications that cited foundational papers. In addition, we reviewed reference lists from recent, authoritative surveys [[26,28,29](#)] to detect publications potentially overlooked by primary database searches. Results from these diverse search methods were integrated, and duplicate entries across sources were systematically identified and removed.

2.2. Survey scope

[Table 3](#) details the inclusion and exclusion criteria used to assess study eligibility. Applying these criteria, we selected a final corpus that

2.1. Paper collection

The paper collection process consists of two stages: keyword-based database searches and reference-based supplementary searches.

Primary sources: We conducted extensive searches across multiple digital libraries and online repositories to gather candidate publications. The primary data sources include major computer science and NLP databases such as the ACL Anthology (covering ACL, EMNLP, NAACL, etc.), ACM Digital Library, IEEE Xplore, SpringerLink, and academic search engines such as Google Scholar. These sources were selected to cover both conference proceedings and journal articles in NLP, AI, and legal informatics venues. The searches were limited to English-language publications released between January 2019 and May 2025 to align with the significant advances introduced by Transformer-based language models.

Search string formulation: We constructed a structured boolean query to capture papers involving both the legal domain and LLMs. The query was organized into three components: (1) LLM-related terms, (2) legal context keywords, and (3) legal task keywords. Synonyms within each component were combined using OR operators, and the three components were joined with AND operators. The general structure of the query is as follows:

Table 3
Inclusion and exclusion criteria for study selection.

Inclusion criteria	Exclusion criteria
IC1: Peer-reviewed publications from 2019 to 2025, and high-quality arXiv preprints.	EC1: Non-English language publications.
IC2: Primary focus on legal texts or tasks (e.g., case law, legislation, contracts, legal QA).	EC2: Studies not focused explicitly on legal NLP.
IC3: Employs LLMs.	EC3: Methods not utilizing LLMs.
IC4: Presents a concrete technical contribution (model, method, dataset, benchmark, or empirical evaluation).	EC4: Grey literature, technical reports, theses, or non-peer-reviewed articles.
IC5: Addresses defined key legal AI tasks (e.g., document summarization, judgment prediction, legal QA, contract analysis/drafting).	EC5: Out-of-scope topics such as purely theoretical discussions without computational analysis, or non-textual AI applications.

highlights the most significant and technically robust advancements in legal AI involving LLMs.

2.3. Statistics of collected papers

Keyword searches were conducted iteratively, with the most recent search completed on May 1, 2025. Each identified paper was independently reviewed by the authors for relevance based on predefined scope criteria. This screening process yielded a total of 166 relevant publications. [Fig. 2](#) illustrates the cumulative publication trend from January 2019 through May 2025. A consistent upward trend indicates increasing research interest in the application of LLMs to legal tasks. [Fig. 3](#) presents the distribution of publication venues, highlighting strong representation from major NLP and AI conferences (ACL, EMNLP, AAAI, IJCAI, COLING), specialized AI and Law workshops (ICAIL, NLLP), and leading journals (e.g., Artificial Intelligence and Law). Notably, a

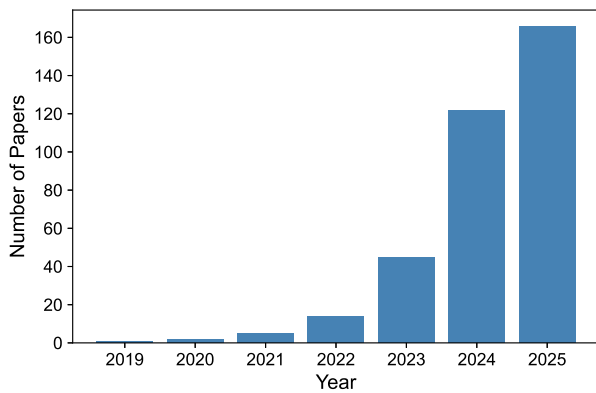


Fig. 2. Cumulative number of papers over time.

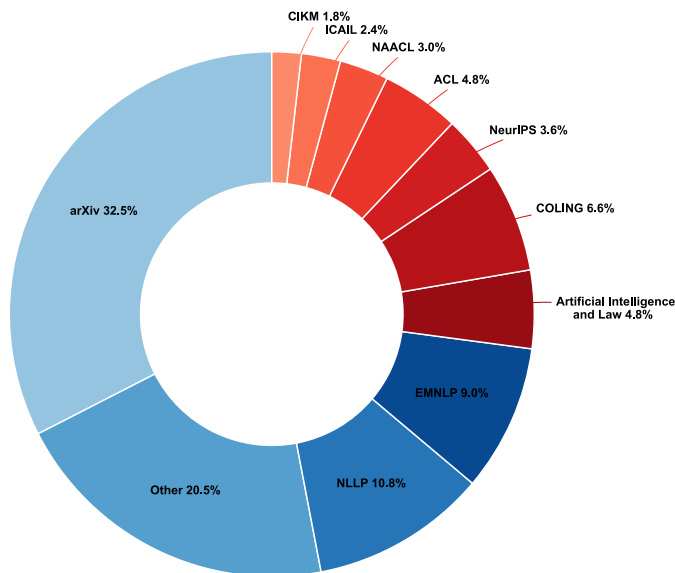


Fig. 3. Distribution of publication venues of all papers.

substantial proportion originated from arXiv, highlighting the emergent and dynamically evolving nature of this research domain.

3. Models for legal domains

3.1. Collections of models

Legal models have evolved significantly to address the challenges presented by legal tasks. These models can be categorized into general models for legal domains, domain-adaptive legal models pretrained on legal corpora, and tool-augmented legal LLMs that integrate external tools. Fig. 4 illustrates an overview of legal models introduced since 2020, grouped by foundational architectures.

3.1.1. General models for legal domains

Early legal NLP research leveraged pre-trained language models (PLMs) (e.g., BERT [30] and RoBERTa [31]), which have been widely used as foundational models for legal tasks [26]. These transformer-based PLMs exhibit strong baseline language understanding due to extensive pretraining on large-scale corpora.

Recently, LLMs such as OpenAI's GPT-3.5/4 and Meta's LLaMA have further extended this capability [6,32]. Although these models were not specifically trained on legal data, their robust reasoning abilities enable them to perform effectively in legal tasks, including question answering, document drafting, and offering preliminary legal advice. For instance,

GPT-4 has demonstrated the ability to answer complex legal questions and outperform average human examinees on the bar exam, highlighting the strength of LLMs in legal reasoning [10]. Similarly, ChatGPT has proven effective in drafting legal documents and providing advice on complex legal cases [33].

However, these models still have limitations in the legal domain. While LLMs such as GPT-4 and ChatGPT demonstrate strong reasoning skills, they lack real-time access to legal databases and recent case law, preventing them from incorporating up-to-date statutes or specialized legal terminology. This leads to potential inaccuracies, as they cannot “know” new case law or niche regulations. Furthermore, these models remain susceptible to hallucinations and oversimplifications when reasoning within the legal domain [34,35]. Therefore, while general LLMs (e.g., GPT-4 and ChatGPT) possess strong language and reasoning abilities, their “out-of-the-box” models may still provide plausible yet unreliable legal analyses without further domain adaptation.

3.1.2. Domain-adaptive legal models

To bridge the gap between general language models and domain-specific expertise, early legal language models evolved from PLMs to more specialized legal-domain models. In the BERT era, legal-domain PLMs were pretrained on law-specific datasets, resulting in improvements over vanilla BERT on downstream legal tasks. LegalBERT [36] and other legal BERT variants (e.g., LexLMs [37], CaseLawBERT [11]) are pretrained on large legal corpora and consistently outperform general models on tasks such as contract analysis, and document classification. Specialized variants focus on specific jurisdictions and languages, including JuriBERT (French) [38], AraLegal-BERT (Arabic) [39], Japanese Legal BERT [40], Indian Legal BERT [41], LegalBERT-pt (Brazilian Portuguese) [42], and ITALIAN-LEGAL-BERT [43]. These models capture specialized legal terminology, stylistic regularities, and structural features of legal text. Furthermore, studies have addressed long-document processing: LegalLongformer [44] and Lawformer [45] handle inputs of thousands of tokens, enabling full-text retrieval and judgment prediction. BudgetLongformer, trained from scratch on legal corpora [46], and LEGALRELECTRA, which utilizes Reformer [47,48], also extend the effective context windows for legal inputs.

Building on the success of PLMs, the advent of LLMs has ushered in a new era for legal NLP. These models are pretrained or fine-tuned on legal-specific corpora, enabling them to handle complex legal tasks. In the U.S., LawLLM [49] is trained on legal corpora and integrated with case-law retrieval, supporting tasks such as precedent retrieval and judgment prediction. In China, LawGPT [12] combines legal-oriented pre-training with a large-scale legal corpus and employs legal-supervised fine-tuning on legal instruction datasets to enhance its effectiveness in handling Chinese legal tasks. Lawyer LLaMA [13] injects legal knowledge through continual pre-training on a large-scale legal corpus, followed by supervised fine-tuning on expert-generated legal datasets. InternLM-Law [14] further demonstrates strong performance on LawBench through a two-stage supervised fine-tuning pipeline. In addition, SaulLM (SaulLM-54B/141B) achieves large-scale domain adaptation with 500B+ legal tokens and human-aligned tuning [50]. LawMate Romania² is fine-tuned on Romanian legal texts for tasks under Romanian law. LexGPT, pretrained on the Pile of Law dataset, supports no-code fine-tuning for legal tasks [51]. Lawma offers open-weight LLaMA-3 derivatives fine-tuned on the 260-task CaseLawQA benchmark [52]. These legal-domain LLMs pair generative capacity with domain specialization, laying the groundwork for the next stage of tool-augmented and interactive legal systems.

Table 4 presents an overview of representative LLMs in the legal domain, highlighting their base architectures and models.

² <https://github.com/DorobantuDiana/Legal-LLM>.

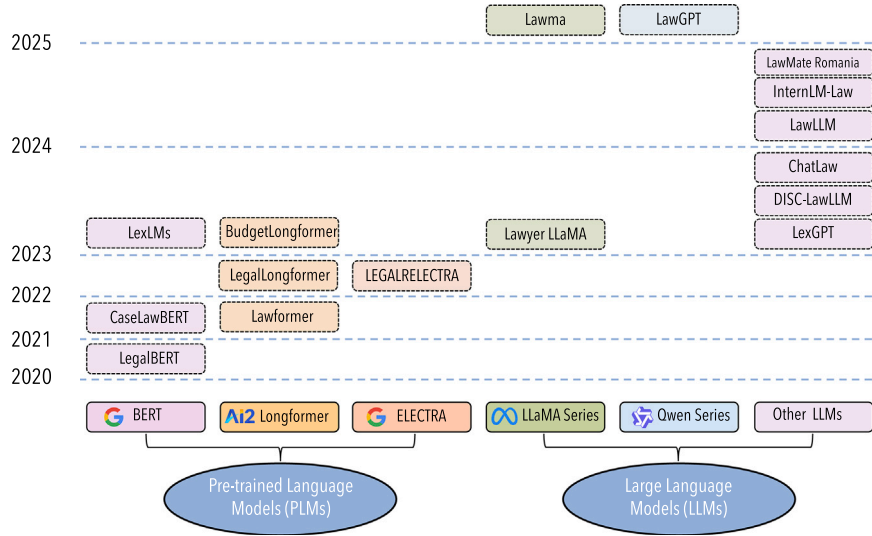


Fig. 4. An overview of legal models introduced since 2020, grouped by foundational architectures.

Table 4

Representative LLMs in the legal domain.

Legal LLMs	Base LLMs	#Parameters	Open source
LawLLM [49]	Gemma	7B	Yes ^a
Lawyer LLaMA [13]	Chinese-LLaMA	13B	Yes ^b
LawGPT [12]	Qwen-2.5	0.5B / 1.5B / 3B	Yes ^c
ChatLaw [53]	InternLM	13B / 33B	Yes ^d
InternLM-Law [14]	InternLM2	7B	Yes ^e
DISC-LawLLM [54]	Baichuan	13B	Yes ^f
LawMate Romania	Saul	7B	Yes ^g
SaulLM [50]	Mixtral	7B	No
LexGPT [51]	GPT-J	6B	Yes ^h
Lawma [52]	LLaMA 3	8B / 70B	Yes ⁱ

^a https://github.com/Tizzzy/Law_LLM

^b <https://github.com/AndrewZhe/lawyer-llama>

^c <https://github.com/pengxiao-song/LaWGPT>

^d <https://github.com/PKU-YuanGroup/ChatLaw>

^e <https://github.com/InternLM/InternLM-Law>

^f <https://github.com/FudanDISC/DISC-LawLLM>

^g <https://github.com/DorobantuDiana/Legal-LLM>

^h <https://github.com/jiehsheng/LexGPT>

ⁱ <https://github.com/socialfoundations/lawma>

3.1.3. Tool-augmented legal LLMs

Recent research demonstrates that augmenting LLMs with tools significantly enhances performance in the legal domain. A common approach involves integrating external retrieval or knowledge systems into legal LLMs to improve accuracy and reduce hallucinations. For instance, Lawyer LLaMA [13] integrates a legal statute retriever that dynamically fetches up-to-date laws for marriage-related queries, grounding the model's responses in authoritative texts. Similarly, DISC-LawLLM [54] employs retrieval modules alongside specialized “syllogism” prompts, enabling it to cite actual legal provisions when drafting advice. By augmenting the prompt with retrieved passages, these systems generate more factually accurate responses than those produced by a basic LLM. In practice, tool-augmented models significantly reduce hallucinations. One study found that a custom-trained case-law model used by Harvey provided 83% more factual answers than vanilla GPT-4 and was preferred by attorneys 97% of the time.³ In contrast, non-augmented chatbots (e.g., GPT-4) often cite nonexistent cases or outdated laws.

Beyond retrieval augmentation, recent research on human-LLM collaboration regards LLMs as complements rather than substitutes for legal expertise. Studies have shown the increasing integration of human-LLM collaboration into judicial and legal information workflows, encompassing decision support, opinion drafting, and legal retrieval. This collaboration has resulted in measurable improvements in both efficiency and coverage [55–57]. These advancements also highlight concerns related to reliability, interpretability, and governance, driving the design of logically disciplined models and rigorous validation protocols [56]. Furthermore, in legal QA, human-LLM collaboration that incorporates interactive clarification (e.g., jurisdiction and key facts) and user-guided retrieval has been shown to enhance accuracy, transparency, and practical utility for both legal professionals and non-experts [58,59]. This suggests that human-in-the-loop workflows offer a promising approach for developing trustworthy, practitioner-centered legal AI.

In parallel with academic research, legal tech companies are deploying RAG-style LLM platforms for law firms. For example, KL3M⁴ and Harvey⁵ are commercial legal AI platforms that integrate RAG systems. KL3M, trained on a curated corpus of legal, regulatory, and financial texts, offers enterprise-focused models optimized for on-site deployment and fine-tuning, ensuring compliance with legal data privacy regulations. Similarly, Harvey, built on OpenAI's GPT models, processes billions of case-law tokens and combines retrieval and generation to draft analyses and summaries from uploaded contracts or litigation files. Both systems emphasize seamless integration into legal workflows, enhancing tasks such as due diligence, contract review, and multi-document comparison.

3.2. Zero-shot vs. fine-tuning

Zero-shot and fine-tuning are two distinct adaptation methods for applying LLMs to legal tasks. Zero-shot or few-shot learning refers to a model's ability to perform tasks it has not been explicitly trained on, relying on its pre-existing knowledge and generalization capabilities [5]. In contrast, fine-tuning involves further training a pre-trained model on a task-specific dataset to improve its performance on the particular task [60]. Each approach presents trade-offs in terms of data requirements, implementation effort, and overall performance.

³ <https://openai.com/index/harvey/>.

⁴ <https://huggingface.co/alea-institute/kl3m-002-170m>.

⁵ <https://www.harvey.ai/legal>.

Zero-shot and Few-shot for Legal Tasks. LLMs such as GPT-4 possess extensive general knowledge, enabling them to perform effectively in zero-shot settings. For example, one can directly prompt ChatGPT with queries such as “summarize this contract” or “is this clause enforceable under California law?” and obtain a reasonable response [6]. Prompting enables rapid prototyping of legal AI systems without the need for task-specific labeled data.

Early studies explored the use of LLMs in zero-shot settings for legal question-answering, document drafting, and reasoning [61–64]. These models show mixed results, achieving strong performance in certain tasks while exhibiting limitations in others. Conversational LLMs excel at understanding legal texts and often outperform smaller supervised models in tasks like document classification [65]. However, they frequently underperform on challenging legal benchmarks, revealing limitations in handling complex legal reasoning. For example, ChatGPT performed well on LexGLUE but lagged behind specialized models on several tasks [65]. Similarly, GPT-3 outperformed earlier systems on the SARA benchmark but struggled with precise legal logic and unfamiliar scenarios [66]. These findings suggest zero-shot LLMs may provide plausible but unreliable legal analyses. Prompt engineering, such as providing legal definitions or reasoning steps, can improve results [64], but core knowledge remains limited by pretraining.

Fine-Tuning for Legal Tasks. Fine-tuning involves training LLMs on domain-specific data or task-specific data. This process can significantly boost performance when high-quality legal datasets are available. For instance, LegiLM [67] fine-tuned the SaulLM-7B model on a GDPR compliance dataset and evaluated it on a legal QA benchmark, significantly outperforming both the zero-shot SaulLM-7B and GPT-4. To further enhance fine-tuning in the legal domain, an “adapt-retrieve-revise” fine-tuning framework [68] trains a smaller LLM on a corpus of legal cases, reducing the model’s tendency to generate irrelevant or inaccurate content in Chinese legal advice. These results demonstrate the substantial performance gains achievable through domain-specific fine-tuning on high-quality legal data.

Fine-tuning can also be performed via reinforcement learning. For instance, Nguyen et al. [69] used ChatGPT’s outputs as pseudo-labels (treated as noisy answers) and trained a label model to refine them, achieving state-of-the-art performance in legal textual entailment. This strategy effectively adapts general LLMs using task-specific supervision signals.

Advantages and Limitations. Zero-shot prompting does not require task-specific training data, relying instead on the model’s general knowledge from pretraining. If public laws and legal cases are included in the training corpus, the model may possess substantial legal information and handle legal tasks without extra supervision. However, these models often fail to apply legal reasoning reliably and may generate plausible but incorrect facts or citations. For example, a Stanford study found that general LLMs produced hallucinated responses to legal queries in 58% to 82% of cases, highlighting the risks of using them in critical legal contexts [70]. In contrast, fine-tuning can better align the model with domain requirements, improving factual accuracy and legal terminology. For instance, Hu et al. [71] proposed a two-stage fine-tuning framework that reduced hallucinations and improved statute relevance and factual consistency. However, fine-tuning requires high-quality annotated data, which is often scarce and expensive in law, and is computationally intensive for LLMs.

Few-shot prompting improves LLM outputs by including a small number of examples, such as question–answer pairs or summaries, in the prompt without retraining the model. Studies show that few-shot prompting significantly boosts performance on legal tasks [72,73]. For instance, Yu et al. [72] found that prompts using structured legal reasoning methods like IRAC achieved the best results on the COLIEE task. Other research also demonstrates that few-shot prompting enhances adaptability and performance in legal text classification, especially when dealing with complex terminology and lengthy documents [73].

To mitigate the high computational cost and overfitting risks associated with traditional fine-tuning, Parameter-Efficient Fine-Tuning (PEFT) techniques have been developed. These methods integrate domain-specific knowledge into the model without modifying its core weights, making the adaptation process more efficient and less prone to overfitting [74–76].

3.3. Why applying LLMs in laws?

LLMs are reshaping the legal field by enabling professionals to process large volumes of data, enhance reasoning, and automate routine tasks, thereby increasing efficiency and reducing costs. This section outlines the main benefits of LLM integration, emphasizing their role in addressing complex legal issues, supporting decisions, and expanding access to legal services.

Handling Volume and Complexity. Legal work often requires processing large volumes of text, such as thousands of pages of evidence, lengthy contracts, and extensive corpora of case law. LLMs can efficiently parse and summarize these materials, significantly reducing manual effort [36]. For instance, LLM-based systems can condense lengthy judgments into concise summaries that highlight key facts and legal holdings, enabling lawyers and judges to access essential information more efficiently [77]. Moreover, LLMs can track complex cross-references and dependencies in legal texts more effectively than manual review [10]. Advances in model architecture and context window length further enhance their suitability for document review and e-discovery tasks.

Legal Reasoning and Decision Support. Recent studies demonstrate that LLMs possess reasoning abilities such as logical inference, analogical reasoning, and step-by-step problem solving [78]. In the legal domain, these capabilities support reasoning tasks such as applying rules to facts and comparing case elements with relevant precedents [79]. Although current models remain limited, studies have shown that they can identify applicable legal principles and predict case outcomes based on factual descriptions [23]. The application of LLMs in judicial decision-making (e.g., forecasting a court’s decision) is detailed in Section 4.3. These models support decision-making by rapidly aggregating relevant precedents and statutory provisions, and proposing plausible reasoning paths. Notably, COT prompting allows LLMs to generate explanations and predictions.

Efficiency and Cost Reduction. LLMs improve legal service efficiency by automating routine tasks such as contract drafting, proofreading, formatting, and clause extraction. For instance, LLMs can identify inconsistencies or ambiguous language in contracts and suggest clearer alternatives, reducing manual workload and costs. Tools like Harvey AI,⁶ used by major firms such as Allen & Overy, can draft memos and contracts quickly while maintaining quality. Early adoption shows that LLMs enable faster legal work without compromising standards. In the public sector, governments are using LLMs to automate form processing, compliance checks, and document generation, further reducing costs and accelerating administrative processes.

Improved Legal Research and Retrieval. Traditional legal research relies on keyword-based searches in databases such as Westlaw⁷ and LexisNexis,⁸ which can be time-consuming and inefficient. LLMs enable more semantic and conversational research, allowing users to ask complex questions and receive direct answers or summaries. Compared to Boolean searches, LLMs retrieve and summarize relevant cases more effectively, streamlining research and reducing the risk of missing key precedents. LLM-based tools also support follow-up queries and clarifications, serving as AI research assistants and further enhancing efficiency [80].

⁶ <https://www.harvey.ai>.

⁷ <https://www.westlaw.com>.

⁸ <https://www.lexisnexis.com>.

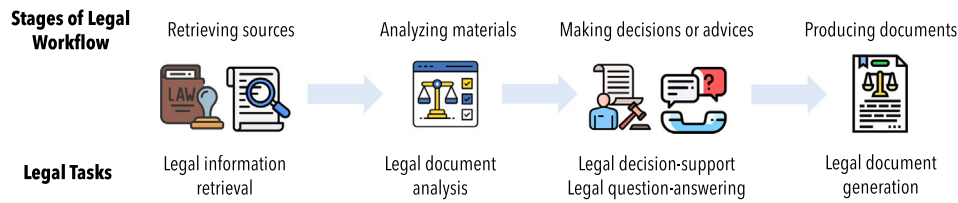


Fig. 5. Typical workflow of legal practice and the corresponding LLM-based legal tasks.

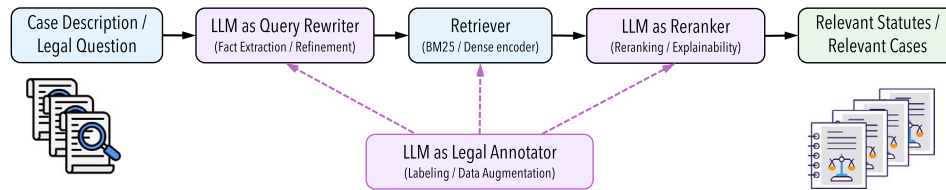


Fig. 6. An overview of an LLM-based pipeline for legal retrieval, including query rewriting, retrieval, reranking, and data annotation, which can be adapted for both legal case retrieval and statutory article retrieval.

Accessibility and Legal Assistance. LLMs hold significant promise for improving access to justice for non-lawyers. On the one hand, individuals often struggle to interpret complex legal language or obtain basic legal assistance due to financial or informational barriers. On the other hand, conversational LLMs can serve as interactive aides, offering preliminary guidance on routine matters. For example, chatbots such as DoNotPay⁹ help users draft wills, dispute tickets, or navigate court procedures with clear explanations and tailored questions. These tools can narrow the justice gap by making legal information more accessible to those unable to afford professional representation.

4. Applications of LLMs in legal tasks

This section surveys the main applications of LLMs in legal tasks and organizes them along the typical workflow of legal practice. In real-world settings, legal work progresses from retrieving authoritative sources and facts, to analyzing and structuring materials, applying rules to support decisions or advice, and producing formal documents. Aligned with this workflow, we categorize LLM-based legal tasks into five groups: (i) legal information retrieval, (ii) legal document analysis, (iii) legal decision-support, (iv) legal question answering, and (v) legal document generation. Fig. 5 illustrates the mapping between each stage of the legal workflow and the corresponding group of LLM-based legal tasks. Furthermore, we review legal agent-based modeling, which integrates these tasks into multi-stage workflows through autonomous, specialized agents. Finally, we review LLM-based tasks in legal education and training, where models are used to support exam preparation, interactive case analysis, and professional development for legal practitioners.

4.1. Legal retrieval tasks

Legal retrieval is a core task in legal research and practice, as it helps lawyers and judges quickly locate relevant case law and statutes. For example, in common law systems, case retrieval is crucial for maintaining judicial consistency and supporting reasonable judgments, with lawyers spending approximately 30% of their time searching for relevant precedents to construct arguments. In civil law systems, while case law is less authoritative than in common law, retrieving relevant statutes and regulations remains a key task.

Recent studies have introduced LLM-based methods to enhance the efficiency of information retrieval (IR), including query rewriting [81–

83], retrievers [84–88], reranking [84,89–93], and generative retrieval techniques [94–98]. Although originally developed for open-domain IR, these techniques can be adapted for legal retrieval with task-specific adjustments, including query reformulation, dense-encoder retrieval, and reranking. Fig. 6 illustrates an LLM-based legal retrieval pipeline, where case descriptions or legal questions are transformed into structured queries, retrieved from case or statutory corpora, reranked using LLM-based reasoning, and further enhanced with LLM-generated relevance annotations.

4.1.1. Legal case retrieval

Legal case retrieval (LCR) is the task of identifying relevant prior court cases (precedents) given a new case or legal query [2]. Traditionally, lawyers have relied on keyword searches in case law databases to locate decisions with similar facts or legal issues (e.g., using court databases like Wenshu in China¹⁰). This task is foundational to legal research, as retrieving relevant precedents promotes consistency in judicial reasoning and strengthens arguments through authoritative support. However, LCR remains challenging: legal queries are often lengthy narratives containing domain-specific terminology, and relevant precedents may be phrased differently.

Early neural models and PLMs established semantic and context-aware retrieval in the legal domain. For instance, SAILER [99] incorporates document structure and legal-specific indicators during pretraining to better capture legal nuances and enhance retrieval effectiveness. A recent survey [2] highlights key advances in LCR methods and reinforces their foundational role in legal AI. Building on PLM-era methods, LLMs further advance legal retrieval by improving query understanding, long-context modeling, and reranking.

LLM as Query Rewriter. LLM-based methods extend general IR techniques to LCR by adapting to the structural and linguistic features of legal texts. A common strategy involves leveraging LLMs as query rewriters to process lengthy case descriptions and generate focused queries that capture essential factual or legal elements. For instance, prompting GPT-style models to summarize or extract key points from case narratives yields concise queries that significantly improve retrieval performance. Zhou et al. [100] demonstrated that prompting an LLM to distill long case texts into critical facts enhances both sparse and dense retrieval outcomes. Similarly, the knowledge-guided KELLER system [101] reformulates complex legal queries into structured sub-facts by leveraging LLMs to extract crimes and relevant legal articles

⁹ <https://en.wikipedia.org/wiki/DoNotPay>.

¹⁰ <https://wenshu.court.gov.cn/>.

from the case. This approach enhances retrieval performance by focusing on key case elements and improves interpretability through the use of structured sub-facts.

LLM as Reranker. LLM-based retrievers and rerankers have demonstrated strong performance in general IR [84,89–93]. For instance, LLMs adapted as dense bi-encoders (e.g., RepLLaMA [84]) enhance long-document retrieval by encoding full texts without aggressive segmentation. LLM rerankers (e.g., RankLLaMA [84]) outperform earlier T5-based models on standard ranking tasks. However, existing research has not focused on training LLMs for LCR. Most LCR pipelines still rely on a combination of lexical recall (e.g., BM25), PLM-based dense retrieval, and cross-encoder reranking. Developing and fine-tuning LLMs for legal case retrieval and ranking remains a promising area for future work.

LLM as Legal Annotator. LLMs have also been employed as surrogate legal experts to address the scarcity of high-quality training data for case retrieval, as manual labeling is resource-intensive and requires legal expertise. Ma et al. [102] introduced a few-shot prompting framework that guides general LLMs through a series of relevance assessment steps, simulating human annotators. By gradually providing more context and criteria, the model generates more consistent and accurate judgments. Empirical results show that LLM-generated labels closely align with expert annotations and significantly improve retrieval performance, demonstrating the cost-effectiveness of LLM-generated relevance annotations, particularly in low-resource legal domains.

4.1.2. Statutory article retrieval

Statutory article retrieval (SAR) involves retrieving relevant statutes or regulations based on a question or fact pattern. SAR is crucial for answering legal questions and supporting compliance checks. For example, a query like “Can an employee be terminated without notice under the Labor Code?” requires identifying the relevant statutory provisions. Since statutory documents are concise and written in formal language, effective SAR involves bridging the lexical gap and understanding context to retrieve relevant statutory provisions.

LLM as Query Rewriter. Recent studies in SAR primarily leverage LLMs for enhancing query understanding and expanding queries to improve retrieval performance. For instance, Nguyen et al. [103] used GPT-style prompting to infer implicit legal concepts in the input question and insert the corresponding statutory terms before retrieval. This LLM-guided expansion improves recall and yields competitive results on COLIEE statutory retrieval tasks. Wang et al. [104] proposed an LLM-augmented framework that generates hypothetical answers and statute-like passages as pseudo-documents for semantic expansion, improving both recall and precision without task-specific fine-tuning. Similarly, Chen et al. [105] evaluated LLM-based retrieval baselines, including HyDE (LLM-generated pseudo-documents embedded for vector search) and Query2Doc (LLM-generated expansions concatenated with the original query), and reported consistent gains over vanilla sparse and dense retrievers, demonstrating the effectiveness of LLM-aided expansions as baselines for SAR.

LLM as Reranker. In SAR pipelines, LLMs are effective during the retrieval and reranking stages, where they reassess candidate statutes based on the query intent. For example, Nguyen et al. [106] proposed a three-stage framework that incorporates BM25 for recall, a fine-tuned BERT reranker, and a GPT-4-based final reranking step via prompting. This approach improves performance on the COLIEE benchmarks, demonstrating the potential of LLMs as reasoning tools in retrieval pipelines. Another approach [107] used a “Retrieve, Revise, and Refine” framework: small fine-tuned models generate a broad set of candidates, an LLM checks and combines these into a shorter set, and small models then refine the results for precision. Trung et al. [108] introduced a LLaMA-based dual-encoder retriever trained in two phases: Phase I learns global context from positive examples and easy or general

negatives, while Phase II trains only on hard negatives to improve discrimination on challenging Japanese legal queries.

4.1.3. Limitations for legal retrieval

Despite the significant potential of LLMs in legal retrieval, several limitations remain, particularly in cross-jurisdictional applications and complex legal reasoning tasks. First, the scarcity of high-quality, domain-specific, and annotated data across jurisdictions limits the transferability of LLMs between legal systems. For instance, case retrieval in common law systems and statute retrieval in civil law systems require distinct training distributions that current models rarely capture. Second, generative retrieval techniques, such as predicting document identifiers or producing candidate snippets, are still in the early stages within the legal domain. Although some models can generate text relevant to legal queries, they often struggle to produce precise legal citations and perform zero-shot reranking for complex case retrieval. Finally, LLMs offer limited explainability: legal professionals require explicit citation grounds and reasoning for retrieved results, yet systems frequently return cases or statutes without adequate justification. Therefore, LLM-based retrieval still needs to be coupled with traditional retrieval techniques and human review to ensure accuracy and reliability.

4.2. Legal document analysis tasks

Legal document analysis is a critical task in the legal domain, as it helps legal professionals to efficiently process and extract relevant information from lengthy and complex texts, such as court judgments, contracts, statutes, and pleadings. Tasks such as summarization, argument mining, and contract analysis are particularly crucial for automating legal workflows and enhancing efficiency. In common law systems, where judicial decisions often involve detailed fact patterns, legal arguments, and numerous precedents, effective document analysis is essential for understanding the reasoning behind rulings. In civil law systems, while legal documents are typically more concise and focused on statutory application, document analysis remains vital for identifying relevant legal provisions and ensuring compliance with legal standards. This section reviews recent advancements in using LLMs to address key tasks in legal document analysis. Fig. 7 illustrates the use of LLM-based methods to process legal documents and contracts for legal summarization, argument mining, and contract analysis.

4.2.1. Legal documents summarization

Legal documents (i.e., case opinions, contracts, regulations, and briefs) are often long and complex, making summarization valuable. It helps lawyers, judges, and clients quickly grasp key points, such as a judgment’s holding and reasoning or a contract’s main obligations. Recent studies have assessed the effectiveness of LLM-based summarization in the legal domain. Deroy et al. [109] found that while abstractive models produce more informative summaries than extractive methods, they are still prone to errors and omissions. To enhance reliability, a human-in-the-loop approach with manual fact-checking of LLM outputs has been proposed [109].

To improve factual consistency, researchers are developing techniques to ground LLM-generated summaries in the source text or external knowledge. RELEXED [110] retrieves exemplar summaries of conceptually similar cases to guide the generation process. By anchoring the output to relevant examples, the model reduces omissions and extraneous content, resulting in summaries that more closely reflect human-written style and focus. Sie et al. [111] proposed a two-stage pipeline for summarizing long regulatory documents, first extracting salient sections, then generating abstractive summaries.

Several studies fine-tune LLMs to enhance the performance of legal document summarization tasks. Deroy et al. [112] compared domain-specific and general LLMs for case judgment summarization across

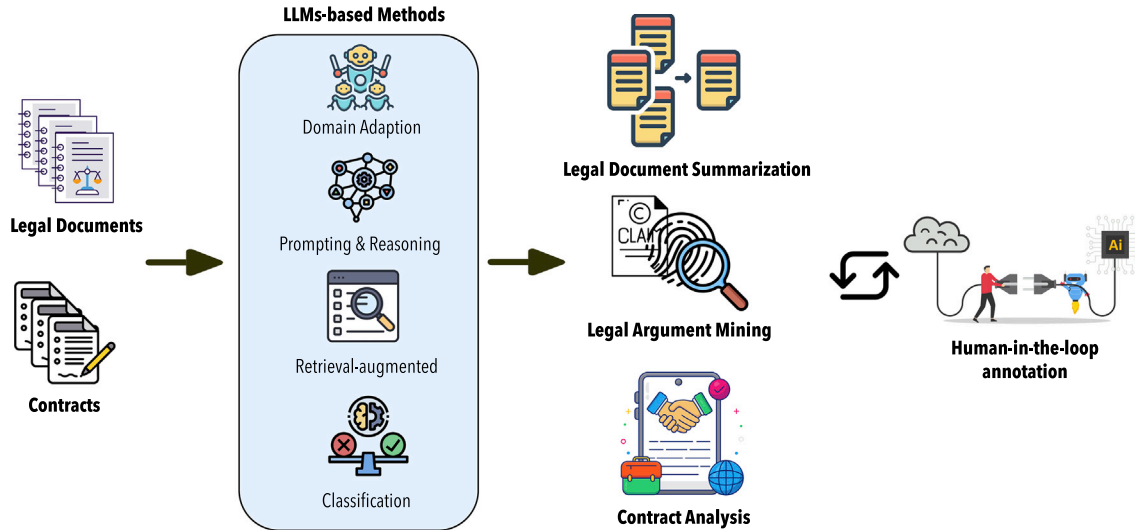


Fig. 7. Overview of an LLM-based pipeline for legal document analysis.

UK, Indian, and U.S. datasets. The study identifies common hallucination and inconsistency patterns and explores mitigation strategies such as domain-specific fine-tuning, targeted prompting, and a semantic similarity-based method. LLaMandement [113], fine-tuned from LLaMA, generates neutral and concise summaries of French legislative amendments and bench memoranda, streamlining amendment processing and improving both efficiency and accuracy in parliamentary workflows. Similarly, LexT5 [114], pre-trained on large legal corpora and fine-tuned on LexSumm, achieves state-of-the-art results in English legal summarization, especially in abstraction and factual accuracy. Benedetto et al. [115] introduced a lightweight LoRA-based fine-tuning framework for Italian seq2seq models (mBART, BART-IT, IT5, Flan-T5, and Llama 2), significantly improving grammaticality, informativeness, and few-shot performance in legal news headline and summary generation.

To address limited training data in low-resource legal domains, Liu et al. [116] leveraged LLMs for data augmentation and evaluation. This approach improves few-shot performance and reduces reliance on large human-annotated datasets for legal summarization.

Legal summarization is increasingly part of broader legal-tech workflows. Mentzingen et al. [117] showed that combining summaries with LLMs significantly improves precedent retrieval from case law databases. Converting lengthy judicial decisions into vector-searchable abstracts enables more efficient retrieval without full-text scanning.

4.2.2. Legal argument mining

Legal argument mining is the task of automatically detecting argumentative components in legal texts (e.g., claims, evidence), classifying their roles, and extracting support-attack relations to reconstruct argument structures. Recent work has introduced specialized benchmarks to evaluate LLMs on these abilities. Chlapanis et al. [118] assessed the performance of LLMs on the legal argument reasoning dataset LAR-ECHR. The task requires the model to select the correct next statement in a chain of legal arguments based on the case facts, thereby testing its ability to follow and predict judicial reasoning steps. Evaluations on LAR-ECHR demonstrate that even GPT-4 achieves only 75.8% accuracy on this multiple-choice reasoning task, indicating considerable potential for further improvement.

Another key aspect is stance classification, which determines whether a cited case or argument supports or refutes a legal position. Gupta et al. [119] introduced the δ -Stance dataset, a large-scale corpus derived from U.S. judicial opinions that pairs court arguments with

cited precedents and labels both polarity and intensity. Compared with zero-shot prompting of state-of-the-art LLMs, fine-tuning smaller LLMs on δ -Stance improves stance classification performance, emphasizing the value of domain-specific fine-tuning.

4.2.3. Contract analysis

Contract analysis involves reviewing contracts for high-risk clauses, extracting structured data (such as parties, dates, and obligations), checking compliance with standards or policies, and comparing clauses against reference templates. Recent advances in automated contract review using LLMs have greatly improved clause extraction, compliance checking, and risk assessment [120]. A foundational dataset, CUAD [62], annotated for risky clauses, has facilitated the development of robust baseline models like BERT and T5. GPT-4-based methods [121] further enhance efficiency by detecting risky clauses and generating explanatory insights.

A key advancement is using LLMs for comparative analysis of draft contracts against templates, achieving high accuracy in identifying deviations [16]. Studies show LLMs can match or surpass human performance in routine legal reviews. For example, GPT-based models match junior lawyers in accuracy while greatly reducing review time and costs [122,123].

To address domain-specific complexities without extensive retraining, researchers have integrated structured knowledge systems and domain-specific prompting strategies. For example, Zheng et al. [124] combined knowledge graphs with LLMs to improve risk identification in construction contracts. Similarly, Wang et al. [125] demonstrated that embedding domain-specific knowledge into prompts improved accuracy in construction contract review tasks, enhancing risk identification without fine-tuning. LegiLM [67] fine-tuned for privacy law compliance, achieves superior accuracy in detecting regulatory violations. Furthermore, complex prompting techniques have been developed to improve LLM performance in complex question-answering tasks within contract review. Structured, multi-stage prompt chaining significantly enhances clause-specific reasoning capabilities and reduces errors when interpreting complex legal texts [126,127].

Efforts have also focused on enhancing the practical usability of automated contract review outputs. Singhal et al. [128] developed systems that identify ambiguous clauses and generate relevant clarification questions, significantly enhancing contract clarity. Similarly, Aspromonte et al. [129] employed LLM-driven explanations to make contract decisions more understandable, translating complex legal content into accessible language.

Despite progress, interpreting nuanced legal language remains challenging. Previous studies note that LLMs still struggle to accurately classify clauses as unfair or void, emphasizing the difficulty LLMs face in interpreting complex legal contexts accurately [130,131].

4.2.4. Limitations for legal document analysis

While LLMs enhance legal workflows, key limitations persist. The legal document summarization task remains prone to factual errors: existing LLM-based models often omit crucial details or generate incorrect facts, requiring human fact-checking. In legal argument mining, even state-of-the-art models make frequent mistakes: for example, GPT-4 scores below 80% on multi-step legal inference (LAR-ECHR), demonstrating that it cannot yet be fully relied upon to follow complex judicial logic [118]. In contract analysis, tasks such as fairness assessment are particularly challenging: no model exceeds an F1 score of 0.54 when classifying voidable clauses in German consumer contracts, and LLMs perform only marginally better than chance in identifying unfair terms in online policies [131]. These weaknesses highlight the difficulty of capturing legal nuance and the risk of overconfidence. LLMs also exhibit biases and gaps in training data, and legal reasoning often requires extra-judicial context (such as precedents and statutory nuance) that models may not capture.

Overall, LLMs are robust tools for document summarization, legal argument parsing, and contract analysis, offering substantial improvements in processing speed and often matching or exceeding human performance in legal document tasks. However, to mitigate hallucinations and enhance reliability, future research is expected to integrate LLM capabilities with expert validation, case law retrieval, and specialized knowledge repositories to address existing limitations.

4.3. Judicial decision tasks

Judicial decision tasks are crucial in the legal field, as they involve applying legal principles, statutes, and precedents to the facts of a case to make a judgment. These tasks include legal statute identification, statutory reasoning, legal judgment prediction, and court view generation. In legal practice, these tasks present significant potential. For instance, statutory reasoning helps determine which legal provisions apply to a particular case, assisting lawyers in preparing legal opinions. Legal statute identification ensures that judges and lawyers do not overlook critical legal references, especially in complex or cross-jurisdictional contexts. Legal judgment prediction can support litigation strategy and risk assessment by predicting case outcomes, such as the probability of a judgment or sentencing range. However, predicting judicial decisions serves as a supplementary tool rather than a substitute for judicial discretion. Therefore, these tasks enhance the consistency of legal analysis by acting as an additional check to identify missed legal references in judgments or uncover factors that may influence the case outcome. Fig. 8 illustrates the overall framework of judicial decision

tasks, summarizing the relevant LLM-based methods involved in each task and their interrelationships.

4.3.1. Legal statute identification

Legal statute identification (LSI), also known as law article prediction, involves identifying the statutory provisions that are relevant and applicable given a case description or fact pattern. Recent studies have explored LLM-based approaches for LSI. Vats et al. [132] assessed GPT-3 and PLM-based methods on the Indian statute prediction task. While GPT-3 demonstrated zero-shot capability, fine-tuned PLM-based methods achieved superior performance. Additionally, Paul et al. [133] evaluated both PLM-based models and zero-shot LLMs (GPT-3.5 and LLaMA-2), finding that PLM-based models achieved significantly higher macro-F1 scores, while zero-shot LLMs prioritized recall at the expense of precision. This suggests that while zero-shot LLMs show potential, fine-tuned PLM-based models outperform them in the LSI task. To address these limitations, Uni-LAP [134] integrates a supervised classifier with an LLM: a smaller model first filters candidate statutes, and the LLM then refines and ranks them. This coupling leverages the precision of specialized models and the broad knowledge of LLMs, improving accuracy across jurisdictions.

4.3.2. Statutory reasoning

Statutory reasoning is the task of determining whether a legal statute, stated in natural language, applies to the text description of a case. Blair-Stanek et al. [66] explored GPT-3's performance on statutory reasoning tasks using the SARA dataset, which involves applying U.S. tax code statutes to facts presented in natural language. The results demonstrate that GPT-3 outperforms BERT-based models, particularly when using 4-shot dynamic prompting and zero-shot prompting. However, GPT-3 still makes significant mistakes when handling complex or unseen statutes, highlighting the necessity for further improvements in its reasoning capabilities within the legal domain.

4.3.3. Legal judgment prediction

Legal judgment prediction (LJP) is the task of predicting the outcome of a court case from the case's facts, such as liability or guilt, charges, and applicable law articles [135,136]. Recent studies enhance LJP with LLMs by integrating relevant legal information retrieval and incorporating structured reasoning and domain knowledge into the prediction process.

Retrieval-Augmented LJP. Retrieval-augmented approaches integrate external legal resources (e.g., statutes, precedents, and doctrinal analyses) into LLMs to enhance accuracy and mitigate knowledge hallucination. PLJP [137] retrieves prior cases ("precedents") with similar facts and provides them to an LLM. A smaller domain model selects candidate precedents and generates initial labels, which are then used as in-context examples by a larger model to produce the final judgment. Similarly, GEAR [21] is a law-guided generative retrieval framework

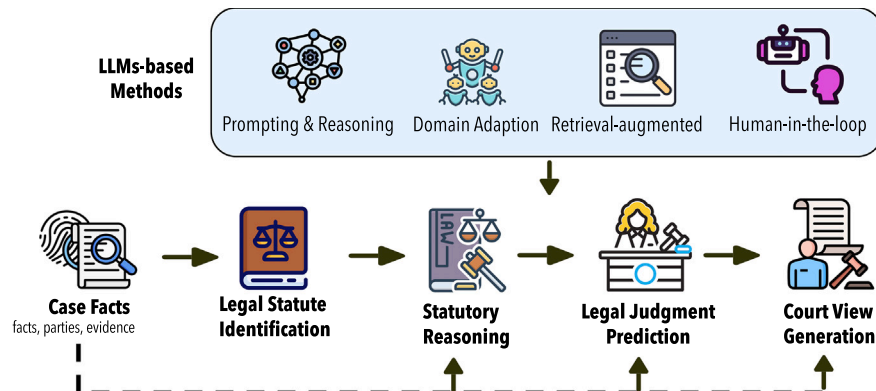


Fig. 8. Overview of judicial decision tasks, summarizing the relevant LLM-based methods involved in each task and their interrelationships.

that incorporates structured legal rationales and hierarchical semantic identifiers, optimized with a joint revision loss for both retrieval and prediction tasks. Athena [138] employs semantic search over vectorized legal accusations to retrieve relevant information, achieving state-of-the-art results on Chinese criminal law datasets. However, Shui et al. [22] found that if the base LLM lacks strong reasoning ability, integrating retrieval systems yields limited improvements. This highlights the importance of a model's capacity to interpret and effectively utilize external legal knowledge.

Reasoning Enhanced LJP. A second line of work strengthens LJP by enhancing the reasoning process of models. Instead of black-box prediction, these methods incorporate legal logic, multi-step inference, and structured analyses into the LLM workflow, aligning reasoning with legal principles and improving predictive accuracy. Legal Syllogism Prompting (LoT) [139] is a zero-shot prompting method that guides LLMs to structure the reasoning as a legal syllogism—specifying the law (major premise), case facts (minor premise), and judgment (conclusion). This simple yet effective prompting strategy outperforms generic COT prompting by focusing on relevant facts and correctly linking legal provisions and justifications to the predicted outcome. In another study, Wei et al. [140] proposed a neuro-symbolic framework that integrates LLMs with first-order logic rules. The model is trained to align with a legal rule knowledge base using a discrepancy loss and employs COT prompting to extract key factual elements as logical variables. Enforcing consistency between neural outputs and symbolic rules improves both prediction accuracy and interpretability.

There has also been interest in simulating how judges logically structure their decisions. In criminal law, judicial reasoning often follows a three-step (trichotomous) structure: (1) determine whether an offense occurred, (2) assess whether it was unlawful, and (3) evaluate the defendant's culpability. Zhang et al. [141] argued that LJP models should reflect this trichotomy to improve robustness. They fine-tune LLMs with a training objective that encodes this stepwise structure (offense, unlawfulness, then culpability). The adapted model better identifies innocent defendants and significantly increases F1 scores for charge prediction. However, absolute performance remains low (F1 below 0.3 in their experiments), indicating that fully capturing legal reasoning in models remains an open challenge. Furthermore, Deng et al. [142] compiled extensive case datasets with structured argument chains, highlighting the necessity of reasoning structures to enhance user trust. Similarly, frameworks like ADAPT [23] and CASAM [143] incorporate structured reasoning and causality-aware mechanisms, significantly improving the performance and robustness of LLMs in LJP tasks.

Domain-Specific LLMs for LJP. A complementary direction for improving LJP is to fine-tune models for the legal domain or specific

jurisdictions. Domain adaptation may include training on legal text corpora, fine-tuning on task-specific data, or developing custom legal LLMs. The intuition is that general-purpose models (such as GPT-3 or GPT-4) may not capture the nuances of a given legal system, whereas a model thoroughly trained on that legal corpus will perform better. For example, Trautmann et al. [63] found that GPT-3 supports effective multilingual zero-shot prediction, but domain-specific supervised models perform better. Similarly, Vats et al. [132] and Kmainasi et al. [144] applied fine-tuning and prompt-based methods to the Indian and Saudi legal domains, respectively, showing that domain specialization improves performance even in smaller models. LawLLM [49], a domain-specific legal LLM trained on large-scale U.S. case law and fine-tuned on multiple downstream tasks, including LJP, consistently outperforms general LLMs on these tasks. This provides further evidence that domain-specific adaptation can substantially enhance LJP performance.

In addition, researchers have developed model architectures for processing lengthy legal cases. MESc [145] is a multi-stage, encoder-supervised fusion clustering framework for LJP that segments long legal documents into overlapping chunks, extracts chunk-level embeddings from a fine-tuned LLM, clusters them to approximate document structure, and integrates these representations with a global Transformer for final classification.

Human-LLMs Collaborative LJP. Recent research highlights the complementary role of LLMs in supporting, rather than replacing, human legal expertise. Han et al. [55] proposed LegalAsst, a human-in-the-loop system that enables judges to iteratively refine model-generated judgments, balancing the efficiency of automation with essential human oversight to ensure reliability and trust. Additionally, Liu et al. [56] documented real-world applications in Shenzhen, China, where judges are increasingly utilizing LLMs to assist in drafting judicial opinions. This trend illustrates the integration of AI-driven tools into judicial workflows, while also highlighting the need for rigorous oversight.

Furthermore, Izzidien et al. [57] demonstrated that Anthropic Claude outperformed expert-crafted keyword searches in identifying “summary judgment” cases within a UK case law corpus, showcasing the effectiveness of LLMs in legal information retrieval and dataset curation tasks that have traditionally relied on manual heuristics. Nevertheless, as highlighted by Liu et al. [56], the increased deployment of LLMs in judicial practice raises critical questions concerning reliability, interpretability, and governance, emphasizing the necessity for logically robust model designs and stringent validation procedures.

Explainability of LJP. A key aspect of judicial judgment is providing detailed explanations for decisions [146]. LLMs can generate hypothetical explanations for predicted outcomes. For example, given case facts, an LLM can produce explanations:

Predicted Outcome: The court will find the defendant liable.

Reasoning: Because the defendant's actions meet all elements of negligence: duty, breach, causation, and harm, as established in [some precedent].

An entity-focused strategy [147] is proposed to enhance the explainability of legal reasoning. The approach first employs legal named entity recognition (NER) to identify and mask details such as roles, laws, and locations within the input text. This guides the LLM to focus on the core legal structure of the case (i.e., the legal entities and their relationships). By reducing distractions from irrelevant information, the method improves logical consistency across similar cases, such as clearly connecting actions to specific statute violations. Another direction integrates prediction and explanation into a unified process. PredEx [148] predicts case outcomes and generates explanations based on legal standards for Indian court cases, such as negligence criteria and constitutional doctrines. While PredEx models typically show slightly lower accuracy

than black-box classifiers, their transparency provides a clear advantage, as legal experts prefer clear, legally grounded explanations over higher, less transparent accuracy.

Furthermore, recent studies extend explainability research to court view generation, where LLMs produce judicial opinions rather than brief rationales. These studies are further discussed in the following subsection.

4.3.4. Court view generation

While LJP focuses on predicting a case's outcome, court view generation (CVG) produces the judge's rationale justifying the decision. CVG involves generating a coherent opinion that explains, based

on the case facts, how the applicable legal rules support the verdict. This task is crucial in the judicial domain, where interpretability and transparency are essential to the legitimacy of legal decision-making [146].

A key approach to improving CVG is structuring the reasoning process. Xu et al. [149] introduced a concept-guided divide-and-conquer framework that decomposes a judgment into sub-rationales aligned with legal elements (e.g., act occurrence, unlawfulness, culpability, and final conclusion). This framework prompts the LLM to generate each part separately, then assembles them into a coherent opinion, enhancing the coverage of required legal reasoning and ensuring alignment with judicial organization. Another approach enhances CVG by incorporating external knowledge. Liu et al. [24] proposed a retrieval-augmented method that combines the LLM's internal knowledge with retrieved authoritative texts from statutes and prior cases. The model first recalls internal knowledge, then applies retrieved content to its reasoning, reducing hallucinations and improving factual consistency. Furthermore, Yue et al. [150] developed a cooperative two-model framework for event-grounded CVG. One model extracts and structures key case events, while the second uses this representation to draft the court's narrative. This approach enhances factual consistency and organization by minimizing the risk of missing or incorrect facts.

4.3.5. Limitations of judicial decision

Despite recent advances, substantial limitations persist in judicial decision tasks. Most existing models are trained on post-judgment texts, which causes them to generalize poorly to pre-decision contexts. High scores on benchmarks such as CAIL2018 [151] or LexGLUE ECHR [79] often do not translate into reliable predictive performance in practice [152]. Even when augmented with statute or precedent retrieval, or fine-tuned under realistic constraints, LLMs still lag behind human judges, particularly in the coherence and consistency of their reasoning. This limits their practical value for litigation strategy and decision support [153].

These performance gaps reflect the structural challenges in replicating human legal reasoning. Law is interpretive and context-sensitive, and precedent requires subjective similarity judgments. Legal practice blends analogical case-based reasoning with deductive rule-based reasoning across diverse sources. Moreover, legal concepts are open-textured, with evolving and often fuzzy boundaries, making purely mechanical rule application inadequate [154]. Judicial decision-making is adversarial and defeasible, and justification, which involves reason-giving that synthesizes authorities, addresses counter-positions, and explains why alternatives are not considered within a social or procedural context, is as crucial as prediction. Current LLMs struggle to produce stable, value-aware, and source-aligned arguments, often delivering surface-level narratives instead. This limitation contributes to the ongoing explanation and judgment gap [155].

4.4. Legal question-answering tasks

Legal question-answering tasks are crucial in the legal domain, utilizing LLMs to answer natural-language queries regarding statutes, case law, procedures, and practical guidance. These systems benefit both professionals and the public by improving access to authoritative information and reducing time spent on manual research. For practitioners, LLMs provide relevant provisions and case summaries, while for non-experts, they offer basic guidance that lowers the barrier to legal information. Differences across legal traditions are significant, as common-law queries often rely on precedent and analogical reasoning, whereas civil-law queries focus on precise statutory interpretation.

4.4.1. Legal question-answering

Legal question-answering (QA) involves providing accurate legal answers by analyzing legal texts and retrieving relevant information [3,156]. Recent studies have enhanced legal QA with LLMs by combining retrieval augmentation, structured reasoning, prompt design, domain-specific adaptation, and human-in-the-loop verification [59, 126,157–161]. Fig. 9 illustrates several representative approaches in legal QA.

Retrieval-Augmented legal QA. RAG has emerged as a crucial method for enhancing the factual accuracy and interpretability of legal QA systems [18–20]. RAG-based legal QA first retrieves relevant statutes, case law, or other authoritative legal documents, providing these as explicit context to an LLM, which generates answers based on real legal sources. Several studies and benchmarks have highlighted the significant advantages of this approach. For example, Su et al. [157], using the STARD dataset, demonstrate that retrieval integration outperforms non-retrieval-based LLM baselines. Similarly, Zheng et al. [158], using the Bar Exam QA and Housing Statute QA benchmarks, show significant performance improvements with retrieval integration in legal QA tasks. RAG is particularly effective given the dynamic nature of law, as it can continuously update the retrieval corpus to reference the most current legal texts without the need for constant retraining.

Several studies further highlight the practical value of retrieval-augmented legal QA. The ELLA system [59] retrieves legal articles and cases for each query, presents clear correspondences between answers and legal sources, and enables users to interactively select the most relevant statutes, demonstrating how retrieval and human-in-the-loop workflows improve trust and reliability. EuropeanLawAdvisor [162] combines small open-source LLMs with robust retrieval modules, achieving accuracy and faithfulness comparable to or surpassing GPT-4 on EU legal queries, with every answer linked to authoritative citations. Similarly, Redelaar et al. [163] and DeliLaw [164] integrate statutory and case-law retrieval with LLMs to ensure that generated advice in Dutch and Chinese legal domains is verifiable and trustworthy, significantly reducing hallucinations.

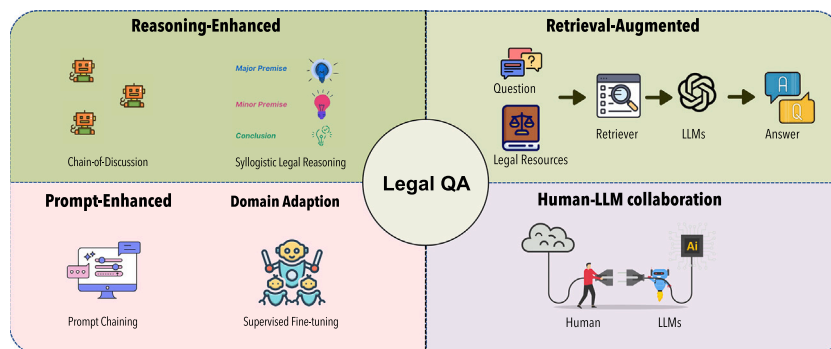


Fig. 9. Illustration of various approaches in legal question-answering.

These studies highlight that RAG-based methods reduce the reliance on LLMs to memorize extensive legal knowledge, keeping model architectures lightweight and adaptable while leveraging up-to-date external sources. By updating the retrieval corpus, these systems remain responsive to evolving laws without the necessity for costly retraining. Moreover, since answers are grounded in verifiable source texts, RAG-based systems enhance the auditability and trustworthiness of legal QA outputs.

Reasoning Enhanced Legal QA. Recent advances in legal QA have focused on improving the reasoning capabilities of LLMs to address the complexities of legal analysis. Legal QA often requires complex reasoning, such as applying abstract legal rules to concrete facts, interpreting ambiguous clauses, and integrating evidence from multiple sources. To tackle these challenges, researchers have proposed structured reasoning frameworks that guide the model's thought process, ensuring greater transparency and reliability in the generated answers.

A representative approach is the "reasoning-before-responding" framework [159], where the method first generates a step-by-step reasoning trace or a structured argument outline before reaching a final answer. The separation of reasoning from answering not only enhances the interpretability of the model's decisions but also allows for logical inspection and verification by human users or downstream modules. Building on this, the "Chain-of-Discussion" framework [160] extends chain-of-thought prompting to handle complex, evidence-intensive legal questions. By decomposing the problem into a sequence of structured prompts and intermediate reasoning steps, these frameworks outperform simple one-shot prompting and exhibit greater robustness in challenging cases. Recent work has also introduced multi-stage pipelines for legal reasoning. For example, LSIM [165] employs a three-stage architecture. First, reinforcement learning generates fact-rule reasoning chains tailored to each legal question. Next, a supervised deep structured semantic model retrieves relevant question-answer pairs by jointly encoding both semantic and logical features. Finally, the system leverages these retrieved exemplars for in-context learning, enabling the LLM to generate highly accurate and contextually appropriate answers. LSIM has been shown to outperform both RAG-based and vanilla LLM approaches, particularly in scenarios requiring intricate legal reasoning and contextual adaptation.

Moreover, several methods focus on the formalization and verification of legal logic. SyLeR [166] enhances LLM syllogistic reasoning by integrating hierarchical retrieval of statutes and precedents with a two-phase fine-tuning strategy. The first phase involves supervised training with reasoning paths, followed by reinforcement learning with structure-aware rewards to enhance logical consistency and transparency. Structured reasoning modules are crucial for developing interpretable, trustworthy, and reliable legal AI systems. Other studies have explored ensemble reasoning strategies. For example, Cross Examine [167] is an ontology-driven pipeline that manages multiple LLMs to analyze and cross-validate answers for Indian court cases. An initial answer generated by one module is iteratively verified, refined, or expanded by subsequent modules, with all outputs checked against both the source text and a structured legal ontology. The framework also constructs knowledge graphs for consistency validation and detailed explanations, enabling rigorous and transparent legal reasoning, even in the absence of extensive task-specific training data.

Prompt-Enhanced Legal QA. Prompt engineering is crucial for improving the accuracy and reliability of legal QA with LLMs. Roegiest et al. [126] showed that conventional semantic similarity or simple chat-style prompting is inadequate for answering detailed questions about specific contract clauses, often leading to inconsistent outputs. In contrast, using focused prompt templates, such as multiple-choice formats, enables LLMs to generate concise, accurate responses that integrate seamlessly into legal workflows. In addition, Roegiest et al. [127] proposed a prompt-chaining method in which the model first summarizes a relevant clause and then generates an answer, improving precision and consistency compared to one-shot methods.

Prompt engineering helps reduce hallucinations and improve answer accuracy in legal QA by requiring models to reference or quote relevant legal texts. Trautmann et al. [161] demonstrated that multi-step prompting and explicit citation requirements improve the reliability of legal QA systems, consistently outperforming simpler prompting methods across various datasets and jurisdictions.

The main advantage of prompt engineering is its ability to align LLM outputs with real-world legal tasks, ensuring accuracy, structure, auditability, and interpretability. By using clear, reusable templates or multi-stage prompt chains, legal AI systems can handle the complexity of legal language, making prompt-enhanced legal QA a scalable and reliable solution.

Domain-Specific LLMs for Legal QA. Domain-specific LLMs have become essential for improving accuracy and adaptability in legal QA. For example, Hu et al. [71] proposed a two-stage fine-tuning framework to reduce hallucinations. In the first stage, the base LLM is fine-tuned with legal knowledge, including statutes, case law, and expert QA pairs. The second stage involves evaluating model outputs for hallucinations and training the model on more challenging cases. Empirical results demonstrate that legal domain-specific LLMs outperform general-purpose models in tasks such as statute retrieval, case law synthesis, and expert consultation, particularly for complex, cross-domain legal issues.

Human-LLMs Collaborative Legal QA. To improve the usability and user experience of legal QA systems, substantial progress has been made in Human-LLM collaborative legal QA. These systems integrate interactive clarification and user participation, enhancing answer quality, interpretability, and real-world applicability. For example, Yao et al. [58] developed an Intelligent Legal Assistant that interactively clarifies user queries before answering. Since legal questions from non-experts often lack crucial details (e.g., jurisdiction or specific facts), the system first asks the user for their location to determine the correct jurisdiction. Subsequently, the LLM formulates targeted clarifying questions, typically presented as multiple-choice options, to capture missing details from the initial inquiry. Once the query is fully specified, the system provides a structured answer including a conclusion, legal reasoning with references, and actionable suggestions. This approach mirrors how lawyers explain issues, improving interpretability and usefulness for users. Similarly, ELLA [59] provides interpretable and accurate legal advice through a user-guided retrieval process. It retrieves relevant legal articles for a query, then allows the user to select those they deem most relevant. The system uses these articles to guide the LLM's response, improving relevance and avoiding misleading references. ELLA also highlights the legal basis behind each sentence in the answer and retrieves similar legal cases, marking key sentences for reference. These systems not only improve answer accuracy and user trust but also support the broader adoption and deployment of legal AI across various scenarios. Human-LLM collaborative approaches are emerging as essential foundations for the next generation of accurate, adaptable, and user-centered legal QA systems, driving the evolution of legal AI from specialized automation tools to widely accessible and trustworthy intelligent services.

4.4.2. Limitations of legal QA

Legal QA systems face significant challenges due to the dynamic and jurisdiction-specific nature of the legal domain. Statutes and case law are constantly evolving, requiring QA systems to retrieve the most current and relevant legal texts. Inaccurate or outdated information can arise if models rely on static knowledge bases, which may cite irrelevant statutes or overlook recent amendments. To address this, modern retrieval-augmented legal QA approaches prioritize jurisdiction-specific grounding and continuous updates to retrieval corpora. Without these adaptive features, legal QA systems risk providing outdated or incorrect answers.

Another critical limitation is the problem of hallucinations in current LLM dialogue systems, such as fabricating laws or citing non-existent

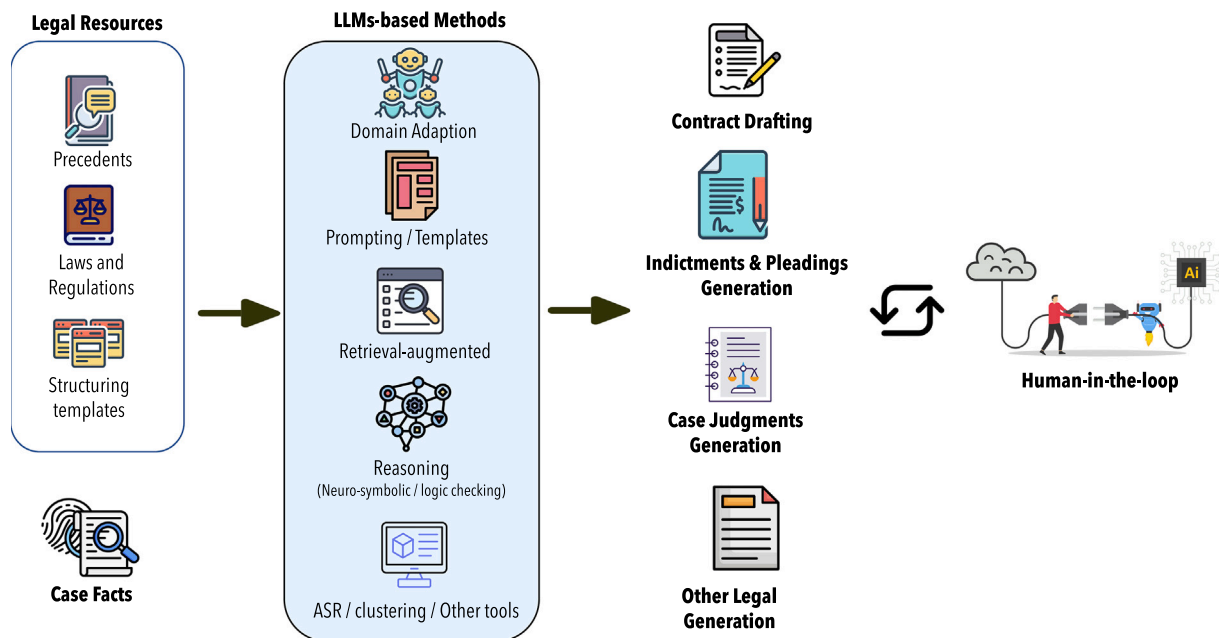


Fig. 10. Overview of an LLM-based pipeline for legal document generation.

cases. A notable real-world incident in 2023 involved lawyers submitting a legal brief generated with the help of ChatGPT, which referenced fictitious precedents — complete with invented case names and quotations.¹¹ This highlights the importance of integrating reliable document retrieval and rigorous citation verification mechanisms into legal chat-bots to prevent plausible yet erroneous legal references.

Additionally, accurately interpreting user queries remains a challenge, particularly when legal questions involve ambiguous terms or unclear jurisdiction. LLMs may provide general answers that are incorrect for the user's specific region, emphasizing the importance of systems that can either identify the relevant jurisdiction or clearly state jurisdictional caveats when providing responses. This issue is further compounded by layperson-generated queries, which often lack the precision and legal terminology required for accurate retrieval. The STARD dataset [157], which includes over 1500 real legal queries from non-professionals in China, highlights this challenge. These queries are often vague and incomplete, making traditional retrieval methods less effective. These challenges highlight the importance of QA systems that can interpret and expand user queries to capture the relevant legal information. Techniques like query clarification and context-aware query expansion could help close this gap and ensure that the retrieved documents are both accurate and relevant.

4.5. Legal generation tasks

Legal generation tasks are crucial in the legal domain, involving the use of LLMs to automatically draft various legal documents. Fig. 10 illustrates an LLM-based legal generation pipeline that takes legal information (such as case facts, statutes, and templates) as input and generates draft documents, which are then reviewed by human experts. These tasks, including the generation of contracts, pleadings, court judgments, and legal opinions, significantly reduce the time spent on repetitive drafting, allowing legal professionals to focus on more substantive work. For example, LLMs can quickly generate initial drafts of contracts and legal documents, saving time on standard clauses and formatting. These tasks are essential at various stages of the legal process, facilitating the efficient creation of key legal documents.

4.5.1. Contract drafting

Contract drafting is a foundational legal task, involving the composition of agreements, clauses, and defined terms. LLMs are leveraged to produce initial drafts of contracts or contract clauses, aiming to save attorneys' time on boilerplate language and clause customization. Lam et al. [168] employed LLMs, such as GPT-4, combined with a validation mechanism that generates contract clauses based on standardized prompts, which are then compared with a repository of precedent clauses. This approach effectively identifies and corrects errors in the generated clauses, improving draft quality before lawyer review.

RAG has emerged as a crucial method for enhancing consistency and efficiency in the contract drafting process. Wang et al. [17] introduced the expert-annotated benchmark dataset ACORD for contract clause retrieval. This dataset supports contract drafting by training models to quickly retrieve and identify relevant contract clauses, thereby enhancing the automation of contract drafting, and improving consistency and compliance. LexDrafter [169] automates the drafting of definition articles in legislative documents using RAG. It extracts term definitions from legal texts and either retrieves or generates new definitions for terms using an LLM, ensuring consistency and efficiency in the drafting process.

Researchers are also exploring neuro-symbolic methods to enhance contract generation. Kant et al. [170] prompted LLMs to output Prolog logic encodings of insurance contract rules, which are then checked with a Prolog interpreter to ensure compliance with policy provisions. This two-step process (text-to-logic, then logic verification) led to significant accuracy gains, demonstrating that integrating LLMs with legal reasoning frameworks can improve reliability. In other words, converting contract language into formal logic and back helps identify errors that a purely text-based model might overlook, resulting in clauses that more closely adhere to legal rules and intent.

4.5.2. Indictments and pleadings generation

Indictments and pleadings are formal documents that present facts and align them with legal charges or claims. Trozze et al. [171] evaluated GPT-3.5's performance in drafting legal complaints for securities fraud cases and identifying violations. They found that while GPT-3.5 could draft a complaint in a style comparable to a junior lawyer, its legal reasoning was weak. It identified some correct violations but often missed additional valid claims. This indicates that current LLMs can

¹¹ <https://www.bbc.com/news/world-us-canada-65735769>.

assist with the basic drafting of an indictment or complaint, but may fail to capture all legal violations in the fact pattern, highlighting their unreliability for complex legal reasoning without human oversight.

For civil pleadings, LLMs have been used to draft documents such as consumer complaints. Hemrajani [172] prompted GPT-4 to draft a detailed consumer complaint using a template from Indian law and compared it with human-drafted versions. The LLM-generated complaints were found to be comparable in quality to junior lawyers' drafts, though generally more concise and less detailed in their explanations. Notably, beyond drafting the initial complaint, LLMs could also assist in generating the answer (defendant's response) to a complaint. This involves addressing each allegation with admissions, denials, or defenses, a challenge that requires the model to map each point in the complaint to a corresponding rebuttal or legal argument. For example, in a consumer case, the defendant's answer (a "written statement of objections") must systematically deny or counter the complainant's allegations.

The main challenge in both indictment and pleading generation lies in ensuring legal coherence and factual accuracy. LLMs must avoid introducing unsupported allegations or charges (thus preventing hallucinations) and must not omit important facts or defenses. Techniques such as structured prompts and templates can help guide the model. By providing a clear outline (e.g., sections for facts, charges, evidence, and legal basis), the LLM produces a legally accurate document that covers all required elements. In practice, combining LLM drafting with human review or iterative refinement yields the best results. The LLM provides a first draft, and a lawyer then checks for completeness, correctness of charges or defenses, and adjusts the legal reasoning.

4.5.3. Case judgments generation

Case judgment generation involves combining case facts, applying legal reasoning, and issuing a final ruling. LLMs face challenges in producing coherent legal text while adhering to the formal structure of judicial decisions. CaseGen [173] decomposes the judgment drafting process into four subtasks: drafting the defense statement, outlining the trial facts, composing the court's legal reasoning, and generating the judgment result. This multi-stage approach reduces error propagation and allows for fine-grained performance evaluation. Lin and Cheng [174] focused on the criminal facts section, fine-tuning LLMs on a large corpus of criminal case facts to generate legally aligned content using structured prompting.

Case judgment generation remains an emerging and challenging task. It requires not only linguistic fluency but also the ability to perform legal reasoning at each step of the decision-making process. However, the outputs of LLMs in this area still require careful human validation to ensure that the facts align with the evidence and the legal reasoning follows the law logically.

4.5.4. Other legal generation tasks

LLMs are increasingly applied to a variety of other legal drafting and reporting tasks to improve efficiency. For instance, LexGenie [175] is an automated pipeline that generates structured legal reports from European Court of Human Rights case law. It retrieves relevant case paragraphs based on user queries, clusters them by theme using techniques such as Maximal Marginal Relevance (MMR) and BERTopic, and expands the content into a coherent report with accurate legal citations.

In police report generation, Kulkarni et al. [176] developed a system that automatically drafts incident reports from bodycam or dashcam audio. The system integrates automatic speech recognition (ASR) and LLMs, ensuring that the generated report remains grounded in the recorded evidence. This process accelerates report writing while maintaining accuracy, with officers reviewing and editing the draft before finalization.

Dallma [177] is a semi-structured drafting framework that combines deterministic rules with LLM-driven free-text generation, allowing experts to define templates for compliance while reducing drafting errors and omissions. Additionally, Nigam et al. [178] proposed a system for

automating the generation of private legal documents in India, utilizing VidhikDastaavej, an anonymized dataset, and NyayaShilp, a fine-tuned LLM. The system incorporates a human-in-the-loop mechanism to refine the generated documents, ensuring compliance with Indian legal practices.

These applications highlight the expanding role of LLMs in automating legal document generation, improving efficiency while maintaining legal rigor. However, continuous refinement is essential to address limitations in reasoning and ensure the accuracy of the outputs.

4.5.5. Limitation of legal generation

Despite significant advancements, LLM-based legal text generation continues to struggle with issues such as accuracy and factual consistency. Models may produce incorrect or outdated legal references, leading to missing clauses, misrepresented facts, or misapplied provisions. Furthermore, LLMs often struggle to maintain internal consistency across long documents. These issues are particularly evident in complex cases involving multiple parties, claims, and defenses. Additionally, adapting to jurisdiction-specific nuances and keeping up with evolving laws remains critical limitations, further increasing the risk of subtle yet consequential errors. Current studies in legal generation tend to focus on stylistic quality or similarity to human drafts but often overlook systematic checks to ensure verifiability and factual consistency.

To mitigate these challenges, several methods from general text generation research can be adapted to improve legal text generation. First, constrained decoding methods limit the search space of the generator to admissible options (e.g., schema-constrained entities, validated citation formats, or label sets aligned with a statute ontology), ensuring that outputs satisfy externally specified constraints, exhibit fewer hallucinated entities, and demonstrate better consistency with underlying knowledge sources [179,180]. Second, RAG conditions the model on retrieved passages from trusted corpora, encouraging it to ground outputs in statutes, case law, or contract repositories rather than in parametric memory. This approach improves traceability and factual faithfulness, and empirically reduces unsupported factual claims in both open-domain QA and legal statute QA [157,181]. Third, structured or template-based generation decomposes outputs into predefined sections and factual "slots" (such as parties, dates, claims, remedies, and applicable provisions), which constrain free-form hallucinations, improve entity-level faithfulness, and reduce drafting omissions or factual slips in both data-to-text systems and semi-structured legal drafting frameworks like Dallma [177,182].

4.6. Legal agent-based modeling

Legal agent-based modeling (LABM) involves using multiple LLM-based agents to integrate legal tasks into multi-stage workflows. LABM decomposes complex legal problems into sub-tasks handled by autonomous, specialized agents, such as statute retrieval, document analysis, document drafting, and compliance review, and coordinates these sub-tasks through multi-stage workflows to produce a final decision or recommendation. This modular, workflow-oriented design aligns with the application taxonomy in Section 4, where retrieval, summarization, judgment prediction, and drafting are treated as building blocks that can be orchestrated into end-to-end legal services. This section examines recent advancements in integrating LLMs with LABM across diverse legal applications, as illustrated in Fig. 11.

4.6.1. Collaborative multi-agent legal services

Collaborative multi-agent legal systems aim to harness multiple specialized agents working in concert to solve legal tasks, thereby reducing errors (such as hallucinations) and improving the robustness of assistance. For example, ChatLaw [53] implements a mixture-of-experts architecture, employing multiple LLM-based "expert" agents, each focusing on a particular legal domain (e.g., criminal law, civil law). These agents are coordinated by a central agent that uses a legal knowledge graph and law firm Standard Operating Procedures to route queries and verify answers. This design enables ChatLaw to reduce error rates

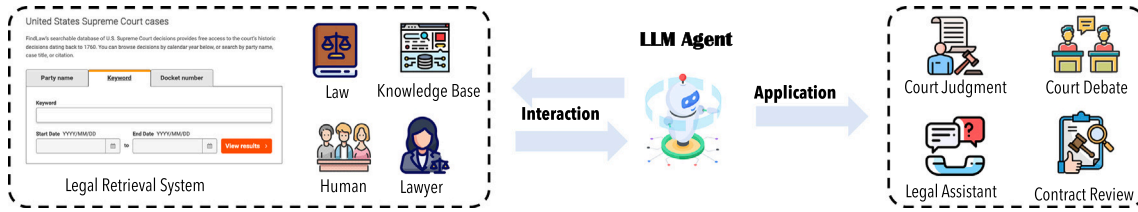


Fig. 11. Illustration of legal tasks related to agent-based modeling.

and achieve higher accuracy compared to a single agent. Similarly, LegalGPT [183] connects dedicated agent modules for distinct functions, such as document review and legal advice generation, all unified by a retrieval component that supplies relevant case information to the agents. LawLuo [184] simulates a mini law-firm environment by assigning different roles to agents, such as a receptionist for client intake, a lawyer for analysis, a secretary for drafting, and a supervising attorney for oversight. This role-play approach mirrors the intake-to-supervision workflow of a real firm, ensuring coherent multi-turn interactions and reducing errors related to ambiguity. These collaborative multi-agent systems demonstrate that dividing legal work among specialized, interacting agents can enhance clarity and reliability in AI-generated legal services.

4.6.2. Adversarial multi-agent courtroom simulation

A significant direction in LABM is courtroom simulation with adversarial multi-agent systems. AgentCourt [25], which employs adversarial role-play among judge, plaintiff's lawyer, and defense lawyer agents, refines their strategies via extensive adversarial interactions. MASER [185], addressing interactive legal scenarios beyond courtroom trials, generates synthetic legal interaction data using multiple agents to simulate diverse legal proceedings, including client-lawyer consultations and negotiations. AgentsCourt [186] integrates adversarial debate with judicial reasoning by coordinating multiple LLM-based agents. A prosecution and defense agent engage in adversarial argumentation, a retrieval agent sources relevant laws and precedents from JudicialKB, and a judge agent synthesizes these inputs into a reasoned judgment. AgentsCourt outperforms baseline models in generating detailed legal rationales. AgentsBench [187] simulates judicial panel deliberation, with multiple judge agents (e.g., lay and career judges) discussing a case and voting on the outcome, mirroring a real court's collegial process. Evaluated on legal judgment prediction tasks, AgentsBench significantly improves decision quality and transparency over single-agent systems.

Several studies have explored adversarial debate between agents as a strategy for improving legal decision-making. Debate-Feedback [188] simulates courtroom deliberation by engaging plaintiff and defendant LLM agents in multi-round debate, with an assistant evaluating argument reliability and refining the judge model's predictions. This method is more efficient and transparent than single fine-tuned models, providing a dialogue trace of decision-making. Courtroom-LLM [189] assigns prosecutor, defender, and judge roles to separate LLMs. For each case, the prosecutor and defender argue for different labels, and the judge evaluates both to decide the outcome. This interplay among agents enables Courtroom-LLM to outperform single-model classifiers and voting ensembles. In addition, the multi-agent legal reasoning (MALR) framework [190] effectively addresses complex legal reasoning tasks without extensive additional training data.

These studies demonstrate that structuring interactions among multiple LLM agents can yield more reliable and interpretable outcomes in complex legal tasks.

4.6.3. Other multi-agent systems

Other legal agent systems have explored applications beyond courtroom simulation and judgment prediction. Legal Agentic Workflows

(LAW) [191] is a multi-agent framework for analyzing complex financial contracts, such as custody and fund services agreements. LAW decomposes user queries into sub-tasks (such as section retrieval, date computation, and term lookup), each handled by a specialized agent or tool. By processing contract segments sequentially and using targeted modules, LAW overcomes context limitations of single-model systems.

4.6.4. Limitations of LABM

Despite their potential, LABM approaches introduce challenges for legal AI. A primary issue is coordination complexity: managing communication and the division of labor among multiple agents (each with its own prompt or objective) is challenging, and errors in one agent's output can propagate through the system if not properly checked. Evaluation is also underdeveloped, as there are few standardized benchmarks or metrics for multi-agent legal systems, making it difficult to compare frameworks or assess their generalizability to real-world settings. Additionally, transparency and traceability remain limited: although agent dialogues can reveal some reasoning steps, the final outcome often depends on long chains of intermediate exchanges that are difficult to audit. Multi-agent architectures also increase computational cost and latency, as multiple models must run and communicate to solve a single query.

4.7. Legal education and training

Legal education and professional training are essential in the legal field. Instead of directly involving dispute resolution or contract drafting, they focus on equipping both current and future legal professionals with the necessary knowledge and skills to effectively perform these tasks. The emergence of LLMs, such as OpenAI's GPT-3.5 (ChatGPT) and GPT-4, has inspired intense interest in their applications within legal education and professional training. Recent studies have explored the performance of these models on law school exams and professional licensure tests. As illustrated in Fig. 12, current applications are grouped into three primary categories: (1) studying for exams, (2) interactive case analysis, and (3) professional training and continuing education.

4.7.1. Studying for exams

Recent studies extensively evaluated LLMs in law school and professional legal assessments, highlighting both their progress and limitations.

Early work by Bommarito II and Katz [192] showed GPT-3.5 achieved approximately 50.3% accuracy on the Multistate Bar Exam, which was better than random guessing but below passing standards. Further exploring LLM capabilities in law school assessments, Choi

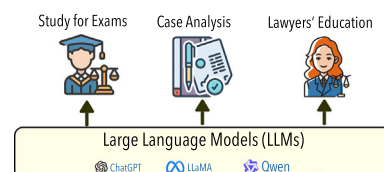


Fig. 12. Illustration of legal application related to legal education and training.

et al. [193] found GPT-3.5 performed at a minimally passing level, similar to a low-performing law student, with basic issue-spotting and coherent writing but lacking depth and analytical rigor. In addition, Hargreaves [194] noted ChatGPT handled broad theoretical essays competently but struggled with problem-solving that required precise legal application.

Subsequently, Katz et al. [10] reported that GPT-4 passed the Uniform Bar Exam, achieving median human-level performance on multiple-choice questions and competent essay scores. However, Martínez [195] found that GPT-4 performed similarly to above-average, but not top-tier, examinees, especially struggling with complex essays. Chalkidis [65] also noted GPT-4's challenges with LexGLUE tasks, indicating that standardized test proficiency does not equate to comprehensive legal reasoning.

Researchers have aimed to improve LLM performance on legal exams using structured prompting and fine-tuning. IRAC-structured prompting [64] significantly improves accuracy on Japanese bar exam tasks. Similarly, fine-tuning smaller models like LLaMA 2 on IRAC-formatted data [196] notably improves accuracy on MBE questions.

4.7.2. Interactive case analysis

Recent literature highlights the interactive use of LLMs to support legal education and the understanding of complex concepts. For instance, Savelka et al. [15] showed that augmenting GPT-4's definitions of open-textured legal terms with authoritative case law improved explanation reliability, suggesting that LLMs can serve as effective educational assistants.

In complex doctrinal tasks, ATRIE [197] employed a retrieval-generation pipeline that integrates relevant precedents to produce doctrinal summaries. These AI-generated interpretations closely match expert analyses, demonstrating practical value for legal education and practice. Furthermore, Jiang et al. [198] improved learner engagement by using LLM-generated storytelling, introducing a dataset that combines doctrines with narratives and enhances comprehension and retention, especially for non-native speakers.

These interactive approaches highlight the value of supplementing LLM outputs with authoritative resources, which greatly improve their reliability, interpretability, and educational effectiveness.

4.7.3. Training and continuing education for lawyers

Recent research explores how LLMs are transforming legal training and continuing education, with the potential to reshape teaching methods, professional practice, and ongoing learning.

Nelson [199] analogized the use of LLMs in legal education to calculators in mathematics, suggesting their integration into teaching frameworks to enhance critical thinking and supplementary research skills. Correspondingly, there is a necessity for AI literacy training [193,194], enabling students to leverage AI's strengths while mitigating its shortcomings.

Critical thinking and ethics remain foundational—now more than ever in the AI era. Peoples [200] cautioned against overreliance on LLMs, highlighting their limitations in logical consistency and depth of reasoning. Similarly, Ajevski et al. [201] advocated shifting educational assessments toward uniquely human skills, such as client interaction and ethical decision-making, to ensure that lawyers remain proficient in areas beyond AI's current capabilities.

LLMs are reshaping the roles of law educators. Oltz [202] showed that AI can handle routine academic tasks, significantly reducing faculty workload in drafting administrative documents and educational materials. Moreover, legal expertise plays a role in shaping AI development. Nay [203] argued that legal norms and ethics can serve as foundational datasets for AI alignment, highlighting the reciprocal relationship between law and technology. Legal professionals and educators thus help train both lawyers to use AI responsibly and AI systems to adhere to legal and ethical standards.

4.8. Other applications

Recent studies highlight the growing scope of legal applications for LLMs. For instance, LLMediator uses GPT-4 for online dispute resolution by reformulating communications and drafting mediator interventions [204]. Breton et al. [205] demonstrated that LLMs can extract legal terms with minimal data using one-shot prompts, reducing manual annotation by over 50% and achieving state-of-the-art performance. In AI-assisted arbitration, Arbitrus.ai rendered decisions on 100 hypothetical cases with reasoning quality comparable to human arbitrators and without hallucinations or omissions [206]. These advances highlight LLM adaptation for mediation, information extraction, and automated decision-making, emphasizing domain adaptation and transparency.

5. Datasets, code and benchmark

Domain-specific datasets, benchmarks, and open-source resources have greatly advanced LLM progress in the legal domain. These resources support model training and fair evaluation on real legal tasks. Below, we summarize key datasets, benchmarks, and repositories for legal NLP research.

5.1. Datasets

High-quality, domain-specific datasets are essential for developing and evaluating legal NLP models. Table 5 provides an overview of widely used datasets for tasks such as legal case retrieval, document summarization, judicial decision, and legal question-answering.

5.1.1. Legal case retrieval datasets

Legal case retrieval datasets generally consist of a query case (with factual descriptions) and historical cases, annotated for relevance to identify similar precedents. Several widely used datasets include:

COLIEE 2018–2024 Task 1 (Case Law Information Retrieval) [207–213] is a series of case law retrieval datasets from the annual COLIEE competitions, based on Federal Court of Canada records. Each COLIEE dataset contains query cases (ranging from 100 to about 1200) with candidate cases, and relevance labels based on citations. For example, COLIEE 2021 comprises 900 queries, each associated with approximately 4.7 relevant precedents.

LeCaRD [214] and **LeCaRDv2** [215] are Chinese criminal case retrieval datasets based on Supreme People's Court records. LeCaRD contains 107 queries and approximately 10,700 candidate cases, with each query paired with an expert-annotated set of relevant precedents (the top 30 relevant candidates are labeled, and the rest are treated as irrelevant). LeCaRDv2 expands to approximately 800 queries and 55,192 candidates from 4.3 million Chinese criminal cases. Crucially, LeCaRDv2 integrates extensive expert vetting, annotating relevance based not only on citations but also on case characteristics, penalties, and procedural elements.

GerDaLIR [216] is a German legal case retrieval benchmark built upon Open Legal Data. It includes approximately 131,000 German case documents paired with around 123,000 query cases. Each query case has at least one relevant annotated case, providing relevance judgments suitable for large-scale standardized evaluations in German legal information retrieval tasks.

CLERC [217] is a U.S. legal information retrieval corpus built on the Caselaw Access Project. It includes approximately 1.84 million full U.S. case documents (CLERC/doc) and around 105,000 “query” passages (CLERC/queries). Each query is a paragraph of legal analysis (approximately 300 words) extracted from case opinions, with the relevant candidates being the prior cases it cites. The citation links (i.e., the true retrieved cases) were extracted and partially verified with the assistance of legal experts.

ECtHR-PCR [218] is based on approximately 15.7K English judgments from the European Court of Human Rights (ECtHR), split chronologically into training, development, and test sets. The dataset simulates realistic retrieval conditions: case “Facts” sections serve as queries,

Table 5
Summary of domain-specific legal datasets.

Dataset	Jurisdiction	Language	Domain	Task Type
COLIEE 2018–2024 (LCR Task)	Canada	English	Civil case law	Legal case retrieval
LeCaRD [214]	China	Chinese	Criminal case law	Legal case retrieval
LeCaRDv2 [215]	China	Chinese	Criminal case law	Legal case retrieval
GerDaLIR [216]	Germany	German	General case law	Legal case retrieval
CLERC [217]	U.S.	English	General case law	Legal case retrieval
ECtHR-PCR [218]	EU	English	Human rights case law	Legal case retrieval
IRLeD [219]	India	English	Supreme court case law	Legal case retrieval
CAIL2019-SCM [220]	China	Chinese	Criminal case law	Similar case matching
COLIEE 2018–2024 (SAR Task)	Japan	Japanese	Japanese civil law	Statutory law retrieval
BSARD [221]	Belgium	French	Belgian legislation	Statutory law retrieval
bBSARD [221]	Belgium	French/Dutch	Belgian legislation	Statutory law retrieval
STARD [157]	China	Chinese	Chinese legislation	Statutory law retrieval
CivilSum [228]	India	English	Court judgments	Abstractive summarization
EUR-Lex-Sum [229]	EU	Multilingual	Regulatory acts	Abstractive summarization
EUR-LexSum [230]	EU	English	Regulatory acts	Abstractive summarization
BillSum [231]	U.S.	English	Legislative bills	Abstractive summarization
ArgLegalSumm [232]	Canada	English	Court judgments	Abstractive summarization
IN-Abs [233]	India	English	Court rulings	Abstractive summarization
IN-Ext [233]	India	English	Court rulings	Extractive summarization
UK-Abs [233]	U.K.	English	Court rulings	Abstractive summarization
CLSum [116]	Multilingual	English	Court judgments	Abstractive summarization
Multi-LexSum [234]	U.S.	English	Civil rights lawsuits	Abstractive summarization
GermanSum [235]	Germany	German	Court rulings	Abstractive summarization
CAIL2020-Sum [236]	China	Chinese	Civil court judgments	Abstractive summarization
CaseSumm [237]	U.S.	English	SCOTUS decisions	Abstractive summarization
Swiss-Lex [238]	Switzerland	Multilingual	Court rulings	Abstractive summarization
Lawsuit [239]	Italy	Italian	Legal verdicts	Abstractive summarization
LexSumm [114]	Multilingual	English	Court judgments	Abstractive summarization
RulingBR [240]	Brazil	Portuguese	Court decisions	Abstractive summarization
PublicHearingBR [241]	Brazil	Portuguese	Transcripts of public hearings	Abstractive summarization
LAR-ECHR [118]	EU	English	European court of human rights	Legal argument reasoning
LAM-ECHR [222]	EU	English	European court of human rights	Legal argument mining
AMELIA [223]	Italy	Italian	Italian tax litigation	Legal argument mining
δ -Stance [119]	U.S.	English	U.S. legal citation system	legal argument classification
CAIL2018 [151]	China	Chinese	China judgment online	Legal judgment prediction
ECHR [242]	EU	English	European court of human rights	Violation prediction
SCOTUS [243]	U.S.	English	US supreme court	Case outcome prediction
Swiss-Judgment-Prediction [244]	Switzerland	Multilingual	Federal supreme court of Switzerland	Case outcome prediction
ILDC [245]	India	English	Supreme court of India	Court decision
LBOX OPEN [246]	South Korea	Korean	Korean supreme court	Explanation for decision
AnnoCaseLaw [247]	U.S.	English	U.S. appeals court negligence case	Legal judgment prediction
PredEx [148]	India	English	Supreme court of India	Statute prediction
LAR-ECHR [118]	EU	English	European court of human rights	Legal judgment prediction
FCCR [248]	France	French	French supreme court	Next-argument prediction
ECtHR [249]	EU	English	European court of human rights	Law area of a case
GermanLJP [250]	Germany	German	German courts	Court decision
ArabicLJP [251]	Saudi Arabia	Arabic	Saudi Arabian personal status cases	Violation prediction
BCD [252]	Brazil	Portuguese	Brazilian courts	Legal judgment prediction
CAIL-LONG [45]	China	Chinese	China judgment online	Legal judgment prediction
Super-SCOTUS [253]	U.S.	English	Supreme court of the US	Court decision
CMDL [254]	China	Chinese	China judgment online	Legal judgment prediction
ILSI [255]	India	English	Indian penal code	Legal statute identification
			India supreme court	
			India high court	

(continued on next page)

Table 5 (continued)

Dataset	Jurisdiction	Language	Domain	Task Type
C3VG [225]	China	Chinese	China judgment online	Court view generation
CCVG [226]	China	Chinese	China judgment online	Court view generation
Li et al. [227]	China	Chinese	China judgment online	Court view generation
LCVG [149]	China	Chinese	China judgment online	Court view generation
COLIEE 2014–2021 (Legal QA Task)	Japan	English/Japanese	Japanese legal bar exams	Legal QA
CaseHOLD [256]	U.S.	English	Legal citations in judicial rulings	Legal QA
LLeQA [257]	Belgium	French	Belgian legislation	Legal QA
LeDQA [258]	China	Chinese	Chinese civil cases	Rationale extraction Legal QA
STARD [157]	China	Chinese	Chinese statutory articles	Rationale extraction Legal QA
JEC-QA [259]	China	Chinese	Chinese judicial examination	Statute retrieval Multiple-choice QA
cLegal-QA [260]	China	Chinese	Chinese legal services network	Legal QA
PIL-QA [261]	EU	English	Private international law	Multiple-choice QA
RepLiQA [262]	Multilingual	English	Synthetic legal scenarios	Reading-comprehension QA
LEGAL-UQA [263]	Pakistan	Urdu/English	Pakistan Constitution	Legal QA
EQUALS [264]	China	Chinese	Law consultation forum	Span-extraction QA
LegalQA [265]	China	Chinese	The ‘bankruptcy’ related forum on Avvo5 in California	Legal QA
AILQA [266]	India	English	India code website	Legal QA
GerLayQA [267]	Germany	German	German legal online forum	Retrieval-based QA
JuRO [268]	Romania	Romanian	Romanian law examinations	Answer generation Multiple-choice QA
CUAD [62]	U.S.	English	Corporate commercial contracts	Clause span extraction
AGB-DE [131]	Germany	German	Consumer contract law	Multi-class classification Clause topic classification
ACORD [17]	U.S.	English	U.S. securities and exchange commission	Validity assessment Clause retrieval
LEDGAR [269]	U.S.	English	U.S. securities and exchange commission	Multi-label classification
UNFAIR-ToS [79]	EU	Multilingual	Contracts	Unfair clause detection
MAUD [270]	U.S.	English	Public merger agreements	Reading comprehension
JuDGE [271]	China	Chinese	China Judgments online	Legal document generation
ClaimGen-CN [272]	China	Chinese	China judgments online	Legal claim generation
VidhikDastaavej [178]	India	Hindi/English	Indian private legal documents	Legal document generation
CaseGen [173]	China	Chinese	China judgments online	Legal document generation
LexGLUE [79]	Multilingual	English	–	Multi-label classification Multi-class classification Multiple choice QA
MultiLegalPile [273]	Multilingual	Multilingual	–	–
LexFiles [37]	Multilingual	Multilingual	–	–
EURLEX57K [274]	EU	English	European court of human rights	Multi-label classification
German Legal NER [275]	Germany	German	Court decisions	Named entity recognition
LeNER-Br [276]	Brazil	Portuguese	Legislative documents	Named entity recognition
VICTOR [277]	Brazil	Portuguese	Brazil’s supreme court	Document classification
Pile of Law [278]	Multilingual	English	–	–
Cambridge Law Corpus [279]	U.K.	English	UK supreme court	Case outcome
LEVEN [280]	China	Chinese	China judgment online	Event extraction
CDD [281]	China	Chinese	Chinese court trials	–
LegalDiscourse [282]	U.S.	English	U.S. state law	Discourse analysis

and relevant precedents are identified from citations within “Legal reasoning” sections.

IRLeD [219], developed for the FIRE 2017 competition, contains Indian Supreme Court cases. It consists of 200 contemporary cases and 2000 prior cases (half cited, half uncited), with citation markers anonymized. The task involves ranking cited precedents higher than irrelevant cases, providing a controlled environment for evaluating Indian legal retrieval systems.

CAL2019-SCM [220]: A Chinese similar-case matching dataset consisting of 8,964 triplets of case fact descriptions. The task is to select, for each anchor case, the more similar case between two candidates.

5.1.2. Statutory article retrieval datasets

COLIEE 2018–2024 Task 3 (Statute Law Retrieval) [207–213] is a series of annual datasets (2018–2024) for Japanese civil law article retrieval. Using Japanese Bar exam questions as queries, the task

provides training and test sets linked to the articles of the Japanese Civil Code.

BSARD [221] is a French-language dataset of Belgian law articles designed for statutory retrieval. It covers Belgium's statutory corpus and includes over 1100 legal questions and more than 22,600 law articles.

bBSARD is a bilingual extension of BSARD for Belgium. It provides parallel French and Dutch versions of Belgian statutory articles, along with the original French questions from BSARD and corresponding Dutch translations.

STARD [157] is a Chinese Statute Retrieval Dataset derived from real-life queries. It contains 1543 user-generated Chinese queries (from legal consultations) and 55,348 Chinese statutory articles. STARD focuses on non-expert queries, reflecting broad variation and absence of legal jargon.

5.1.3. Legal argument mining datasets

LAR-ECHR [118] is a legal argument reasoning dataset based on European Court of Human Rights cases. Each example consists of a chain of argument statements from an ECHR case, where the task is to select the correct next statement based on the case facts.

LAM-ECHR [222] is a large corpus of 373 ECHR decisions annotated with detailed legal arguments. It includes 2.3 million tokens and 15,205 argument spans, each labeled according to a comprehensive legal argumentation scheme.

AMELIA [223] consists of 225 Italian court decisions on Value-Added Tax, annotated at the clause level. Each argumentative component is labeled as either a premise or a conclusion, with premises further classified (e.g., legal vs. factual).

δ -Stance [119] contains stance-annotated argument–argument pairs mined from millions of instances of U.S. judges citing precedent. Each pair is labeled for polarity (whether the cited case strengthens or weakens the argument) and intensity.

Furthermore, Zhang et al. [224] review the availability of annotated legal argument corpora, annotation schemes, and approaches in the literature, providing an effective summary of the datasets and methods developed over the past ten years.

5.1.4. Court view generation datasets

C3VG [225] is a Chinese criminal court-view generation corpus built from China Judgments Online. It aligns case facts with adjudication (ADC) and sentencing (SEC) circumstances, providing paired fact–opinion samples with charge labels for both training and evaluation.

CCVG [226] is a Chinese criminal court-view generation corpus (> 110K cases) built from China Judgments Online. It pairs fact descriptions with court views and charge labels, spanning documents from 1985 to 2021, and supports both court view generation and charge prediction.

Li et al. [227] presented a Chinese dataset for generating the “court view” section of judgments. Given a case's fact description, charges, and applicable law articles, the task is to generate the court's opinion text that explains the verdict.

LCVG [149] is a Chinese criminal court-view generation dataset built from China Judgments Online. It pairs case facts with legal concepts, concept-level rationales, and full court views.

5.1.5. Legal document summarization datasets

Legal documents, including court judgments, statutes, and contracts, are typically lengthy and complex, complicating manual review and analysis. Therefore, legal document summarization datasets consist of full-length texts paired with concise, human-generated summaries or headnotes. These datasets vary significantly in document type, jurisdictional coverage, language, and summarization methods (abstractive or extractive). Despite limited availability, existing legal summarization resources demonstrate considerable diversity, enhancing model development across multiple jurisdictions and linguistic contexts.

CivilSum [228] is an extensive English corpus comprising 23,350 Indian Supreme Court and High Court judgments. Each case includes abstractive summaries spanning diverse civil-law areas such as labor and education.

EUR-Lex-Sum [229] and **EUR-LexSum** [230] focus on European Union regulatory texts. EUR-Lex-Sum, multilingual across all 24 EU languages, includes approximately 1500 EU acts per language, paired with paragraph-aligned summaries, supporting cross-lingual summarization research. In contrast, EUR-LexSum is monolingual (English), comprising 4595 EU regulatory documents (2003–2022) with lay summaries aimed at non-specialist audiences.

BillSum [231], introduced in 2019, covers U.S. legislative documents from Congress and the California State Legislature. The dataset includes a sizable “Congress” subset (18,949 training and 3269 test bills from sessions 103–115) and a “California” test subset (1,237 bills from 2015–2016), each paired with official summaries issued by legislative bodies. The dataset's complexity stems from the extensive length of the documents, necessitating effective compressive summarization techniques.

ArgLegalSumm [232] is a Canadian dataset comprising 1262 court cases from the Canadian Legal Information Institute (CanLII), primarily Supreme Court and appellate decisions. Each case includes official summaries supplemented by sentence-level argumentative annotations (Issues, Reasons, Conclusions), combining summarization and argument mining.

IN-Abs / IN-Ext and UK-Abs [233] focus on Indian and UK Supreme Court judgments. IN-Abs contains abstractive summaries of 7130 Indian judgments, while IN-Ext contains expert-generated extractive summaries with rhetorical annotations. UK-Abs includes 793 UK judgments with official press-release summaries.

CLSum [116] is a multi-jurisdictional corpus designed for summarization research across common-law jurisdictions, including Canada, Hong Kong, the United Kingdom, and Australia. It supports cross-jurisdictional analysis and low-resource summarization by providing lengthy judicial decisions paired with highly abstractive summaries.

Multi-LexSum [234] addresses multi-granularity summarization tasks within complex civil rights litigation contexts, providing summaries at extreme, short, and detailed granularities. The dataset presents realistic legal research challenges, emphasizing extensive document lengths and structural complexity.

GermanSum [235], comprising around 100,000 German court judgments, includes official guiding principles (“Leitsätze”).

CAIL2020-Sum [236] is crucial for Chinese legal NLP, including 9484 civil court decisions from China Judgments Online, each with judicial summaries.

CaseSumm [237] is a large-scale dataset comprising approximately 25,000 U.S. Supreme Court opinions paired with official summaries, designed to support the summarization of lengthy judicial texts.

Swiss-Lex [238] provides multilingual coverage with 18,000 Swiss Federal Tribunal cases in German, French, and Italian, enabling summarization across multiple languages and legal systems.

LAWSUIT [239], a substantial Italian dataset, comprises 14,000 Constitutional Court decisions paired with expert-written abstractive summaries.

RulingBR [240]: A Portuguese legal summarization dataset of 10,623 Brazilian court decisions, aimed at building and evaluating summarization systems for rulings.

PublicHearingBR [241]: A Brazilian Portuguese dataset containing 206 public hearing transcripts from the Brazilian Chamber of Deputies, paired with structured summaries. It supports long document summarization and the evaluation of hallucinations in summaries, with tasks such as Natural Language Inference (NLI) and summarization evaluation using recall and precision metrics.

5.1.6. Legal judgment prediction datasets

Legal judgment prediction involves predicting the outcome or attributes of a case from textual case descriptions (i.e., facts). Datasets for this task are primarily derived from court records annotated with outcomes such as legal violations, appeal results, or sentencing details. Due to jurisdictional differences in public record availability, these datasets vary significantly.

CAIL2018 [151]: A large-scale Chinese dataset comprising over 2.6 million criminal case reports from China's Supreme People's Court. Each record includes detailed factual descriptions, along with annotations of applicable criminal laws, charges, and sentencing information, providing extensive coverage for judgment prediction tasks.

ECJR [242]: An English-language dataset includes approximately 11,500 cases from the European Court of Human Rights. Each document contains the full text, a factual description, and labels indicating the specific articles of the European Convention on Human Rights that were violated, along with binary outcomes (violation/no violation).

SCOTUS [243]: A dataset of U.S. Supreme Court cases spanning 1955–2023, enriched with structured metadata and detailed court-written factual descriptions. Each dataset includes the decision outcome, majority vote counts, and verdicts.

Swiss-Judgment-Prediction [244]: A multilingual corpus comprising approximately 85,000 Swiss Federal Supreme Court cases from 2000–2020 in German, French, and Italian. Each case includes factual and decision descriptions, along with binary labels indicating whether the plaintiffs' claims were granted or dismissed.

ILDC [245]: An English-language dataset containing around 35,000 Supreme Court cases from India, each comprising facts and final judgments. A subset includes expert-annotated rationales explaining judicial decisions, emphasizing explainability.

LBOX OPEN [246]: A Korean corpus of 147,000 court precedents (259 million tokens), supporting tasks such as case-name classification, statute categorization, judgment prediction, and summarization.

Other datasets such as **AnnoCaseLaw** [247], **PredEx** [148], and **LAR-ECHR** [118] extend the scope and annotation depth of legal judgment prediction tasks. These include expert-annotated U.S. appellate negligence cases, Indian Supreme Court cases with detailed rationales, and European Court of Human Rights reasoning tasks. Additional resources like **FCRR** [248], **ECtHR** [249], **GermanLJP** [250], **ArabicLJP** [251], **BCD** [252], and specialized variants including **CAIL-LONG** [45], **Super-SCOTUS** [253], and **CMDL** [254] collectively enrich the legal judgment prediction datasets.

5.1.7. Legal question-answering datasets

Legal QA involves models responding to queries based on legal knowledge, typically derived from statutes, case law, or other legal documents. These datasets often originate from competitions or QA pairs designed to evaluate models' ability to interpret and apply legal texts. Tasks may include retrieval-based challenges (e.g., locating relevant statutes or precedents) or reading comprehension tasks (e.g., answering questions based on provided legal passages). Datasets include:

COLIEE series [207–211,213,283–286]: An annual competition series initiated in 2014. COLIEE datasets evaluate the retrieval of relevant statutes from the Japanese Civil Code and binary entailment decisions for Japanese bar-exam queries. The dataset typically contains around 1200 questions annually.

CaseHOLD [256]: A dataset comprising over 53,000 multiple-choice questions based on U.S. Supreme Court decisions. Each question presents an excerpt from a court case along with four candidate "holding" statements, exactly one of which is correct.

LLeQA [257]: A French-language QA dataset containing 1868 questions answered based on Belgian statutory law. The answers are detailed, provided at the paragraph level, explicitly linked to relevant articles, and include comprehensive annotations.

LeDQA [258]: A Chinese QA dataset comprising 100 case documents, 4800 annotated question-answer triples, and 132,048 sentence-level

relevance annotations. LeDQA emphasizes sentence-level retrieval and QA generation tasks tailored to the Chinese legal context.

STARD [157]: A Chinese statute retrieval dataset consisting of 1543 queries from non-professionals, paired with 55,348 statutory articles. It addresses the challenges of retrieving relevant statutes based on real-world, non-expert queries, supporting the development of legal retrieval models for public legal assistance.

Other notable datasets include **JEC-QA** [259], **cLegal-QA** [260], **PIL-QA** [261], **RepLiQA** [262], **LEGAL-UQA** [263], **EQUALS** [264], **LegalQA** [265], **AILQA** [266], **GerLayQA** [267], and **JuRO** [268]. These datasets enhance diversity in languages, jurisdictions, and task complexity, providing comprehensive evaluation platforms for legal QA systems.

5.1.8. Contract analysis and drafting datasets

These datasets support tasks such as contract analysis, clause identification, and the detection of problematic provisions, essential precursors to contract drafting automation.

CUAD [62]: An English-language corpus of 510 SEC EDGAR contracts (9,283 pages) annotated across 41 clause categories. The task involves extractive clause identification, pinpointing clause locations within contract texts.

AGB-DE [131]: A German dataset comprising 3764 clauses extracted from 93 consumer contracts. Expert annotations include 23 topics, 37 subtopics, and binary validity judgments under consumer protection laws.

ACORD [17]: An English corpus includes 114 attorney-generated queries paired with over 126,000 clause candidates from SEC filings and online Terms of Service contracts, annotated for retrieval tasks based on relevance scores.

LEDGAR [269]: An extensive corpus with 60,540 contracts, offering 846,274 provisions labeled across 12,608 topics, designed for extreme multi-label classification.

UNFAIR-ToS [79]: A primarily English multilingual dataset for detecting unfair clauses in consumer-facing Terms of Service. It comprises 50 ToS documents from platforms such as YouTube, eBay, and Facebook, split into sentences and annotated by legal experts, yielding approximately 9400 sentence-level instances. Each sentence is labeled with zero or more of eight unfairness categories defined under EU consumer law, supporting multi-label classification tasks.

MAUD [270]: The Merger Agreement Understanding Dataset consists of 152 public merger agreements and 47,457 annotations, including 8226 deal-point text annotations and 39,231 question-answer annotations for reading comprehension over transactional documents.

5.1.9. Legal documents generation datasets

JUDGE [271] contains extensive pairs of factual case summaries and full written judgments from Chinese courts. The task involves document-level generation of judgments, including court views and outcomes, based on case facts.

ClaimGen-CN [272], constructed from approximately 207K civil case documents, maps case fact descriptions to plaintiffs' legal claims. The task is to generate clear, factual plaintiff claims from case facts.

VidhikDastaavej [178] contains an anonymized corpus of Indian private legal drafts (e.g., petitions, legal letters, affidavits, leases) totaling 489 documents (469 for training, 20 for testing) across 17 categories for training and 7 categories for testing. The dataset includes LLM-assisted type labeling, verified by legal experts, and NER-based de-identification, and is released for structured legal document generation and evaluation.

CaseGen [173] is a Chinese-language dataset comprising 500 real-world civil case instances. Each case is structured into seven sections—Prosecution, Defense, Evidence, Events, Trial Facts, Legal Reasoning, and Judgment—and annotated by legal experts. CaseGen supports four key generation tasks: drafting defense statements, writing trial facts, composing legal reasoning, and generating judgment results.

5.1.10. Other datasets

LexGLUE [79] aggregates seven legal NLP datasets, covering tasks such as case outcome classification, issue area classification, multi-label legal document classification, and multiple-choice QA.

MultiLegalPile [273] is a 689 GB corpus of legal texts across 24 languages and 17 jurisdictions. It includes materials from the EU, UN, and national sources, aggregated to support the training of multilingual legal LLMs.

LexFiles [37] is a 19-billion-token English legal text corpus spanning six major legal systems (U.S., U.K., E.U., Canada, Australia, and India). These datasets include legislation, case law, and contracts, and address the previous lack of non-English legal training data.

EURLEX57K [274]: A multi-label text classification dataset containing 57,000 EU legal documents from the EUR-Lex platform, annotated with approximately 4700 EuroVoc labels. It supports large-scale evaluation of neural multi-label models in the EU legal domain.

German Legal NER [275]: A German corpus of 750 court decisions annotated for named entity recognition, covering both general classes (person, location, organization) and domain-specific classes (lawyer, judge, court, court decision, regulation, contract), with a total of 53,632 entity mentions.

LeNER-Br [276]: A Brazilian Portuguese legal NER dataset consisting of 70 documents (66 court documents and 4 legislative documents). It includes standard NER tags as well as two domain-specific tags for laws and legal decisions.

VICTOR [277]: A large-scale Brazilian corpus for document classification of appeals, covering 44,855 suits and 628,820 documents, with labels including initial lower-court judgment, appeal petition, judgment, and court order.

Pile of Law [278]: An open-source pretraining corpus sourced from 35 legal materials, including court filings and opinions, legal analyses, regulations, and contracts. It is designed to guide data filtering and ensure the responsible training of large language models in the legal domain.

Cambridge Law Corpus [279]: A UK corpus containing over 250,000 cases from the courts of England and Wales, with an annotated subset of 638 cases for case outcome extraction. The corpus is designed for benchmarking and fine-tuning legal-domain LLMs.

LEVEN [280]: A Chinese legal event detection dataset containing 8116 documents annotated for 108 event types and 150,977 event mentions, supporting event extraction and downstream legal reasoning.

CDD [281]: A Chinese corpus of 30,481 private-lending cases, comprising dialogues among judges, plaintiffs, and defendants, annotated with domain aspects (e.g., litigation period expiration, agreed interest rate, private loan) as well as details such as loan amount, period, and interest payment. It supports classification and summarization tasks.

LegalDiscourse [282]: A U.S. state law dataset containing over 100,000 laws, annotated with eight discourse elements (e.g., subject, object, consequence, exception) to facilitate the extraction of actors, conditions, and exceptions in statutory rules.

5.1.11. Biases and limitations in legal datasets

Legal datasets often exhibit biases, such as sampling bias, where high-profile or published cases dominate, leaving out lower-court or unpublished opinions. Jurisdictional bias is another issue, as many datasets focus primarily on a few legal systems (e.g., the U.S. and China). Additionally, language imbalance is a concern, with datasets predominantly in English and limited resources for other legal languages.

Temporal drift is also a concern, as many datasets contain outdated laws or rulings that no longer align with current legal standards. Furthermore, annotation inconsistencies, due to limited expert involvement and unclear labeling rules, reduce data reliability. Overall, these issues distort model behavior, hinder cross-jurisdictional generalization, promote outdated reasoning, and negatively affect the performance of models in real-world legal tasks.

5.2. Benchmarks and code

Furthermore, we outline the benchmarks used to assess LLM performance in the legal domain. Table 6 summarizes key benchmarks for evaluating LLMs in legal tasks.

5.2.1. Chinese benchmarks

A notable benchmark is **LAiW** [287], which focuses on Chinese legal AI and reflects real judicial practice. LAiW organizes its tasks into three progressive layers: basic legal information retrieval, knowledge inference, and complex application. Specifically, it evaluates tasks such as statute retrieval, named entity recognition, judgment summarization, similar-case matching, charge and sentence prediction, and legal QA—all based on authentic statutes and court case corpora.

Another influential benchmark is **LawBench** [288], which evaluates legal knowledge across memorization, understanding, and application levels. LawBench comprises 20 tasks, including single- and multi-label classification, regression, extraction, and generation. It assesses capabilities ranging from statutory recall to dispute-focus detection and charge prediction. The benchmark evaluated 51 multilingual, Chinese, and legal LLMs, revealing significant performance gaps and emphasizing the importance of domain specialization.

Furthermore, **LexEval** [289] provides the largest Chinese legal benchmark, comprising 23 tasks and over 14,000 questions. It covers a broad spectrum of legal capabilities, including memorization, comprehension, logical inference, discrimination, generation, and ethics. The tasks range from element extraction and multi-hop reasoning to bias detection and privacy evaluation. LexEval combines public datasets, national exam items, and expert-annotated cases, providing a comprehensive evaluation platform.

5.2.2. English benchmarks

LEGALBENCH [290] is an English-language benchmark comprising 162 tasks that reflect six distinct modes of legal reasoning, including issue spotting, rule recall, rule application, and statutory interpretation. These tasks were manually reviewed by practicing lawyers and legal scholars, capturing real-world contractual, tort, and procedural scenarios—thereby bridging academic evaluation with professional relevance.

In U.S. jurisprudence, **LawLLM** [49] presents a multi-task benchmark assessing similar case retrieval, precedent recommendation, and legal judgment prediction, demonstrating the superiority of integrated retrieval-generation pipelines over standalone generative models.

5.2.3. Multilingual and cross-lingual benchmarks

Addressing Korean legal understanding, **Kim et al.** [291] aggregated seven legal-knowledge multiple-choice sets, four lawyer-curated reasoning tasks, and extensive Korean Bar Exam problems, thereby covering statutory recall, case reasoning, and professional qualification scenarios. Addressing India's multilingual landscape, **IL-TUR** [292] includes monolingual (English, Hindi) and multilingual tasks across nine languages, covering case retrieval, document classification, and logical reasoning over statutory and judicial texts. Its publicly available leaderboard highlights performance gaps in multilingual models. **Rasiah et al.** [293] addressed Switzerland's multilingual legal system, benchmarking tasks across German, French, Italian, Romansh, and English, emphasizing long-document retrieval, judgment summarization, and multilingual text classification. **LegalBench.PT** [294] introduces a Portuguese-language benchmark derived from bar exam questions, which have been converted and verified for multiple-choice and matching tasks, addressing linguistic gaps and establishing robust accuracy benchmarks.

5.2.4. Biases and limitations in legal benchmarks

Legal LLM benchmarks often oversimplify tasks by reducing complex, multi-step legal reasoning to single-step classification or

Table 6

Summary of benchmarks for evaluating LLMs in the legal domain.

Name	Year	Task(s)	Jurisdiction	Language	Open
LAiW [287]	2024	Court judgment prediction Legal case retrieval Legal QA	China	Chinese	Yes ^a
LawBench [288]	2024	Legal summarization Contract review Court judgment prediction Legal QA	China	Chinese	Yes ^b
LexEval [289]	2024	Contract review Court judgment prediction Legal QA Legal case retrieval Legal document drafting Legal summarization Legal translation Ethics	China	Chinese	Yes ^c
LEGALBENCH [290]	2023	Court judgment prediction Legal case retrieval Legal QA Contract review	Multilingual	English	Yes ^d
LegalAgentBench [290]	2024	Legal agents	China	Chinese	Yes ^e
CitaLaw [296]	2024	Citation-grounded QA Legal case retrieval	China	Chinese	No
LawLLM [49]	2024	Legal case retrieval Court judgment prediction	U.S.	English	Yes ^f
[298]	2024	Ethics evaluation	China	Chinese	No
[291]	2024	Legal QA	Korea	Korean	Yes ^g
DISC-Law-Eval [54]	2023	Bar-exam tasks Legal case retrieval Court judgment prediction Legal summarization Legal QA Bar-exam	China	Chinese	Yes ^h
IL-TUR [292]	2024	Court judgment prediction Legal case retrieval Legal summarization Legal translation	India	Multilingual	Yes ⁱ
[293]	2024	Legal document drafting Legal summarization Court judgment prediction Legal case retrieval	Multilingual	Multilingual	No
LexSumm / LexT5 [114]	2024	Legal summarization	Multilingual	Multilingual	Yes ^j
LegalBench-RAG [295]	2024	Legal document retrieval	Multi	English	Yes ^k
SwiLTra-Bench [297]	2025	Legal translation	Switzerland	Multilingual	Yes ^l
JuDGE [271]	2025	Legal document drafting	China	Chinese	Yes ^m
LexRAG [80]	2025	Legal QA	China	Chinese	Yes ⁿ
REPLIQA [262]	2024	Legal QA	Multilingual	English	Yes ^o
LegalBench.PT [294]	2024	Legal QA	Portugal	Portuguese	Yes ^p

^a <https://github.com/CSHaitao/LexiLaw>^b <https://github.com/open-compass/LawBench>^c <https://github.com/CSHaitao/LexEval>^d <https://github.com/HazyResearch/legalbench>^e <https://github.com/CSHaitao/LegalAgentBench>^f https://github.com/Tizzzy/Law_LLM^g <https://github.com/lbox-kr/kbl>^h <https://github.com/FudanDISC/DISC-LawLLM>ⁱ <https://exploration-lab.github.io/IL-TUR>^j <https://github.com/TUMLegalTech/LexSumm-LexT5>^k <https://github.com/zeroentropy-cc/legalbenchrag>^l <https://huggingface.co/collections/joelniklaus/swiltra-bench-67c569a2ada47e4549733deb>^m <https://github.com/oneal2000/JuDGE>ⁿ <https://github.com/CSHaitao/LexRAG>^o <https://huggingface.co/datasets/ServiceNow/repliq>^p <https://huggingface.co/datasets/BeatrizCanaverde/LegalBench.PT>

multiple-choice formats, creating a gap between the benchmarks and real-world workflows (e.g., IRAC reasoning, drafting, compliance). Coverage is typically biased towards certain jurisdictions and languages (e.g., English or common-law systems), limiting cross-system and cross-lingual transfer, while static test sets fail to account for the evolving nature of law (temporal drift). As a result, high scores may overstate a model's real-world applicability. Models can still hallucinate sources,

mis-cite authorities, or generate legally unacceptable content outside the benchmark's niche. Future benchmarks should broaden jurisdictional and linguistic coverage, enhance task realism (multi-hop, end-to-end, human-in-the-loop settings), and incorporate evidence-linked evaluations (retrieval recall/precision, citation precision/recall, support overlap), along with robustness and calibration metrics, to provide more reliable assessments.

5.3. Task-oriented benchmarks

DISC-LawLLM [54] is a benchmark that pairs a retrieval-augmented Chinese chatbot with evaluation tasks assessing both objective accuracy and subjective user satisfaction. The model is trained using legal-syllogism prompts and supervised fine-tuning, while the evaluation combines automated metrics with human ratings to provide a comprehensive assessment of service quality. **LegalBench-RAG** [295] isolates the retrieval component within RAG tasks, providing precision-oriented metrics for evaluation over a large corpus.

RepliQA [262] extends QA evaluation beyond memorization by using novel reference documents and synthetic QA pairs to rigorously assess contextual understanding and retrieval accuracy. The conversational aspect of legal consultation is explored in **LexRAG** [80], offering a dataset and evaluation framework for multi-turn consultations, emphasizing conversational retrieval and answer generation using the LexIT toolkit.

LegalAgentBench [290] evaluates multi-step reasoning tasks for agent-based LLMs in the Chinese legal domain. It comprises 17 real-world corpora and 37 external knowledge tools across 300 multi-step tasks that require tool use, multi-hop reasoning, and legal drafting at varying levels of difficulty. This benchmark incorporates a progressive scoring mechanism to assess intermediate reasoning steps, enabling detailed diagnostics for autonomous legal agents.

CitaLaw [296] focuses on citation accuracy, assessing models' ability to generate legally justified answers accompanied by relevant authoritative citations. It integrates diverse questions from both laypersons and legal practitioners, along with a corpus of statutes and precedent cases. Systems are required to not only answer the questions but also provide correctly matched citations that satisfy a syllogistic alignment test—comprising a major premise, minor premise, and conclusion. The benchmark's metrics evaluate both answer quality and citation relevance, offering the first systematic assessment of “answer-with-proof” generation in the legal domain.

LexSumm [114] evaluates generative summarization quality across multiple English jurisdictions (U.S., U.K., EU, India), accompanied by a fine-tuned T5-based model (LexT5), employing ROUGE metrics and human evaluations to assess summarization.

SwiLTra-Bench [297] is a cross-lingual legal translation benchmark that aligns over 180,000 sentence pairs across German, French, Italian, Romansh, and English legal texts. It uses BLEU, COMET, and an LLM-based evaluator (SwiLTra-Judge [297]) to analyze translation accuracy and fluency.

JUDGE [271], focusing on document-level generation, assesses Chinese LLMs on drafting court judgments from factual summaries, with extensive data and comprehensive evaluation metrics that highlight factual consistency, legal reasoning, and stylistic quality.

Zhang et al. [298] evaluated the ethical robustness of LLMs in the legal domain, outlining protocols to benchmark models on ethical compliance, bias detection, and the legality of advice, thereby promoting responsible AI usage.

These benchmarks enable rigorous, transparent, and multilingual evaluation of legal LLMs, encompassing a broad range of tasks, including retrieval, reasoning, generation, translation, and ethical compliance.

6. Challenges and future directions

6.1. Challenges

Despite progress, integrating LLMs into the legal domain still faces several limitations. Addressing these challenges is essential for ensuring trustworthiness, reliability, and ethical compliance. Key challenges include data quality, model limitations, ethical concerns, regulatory issues, jurisdictional complexities, interpretative uncertainty, and the diversity of legal reasoning across different traditions.

6.1.1. Data issues

Legal LLMs depend heavily on extensive, high-quality textual datasets. However, these datasets face significant limitations in the legal domain. A major issue is temporality: legal data can become outdated as the law evolves. Statutes are amended, and new case precedents continually emerge, meaning static training corpora risk becoming obsolete over time. A model trained on outdated law may generate answers inconsistent with current statutes or recent rulings, increasing the risk of outputs based on superseded legal standards. As Bex [299] emphasizes, purely data-driven approaches are often insufficient, and legal AI systems must stay aligned with up-to-date legal norms, rather than depending solely on static data. In other words, an LLM requires “knowing” when laws change and adapting accordingly, a challenging requirement not found in many other domains.

Another challenge lies in the specificity and scarcity of legal text. Legal documents are often written in domain-specific jargon, and models require specialized datasets covering diverse areas of law. However, constructing datasets that satisfy these requirements is difficult. Many public legal NLP corpora focus on narrow jurisdictions or case types, leading to biased representations in models and affecting their performance. For example, an LLM might perform well on U.S. federal cases but struggle with state or international law due to underrepresentation. Additionally, privacy and confidentiality concerns permeate legal data: court filings and contracts contain sensitive personal information, meaning their use for training necessitates careful anonymization or privacy-preserving techniques. Prior studies have shown that LLMs can memorize and regurgitate private data encountered during training, raising significant concerns when handling client information or confidential evidence [300]. Finally, much of the legal text required for robust training (e.g., proprietary case reporters, commercial databases, private contracts) is inaccessible due to copyright or confidentiality restrictions. This lack of open data further limits LLM training. In summary, constructing balanced, current, and secure legal datasets remains a persistent challenge. Specific data issues include:

- **Domain specificity and scarcity:** Effective legal model training requires extensive and specialized datasets, which are often difficult to compile, incomplete, and biased toward common legal domains, leading to poor performance on underrepresented legal topics.
- **Data bias and imbalance:** Historical biases and selection biases (such as the overrepresentation of specific demographics or case types) can significantly skew dataset representations, potentially amplifying these biases during model training [301].
- **Privacy and confidentiality:** Legal texts often contain sensitive or privileged information [302]. Without privacy-preserving mechanisms such as differential privacy, LLMs risk leaking confidential content. For instance, a language model may reproduce verbatim names or addresses encountered during pretraining [303]. Consequently, court decisions and contracts pose significant privacy risks if used naively, emphasizing the importance of privacy-preserving training methods.
- **Data freshness:** LLMs trained on outdated data may reflect obsolete standards, yet regularly updating models is difficult given the volume of new legal documents and high retraining costs.

Thus, constructing balanced, updated, and privacy-secured legal datasets is critical for reliable and ethical legal AI applications.

6.1.2. Model issues

Even with sufficient data, model-level challenges hinder the deployment of LLMs in legal settings. Modern LLMs are highly complex neural networks, often referred to as “black boxes” because their internal reasoning is not transparent. This opacity conflicts with the expectations of the legal domain, where decisions must be explained and justified. Attorneys, judges, and clients justifiably hesitate to trust a legal answer that cannot be traced back to understandable legal reasoning.

Explanation has been a central concern in AI & Law since its inception: from early expert systems to modern machine learning models, researchers have long recognized that legal AI must provide reasoned explanations for its conclusions [155]. Atkinson et al. [155] reviewed the evolution of explanation in AI & Law, emphasizing that effective legal AI systems should be able to articulate their reasoning in terms of arguments, cases, and principles, much like a human lawyer. This remains an unsolved problem for LLMs. While an LLM can produce fluent answers, it does not inherently follow the structured reasoning of jurisprudence (e.g., citing relevant precedents or statutes for each conclusion). The complexity of legal explanation thus represents a critical gap: current models struggle to generate the transparent, legally grounded justifications that would make their outputs fully acceptable in court or advisory settings.

Another modeling challenge is the tendency of LLMs to hallucinate or generate plausible-sounding but incorrect information. In law, such errors are particularly harmful — for example, an LLM might fabricate a statute or misstate a case holding, which can have serious consequences. A well-known incident in 2023 saw attorneys sanctioned after relying on ChatGPT to draft a brief that cited non-existent cases, highlighting this risk in stark terms. Beyond factual accuracy, the lack of true legal understanding limits LLM performance. Models do not possess judgment, cannot appreciate nuanced values or equities, and may apply rules rigidly. Rissland et al. [154] noted that AI lacks the “nuanced understanding of legal and social contexts” that human lawyers bring. Legal reasoning often involves interpreting subtle language and balancing competing principles in light of social dynamics. These tasks go beyond the pattern recognition of current models. This can lead LLMs to miss context or produce one-size-fits-all answers that a trained jurist would know to adjust. Key limitations include:

- **Lack of interpretability:** The “black-box” nature of deep neural models hinders users’ ability to trust and verify the reasoning behind LLM outputs, necessitating greater interpretability and explainability. Judges and lawyers cannot easily assess LLM’s reasoning, and users may be unwilling to rely on advice generated by a “black box”. Efforts to reduce the “black-box” nature of models are vital.
- **Hallucinations and factual errors:** LLMs can generate plausible but incorrect or fabricated content, undermining reliability in critical legal situations [304].
- **Limited legal reasoning:** LLMs lack genuine legal judgment, empathy, and the ability to factor subjective or moral considerations, potentially producing insensitive or rigid outputs [305].
- **Domain transfer and maintenance:** Legal models often fail to generalize across jurisdictions and require costly and complex updates as laws evolve.

Addressing these modeling challenges is essential for ensuring accuracy, transparency, and accountability in legal AI.

6.1.3. Ethical challenges

Ethical considerations are crucial when deploying LLMs in legal applications. Key ethical challenges include bias, fairness, transparency, accountability, and intellectual property rights.

- **Bias and Fairness:** LLMs trained on biased legal records may persistently reproduce or even amplify existing patterns of discrimination [132]. For instance, prior studies have shown that predictive systems such as COMPAS have made systematically different errors for Black and white defendants [306].
- **Transparency and Accountability:** The lack of transparency regarding model outputs complicates determining responsibility for errors and makes governance challenging.
- **Privacy of Users:** Ensuring confidentiality during model inference is critical to prevent misuse or logging of sensitive client data.
- **Intellectual Property:** Generating content from copyrighted materials poses risks of plagiarism and infringement [304].

- **Misinformation:** Hallucinations are not only model errors but also ethical concerns. Disseminating false legal “facts” misleads users and undermines trust [304]. Legal practitioners depend on accuracy, and the unchecked dissemination of misinformation by LLMs is ethically unacceptable.

6.1.4. Legal and regulatory issues

Legal LLMs must comply with data and consumer protection laws (e.g., the GDPR’s restrictions on automated decision-making), as well as professional ethical standards. For example, the European Union’s Artificial Intelligence Act proposes a risk-based regulatory framework, in which systems that provide legal advice are classified as “high-risk” AI. These systems are subject to strict compliance obligations, including detailed documentation, rigorous testing, and human oversight [307]. Other regulatory concerns include:

- **Liability:** Existing legal frameworks do not clearly establish who bears responsibility for faulty legal advice provided by LLMs.
- **Unauthorized practice of law:** Jurisdictions restrict who is authorized to provide legal advice, leading to potential regulatory barriers for unsupervised LLM applications.
- **Cross-sector compliance:** Balancing AI-specific regulations with existing legal norms is crucial to ensure regulatory adherence and successful adoption.

Thus, legal LLMs face multi-layered regulation: not only AI governance rules, but also laws governing legal practice and data use. Complying with these requirements, such as implementing audit trails or incorporating human-in-the-loop oversight, presents significant challenges.

6.1.5. Jurisdictional issues

Legal systems are jurisdiction-specific, posing significant challenges for globally trained LLMs. Training on cross-jurisdictional data may lead to outputs inconsistent with local legal norms and regulations [308]. For example, LLMs trained on case law from one country might apply incorrect legal principles that fall outside the acceptable boundaries of the affected jurisdiction. Other jurisdictional challenges include:

- **Data Sovereignty:** Restrictions on cross-border data transfers complicate the training and deployment of global LLMs.
- **Legal Diversity:** Differing legal systems (e.g., civil vs. common law) necessitate jurisdiction-specific customization of LLMs.
- **Border Enforcement:** Variations in AI regulatory enforcement across jurisdictions require careful compliance to avoid legal conflicts.

Addressing jurisdictional diversity requires localization and validation of LLMs to ensure compliance and reliability in each jurisdiction [308].

6.1.6. Interpretative uncertainty

A fundamental challenge in the legal domain is the indeterminacy of legal interpretation: statutes and contracts can be ambiguous, and multiple interpretations may be reasonable. LLMs currently struggle with such ambiguity, as they are inclined to provide a single answer even when a question of law admits more than one plausible interpretation. Moreover, studies have found that an LLM’s answer to a legal question can be highly sensitive to how the prompt is phrased, suggesting a lack of robust interpretive strategy. This instability undermines trust in LLM-driven analysis of legal texts. For instance, a subtle rewording of a query about a statutory term might yield a different “ordinary meaning” response, indicating that the model has not truly internalized consistent interpretative principles. Current models also have no built-in mechanism to signal uncertainty or to request clarification when faced with indeterminate language, in contrast to human lawyers who might acknowledge ambiguity or seek additional context. Key aspects of this interpretative challenge include:

- **Semantic ambiguity:** Many legal provisions (and even court judgments) use broad or vague terms that lead to multiple reasonable interpretations. LLMs often lack the nuanced understanding to distinguish fine-grained meanings or to identify all plausible interpretations of an ambiguous term.
- **Context sensitivity:** The meaning of legal language can depend heavily on context—such as the specific facts of a case, the purpose of a statute, or jurisdiction-specific definitions. Without explicit contextual cues, LLMs may misinterpret a provision by assuming a default or generic meaning that is inappropriate for the situation at hand.
- **Prompt instability:** LLM outputs on legal interpretation tasks have been shown to vary significantly with slight changes in prompt wording. This sensitivity indicates that models are not reliably capturing stable legal meanings, but rather are influenced by superficial prompt cues.
- **Lack of interpretive feedback:** Unlike a trained lawyer who refines interpretations through critique and counter-arguments, a vanilla LLM has no iterative feedback mechanism to refine its understanding. Without guidance, the model might confidently present one interpretation without recognizing competing interpretations that might be equally or more valid.

6.1.7. Legal reasoning across traditions

Legal reasoning styles vary widely across different legal traditions, most notably between common law and civil law systems. For instance, a common law model trained on Anglo-American case reports may prioritize analogical reasoning with extensive citations to precedent, whereas a civil law-oriented model focuses on code interpretation, principles, and doctrinal exposition. If an LLM is not adapted to these differences, it may produce analyses that seem cogent in one jurisdiction yet are unpersuasive or even incorrect in another. For example, a response that heavily cites case precedents might be suitable for a U.S. legal question but would appear out of place in a French civil law context, which values statutory authority and scholarly commentary over case law. Conversely, a purely code-based explanation might overlook the importance of case precedents in a common law setting. These discrepancies can undermine the performance and trustworthiness of LLMs in legal practice. Key differences in legal reasoning include:

- **Source of authority:** Common law reasoning relies heavily on judicial precedents and past case analogies, whereas civil law reasoning emphasizes codified statutes, codes, and regulations as primary sources of authority. An LLM requires recognizing which sources carry weight in a given jurisdiction.
- **Argumentative style:** In common law systems, legal arguments often focus on detailed comparisons of specific cases and the development of legal principles. In contrast, civil law arguments tend to be more abstract and principle-driven, often structured as applications of general code provisions to the facts.
- **Terminology and format:** Different traditions use distinct legal terminology and document formats. For instance, legal writing in common law jurisdictions includes citations to reporters and uses terminology inherited from English law, while civil law jurisdictions may refer to articles of codes and employ different phrasing. LLMs require adaptation to these language variations to generate contextually appropriate content.

6.2. Future directions

6.2.1. Development of legal data resources

The next generation of legal LLMs relies on larger, more diverse data resources, making large-scale, multilingual legal corpora a priority. For example, the MultiLegalPile dataset [273] offers 689GB of legal texts in 24 languages from 17 jurisdictions, using publicly available statutes and

case law. This work demonstrates that openly licensed government legal texts can provide high-quality training data for LLMs, helping to mitigate the over-representation of English-language and common-law sources. Chalkidis et al. [79] highlighted that the scarcity of non-English legal datasets is a major problem for the globalization of legal AI. Expanding multilingual and cross-jurisdictional corpora will be crucial for enabling LLMs to handle international and comparative legal tasks.

Beyond raw text corpora, structured legal knowledge databases are crucial for enhancing LLM performance. Open-access initiatives, such as Harvard Law School's CaseLaw Project, already provide hundreds of thousands of annotated decisions. Expanding these repositories and fostering data-sharing partnerships among judiciaries, governments, and academic institutions could significantly enrich training resources. However, such initiatives must navigate privacy and copyright constraints, necessitating the adoption of privacy-preserving techniques to enable access to protected materials.

6.2.2. Coordination with the traditional judicial system

While LLMs are unlikely to replace judges or lawyers, they can significantly enhance legal practice. A key direction is the development of human-AI interfaces that integrate into legal professionals' workflows. For instance, Han et al. [55] proposed LegalAsst, a human-AI tool that "provides explainable, traceable, and controllable legal assistance and references" by visualizing the judgment process. LegalAsst employs structured representations and decision trees to make case analysis and legislative logic fully transparent to users. In practice, courts are increasingly employing AI tools to support routine tasks. For example, the American Bar Association outlines permissible uses of AI in judicial settings, including conducting legal research, summarizing briefs and depositions, creating timelines, and checking drafts for errors [309]. These examples demonstrate that LLMs can assume repetitive or labor-intensive responsibilities, thereby enabling judges and clerks to focus on substantive judicial decision-making.

LLMs can serve as supervisors and research assistants for law students and early-career lawyers. Recent studies suggest that AI is capable of addressing basic educational queries and automating routine administrative tasks, thereby allowing professors to concentrate on more critical aspects of legal instruction. One survey concludes that in legal education, the role of AI should be to enhance the learning experience rather than replace rigorous instruction [26] — for example, by generating explanations for practice problems or summarizing complex legal doctrines. In this way, law schools and bar preparation courses can incorporate LLM tools to simulate client queries or assist in drafting pleadings under supervision, thus cultivating a new generation of lawyers proficient in AI-assisted legal practice.

6.2.3. Regulatory and ethical governance

As LLMs become increasingly integrated into legal workflows, robust governance mechanisms are essential. First, accountability frameworks should ensure that AI behavior aligns with legal and ethical standards. In AI policy literature, accountability is defined as a normative obligation to act in a "transparent, fair, and equitable manner," with developers held responsible for the impacts of their systems [310]. Privacy is another key concern. Recent studies highlight that the use of LLMs in legal contexts raises risks of "privacy invasion" and challenges related to interpretability [26]. For example, court records often contain sensitive personal information, necessitating anonymization or strict access controls during both model training and deployment in legal proceedings. In other words, as AI systems assume advisory roles in legal decision-making, regulators may classify them as high-risk applications. This would trigger compliance obligations under emerging regulatory frameworks, particularly regarding transparency, data protection, and human oversight.

7. Conclusion

This survey presents a task-oriented overview of the application of LLMs in the legal domain. We review general LLMs and legal-domain LLMs, adaptation strategies, and applications in core legal tasks, including legal information retrieval, legal document analysis, judicial decision prediction, legal question answering and legal document generation. We also discuss legal agent-based modeling, which integrates these tasks into multi-stage workflows, and LLM-based applications in legal education and training. By automating document analysis, integrating extensive legal information and assisting decision-making workflows, LLMs can improve the efficiency and consistency of legal practice.

This survey further consolidates publicly available datasets, benchmarks and code resources for legal LLMs, providing a basis for reproducible research and more rigorous, task-oriented evaluation. Nevertheless, significant challenges remain, including data quality and coverage, bias and robustness, interpretability and fidelity to legal reasoning, data privacy and confidentiality, and regulatory and jurisdictional constraints. Addressing these issues requires coordinated advances in legal data curation, model design, evaluation methodologies and governance frameworks to enable responsible deployment of LLMs in high-stakes legal settings.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] D. Jain, M.D. Borah, A. Biswas, Summarization of legal documents: where are we now and the way forward, *Comput. Sci. Rev.* 40 (2021) 100388.
- [2] Y. Feng, C. Li, V. Ng, Legal case retrieval: a survey of the state of the art, in: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 6472–6485.
- [3] A. Abdallah, B. Piryani, A. Jatowt, Exploring the state of the art in legal QA systems, *J. Big Data* 10 (1) (2023) 127.
- [4] C. Zhang, Y. Meng, Bridging the divide: technical research and application on legal judgment prediction, *Artif. Intell. Law* (2025) 1–32.
- [5] T.B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D.M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language models are few-shot learners, in: *Proceedings of the 34th International Conference on Neural Information Processing Systems, NIPS '20, Vancouver, BC, Canada, 2020*, pp. 1877–1901, <https://doi.org/10.5555/3495724.3495883>.
- [6] OpenAI, Gpt-4 technical report, arXiv preprint [arXiv:2303.08774](https://arxiv.org/abs/2303.08774), 2023.
- [7] H. Zhang, Y.-N. Cheah, O.M. Alysiri, J. An, Exploring aspect-based sentiment quadruple extraction with implicit aspects, opinions, and chatgpt: a comprehensive survey, *Artif. Intell. Rev.* 57 (2) (2024) 17, <https://doi.org/10.1007/s10462-023-10633-x>.
- [8] S. Minaee, T. Mikolov, N. Nikzad, M. Chenaghlu, R. Socher, X. Amatriain, J. Gao, Large language models: a survey, arXiv preprint [arXiv:2402.06196](https://arxiv.org/abs/2402.06196), 2024.
- [9] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, P.J. Liu, Exploring the limits of transfer learning with a unified text-to-text transformer, *J. Mach. Learn. Res.* 21 (1) (Jan 2020), <https://doi.org/10.5555/3455716.3455856>.
- [10] D.M. Katz, M.J. Bommarito, S. Gao, P. Arredondo, Gpt-4 passes the bar exam, *Philos. Trans. R. Soc. A* 382 (2270) (2024) 20230254, <https://doi.org/10.1098/rsta.2023.0254>.
- [11] L. Zheng, N. Guha, B.R. Anderson, P. Henderson, D.E. Ho, When does pretraining help? Assessing self-supervised learning for law and the casehold dataset of 53,000+ legal holdings, in: *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Law: ICAIL 2021, São Paulo, Brazil, 2021*, pp. 159–168, <https://doi.org/10.1145/3462757.3466088>.
- [12] Z. Zhou, K.-Y. Yu, S.-Y. Tian, X.-W. Yang, J.-X. Shi, P. Song, Y.-X. Jin, L.-Z. Guo, Y.-F. Li, Lawgpt: knowledge-guided data generation and its application to legal llm, arXiv preprint [arXiv:2502.06572](https://arxiv.org/abs/2502.06572), 2025, <https://doi.org/10.48550/arXiv.2502.06572>.
- [13] Q. Huang, M. Tao, C. Zhang, Z. An, C. Jiang, Z. Chen, Z. Wu, Y. Feng, Lawyer llama technical report, arXiv preprint [arXiv:2305.15062](https://arxiv.org/abs/2305.15062), 2023, <https://doi.org/10.48550/arXiv.2305.15062>.
- [14] Z. Fei, S. Zhang, X. Shen, D. Zhu, X. Wang, J. Ge, V. Ng, InternLM-law: an open-sourced Chinese legal large language model, in: O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. Eugenio di, S. Schockaert (Eds.), *Proceedings of the 31st International Conference on Computational Linguistics, Abu Dhabi, UAE, 2025*, pp. 9376–9392.
- [15] J. Savelka, K.D. Ashley, M.A. Gray, H. Westermann, H. Xu, Explaining legal concepts with augmented large language models (gpt-4), arXiv preprint [arXiv:2306.09525](https://arxiv.org/abs/2306.09525), 2023, <https://doi.org/10.48550/arXiv.2306.09525>.
- [16] S. Narendra, K. Shetty, A. Ratnaparkhi, Enhancing contract negotiations with LLM-based legal document comparison, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, D. Preotiuc-Pietro, G. Spanakis (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2024, Miami, FL, USA, 2024*, pp. 143–153, <https://doi.org/10.18653/v1/2024.nllp-1.11>.
- [17] S.H. Wang, M. Zubkov, K. Fan, S. Harrell, Y. Sun, W. Chen, A. Plesner, R. Wattenhofer, Acord: An expert-annotated retrieval dataset for legal contract drafting, arXiv preprint [arXiv:2501.06582](https://arxiv.org/abs/2501.06582), 2025, <https://doi.org/10.48550/arXiv.2501.06582>.
- [18] C. Ryu, S. Lee, S. Pang, C. Choi, H. Choi, M. Min, J.-Y. Sohn, Retrieval-based evaluation for llms: a case study in Korean legal QA, in: D. Preotiuc-Pietro, C. Goanță, I. Chalkidis, L. Barrett, G. Spanakis, N. Aletras (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2023, 2023*, pp. 132–137, <https://doi.org/10.18653/v1/2023.nllp-1.13>.
- [19] H.Q. Pham, Q. Van Nguyen, D.Q. Tran, T.K.-B. Nguyen, K. Van Nguyen, Top 2 at alqac 2024: large language models (llms) for legal question answering, *Int. J. Asian Lang. Process.* 35 (1) (2025) 2450010, <https://doi.org/10.1142/S2717554524500103>.
- [20] P. Akarajaratwong, P. Pothavorn, C. Chaksangchaichot, P. Tasawong, T. Nopparatbundit, S. Nutanong, Nitibench: a comprehensive study of llm framework capabilities for thai legal question answering, arXiv preprint [arXiv:2502.10868](https://arxiv.org/abs/2502.10868), 2025, <https://doi.org/10.48550/arXiv.2502.10868>.
- [21] W. Qin, Z. Cao, W. Yu, Z. Si, S. Chen, J. Xu, Explicitly integrating judgment prediction with legal document retrieval: a law-guided generative approach, in: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2024*, pp. 2210–2220, <https://doi.org/10.1145/3626772.3657717>.
- [22] R. Shui, Y. Cao, X. Wang, T.-S. Chua, A comprehensive evaluation of large language models on legal judgment prediction, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, 2023*, pp. 7337–7348, <https://doi.org/10.18653/v1/2023.findings-emnlp.490>.
- [23] C. Deng, K. Mao, Y. Zhang, Z. Dou, Enabling discriminative reasoning in LLMs for legal judgment prediction, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, Florida, USA, 2024*, pp. 784–796, <https://doi.org/10.18653/v1/2024.findings-emnlp.43>.
- [24] Y. Liu, Y. Wu, A. Li, Y. Zhang, C. Sun, W. Lu, F. Wu, K. Kuang, Unleashing the power of LLMs in court view generation by stimulating internal knowledge and incorporating external knowledge, in: K. Duh, H. Gomez, S. Bethard (Eds.), *Findings of the Association for Computational Linguistics: NAACL 2024, Mexico City, Mexico, 2024*, pp. 2782–2792, <https://doi.org/10.18653/v1/2024.findings-naacl.177>.
- [25] G. Chen, L. Fan, Z. Gong, N. Xie, Z. Li, Z. Liu, C. Li, Q. Qu, S. Ni, M. Yang, Agentcourt: Simulating court with adversarial evolvable lawyer agents, arXiv preprint [arXiv:2408.08089](https://arxiv.org/abs/2408.08089), 2024, <https://doi.org/10.48550/arXiv.2408.08089>.
- [26] W. Qin, Z. Sun, Exploring the nexus of large language models and legal systems: a short survey, arXiv preprint [arXiv:2404.00990](https://arxiv.org/abs/2404.00990), 2024, <https://doi.org/10.48550/arXiv.2404.00990>.
- [27] Z. Chen, J. Ma, X. Zhang, N. Hao, A. Yan, A. Nourbakhsh, X. Yang, J. McAuley, L.R. Petzold, W.Y. Wang, A survey on large language models for critical societal domains: finance, healthcare, and law, *Trans. Mach. Learn. Res.* 11 (2024).
- [28] J. Lai, W. Gan, J. Wu, Z. Qi, P.S. Yu, Large language models in law: a survey, *AI Open* 5 (2024) 181–196, <https://doi.org/10.1016/j.aiopen.2024.09.002>.
- [29] D. Küçük, F. Can, Computational law: datasets, benchmarks, and ontologies, arXiv preprint [arXiv:2503.04305](https://arxiv.org/abs/2503.04305), 2025.
- [30] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: J. Burstein, C. Doran, T. Solorio (Eds.), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Minneapolis, Minnesota, 2019, pp. 4171–4186, <https://doi.org/10.18653/v1/N19-1423>.
- [31] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, Roberta: a robustly optimized bert pretraining approach, arXiv preprint [arXiv:1907.11692](https://arxiv.org/abs/1907.11692), 2019.
- [32] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, et al., Llama: open and efficient foundation language models, arXiv preprint [arXiv:2302.13971](https://arxiv.org/abs/2302.13971), 2023, <https://doi.org/10.48550/arXiv.2302.13971>.
- [33] R. Macey-Dare, Chatgpt & generative AI systems as quasi-expert legal advice lawyers-case study considering potential appeal against conviction of tom hayes, Available at SSRN [4342686](https://ssrn.com/abstract=4342686), 2023.
- [34] K.Y. Iu, V.M.-Y. Wong, Chatgpt by openai: the end of litigation lawyers? Available at SSRN [4339839](https://ssrn.com/abstract=4339839), 2023.
- [35] M. Molinari, I.A. Amantea, M. Quaranta, G. Governatori, Chatgpt unveils its limits: principles of law deliver checkmate, arXiv preprint [arXiv:2510.19261](https://arxiv.org/abs/2510.19261), 2025.
- [36] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, I. Androustopoulos, LEGAL-BERT: the muppets straight out of law school, in: T. Cohn, Y. He, Y. Liu (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020, Online, 2020*, pp. 2898–2904, <https://doi.org/10.18653/v1/2020.findings-emnlp.261>.

- [37] I. Chalkidis, N. Garneau, C. Goanta, D. Katz, A. Søgaard, LeXFiles and LegalLAMA: facilitating English multinational legal language model development, in: A. Rogers, J. Boyd-Graber, N. Okazaki (Eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Toronto, Canada, 2023, pp. 15513–15535, <https://doi.org/10.18653/v1/2023.acl-long.865>.
- [38] S. Douka, H. Abdine, M. Vazirgiannis, R. El Hamdani, D. Restrepo Amariles, JuriBERT: a masked-language model adaptation for French legal text, in: N. Aletras, I. Androustopoulos, L. Barrett, C. Goanta, D. Preotiuc-Pietro (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2021*, Punta Cana, Dominican Republic, 2021, pp. 95–101, <https://doi.org/10.18653/v1/2021.nllp-1.9>.
- [39] M. Al-Qurishi, S. Alqaseemi, R. Souissi, AraLegal-BERT: a pretrained language model for arabic legal text, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanta, D. Preotiuc-Pietro (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2022*, Abu Dhabi, United Arab Emirates (Hybrid), 2022, pp. 338–344, <https://doi.org/10.18653/v1/2022.nllp-1.31>.
- [40] K. Miyazaki, H. Yamada, T. Tokunaga, Cross-domain analysis on Japanese legal pretrained language models, in: *Findings of the Association for Computational Linguistics: ACL-LJCNLP 2022*, 2022, pp. 274–281, <https://doi.org/10.18653/v1/2022.findings-acl.26>.
- [41] S. Paul, A. Mandal, P. Goyal, S. Ghosh, Pre-trained language models for the legal domain: a case study on Indian law, in: *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*, 2023, pp. 187–196, <https://doi.org/10.1145/3594536.3595165>.
- [42] R. Silveira, C. Ponte, V. Almeida, V. Pinheiro, V. Furtado, Legalbert-pt: a pretrained language model for the Brazilian Portuguese legal domain, in: *Brazilian Conference on Intelligent Systems*, Springer, 2023, pp. 268–282.
- [43] D. Licari, G. Comandè, Italian-legal-bert models for improving natural language processing tasks in the Italian legal domain, *Comput. Law Secur. Rev.* 52 (2024) 105908, <https://doi.org/10.1016/j.clsr.2023.105908>.
- [44] D. Mamakas, P. Tsotsi, I. Androustopoulos, I. Chalkidis, Processing long legal documents with pre-trained transformers: modding LegalBERT and longformer, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanta, D. Preotiuc-Pietro (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2022*, Abu Dhabi, United Arab Emirates (Hybrid), 2022, pp. 130–142, <https://doi.org/10.18653/v1/2022.nllp-1.11>.
- [45] C. Xiao, X. Hu, Z. Liu, C. Tu, M. Sun, Lawformer: a pre-trained language model for Chinese legal long documents, *AI Open* 2 (2021) 79–84, <https://doi.org/10.1016/j.aiopen.2021.06.003>.
- [46] J. Niklaus, D. Gíofre, Can we pretrain a SotA legal language model on a budget from scratch? in: *Proceedings of the Fourth Workshop on Simple and Efficient Natural Language Processing (SustainNLP)*, Toronto, Canada (Hybrid), 2023, pp. 158–182, <https://doi.org/10.18653/v1/2023.sustainlp-1.11>.
- [47] W. Hua, Y. Zhang, Z. Chen, J. Li, M. Weber, Mixed-domain language modeling for processing long legal documents, in: D. Preotiuc-Pietro, C. Goanta, I. Chalkidis, L. Barrett, G. Spanakis, N. Aletras (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2023*, Singapore, 2023, pp. 51–61, <https://doi.org/10.18653/v1/2023.nllp-1.7>.
- [48] N. Kitaev, L. Kaiser, A. Levskaya, Reformer: the efficient transformer, in: *International Conference on Learning Representations*, 2020, <https://openreview.net/forum?id=rkgNKKHtvB>.
- [49] D. Shu, H. Zhao, X. Liu, D. Demeter, M. Du, Y. Zhang, Lawllm: law large language model for the us legal system, in: *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management: CIKM 2024*, 2024, pp. 4882–4889, <https://doi.org/10.1145/3627673.3680020>.
- [50] P. Colombo, T. Pires, M. Boudiaf, R. Melo, D. Culver, E. Malaboeuf, G. Hautreux, J. Charpentier, M. Desa, Saullm-54b & saullm-141b: scaling up domain adaptation for the legal domain, *Adv. Neural Inf. Process. Syst.* 37 (2024) 129672–129695.
- [51] J.-S. Lee, Lexgpt 0.1: pre-trained gpt-j models with pile of law, *arXiv preprint arXiv:2306.05431*, 2023, <https://doi.org/10.48550/arXiv.2306.05431>.
- [52] R. Dominguez-Olmedo, V. Nanda, R. Abebe, S. Bechtold, C. Engel, J. Frankenreiter, K.P. Gummadi, M. Hardt, M. Livermore, Lawma: the power of specialization for legal annotation, in: *The Thirtieth International Conference on Learning Representations*, 2025, <https://openreview.net/forum?id=7El7K1DoyX>.
- [53] J. Cui, M. Ning, Z. Li, B. Chen, Y. Yan, H. Li, B. Ling, Y. Tian, L. Yuan, Chatlaw: A multi-agent collaborative legal assistant with knowledge graph enhanced mixture-of-experts large language model, *arXiv preprint arXiv:2306.16092*, 2023, <https://doi.org/10.48550/arXiv.2306.16092>.
- [54] S. Yue, W. Chen, S. Wang, B. Li, C. Shen, S. Liu, Y. Zhou, Y. Xiao, S. Yun, X. Huang, et al., Disc-lawllm: fine-tuning large language models for intelligent legal services, *arXiv preprint arXiv:2309.11325*, 2023, <https://doi.org/10.48550/arXiv.2309.11325>.
- [55] W. Han, J. Shen, Y. Liu, Z. Shi, J. Xu, F. Hu, H. Chen, Y. Gong, X. Yu, H. Wang, et al., Legalasst: human-centered and ai-empowered machine to enhance court productivity and legal assistance, *Inf. Sci.* 679 (2024) 121052, <https://doi.org/10.1016/j.ins.2024.121052>.
- [56] J.Z. Liu, X. Li, How do judges use large language models? Evidence from shenzhen, *J. Leg. Anal.* 16 (1) (2024) 235–262, <https://doi.org/10.1093/jla/laaf002>.
- [57] A. Izzidien, H. Sargeant, F. Steffek, Llm vs. lawyers: Identifying a subset of summary judgments in a large uk case law dataset, *arXiv preprint arXiv:2403.04791*, 2024, <https://doi.org/10.48550/arXiv.2403.04791>.
- [58] R. Yao, Y. Wu, T. Zhang, X. Zhang, Y. Huang, Y. Wu, J. Yang, C. Sun, F. Wang, X. Liu, Intelligent legal assistant: an interactive clarification system for legal question answering, *arXiv preprint arXiv:2502.07904*, 2025, <https://doi.org/10.48550/arXiv.2502.07904>.
- [59] Y. Hu, K. Luo, Y. Feng, ELLA: empowering LLMs for interpretable, accurate and informative legal advice, in: Y. Cao, Y. Feng, D. Xiong (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, Bangkok, Thailand, 2024, pp. 374–387, <https://doi.org/10.18653/v1/2024.acl-demos.36>.
- [60] X.-Y. Fu, M.T.R. Laskar, E. Khasanova, C. Chen, S. Tn, Tiny titans: can smaller large language models punch above their weight in the real world for meeting summarization? in: Y. Yang, A. Davani, A. Sil, A. Kumar (Eds.), *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 6: Industry Track)*, Mexico City, Mexico, 2024, pp. 387–394, <https://doi.org/10.18653/v1/2024.naacl-industry.33>.
- [61] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, J. Steinhardt, Measuring massive multitask language understanding, in: *International Conference on Learning Representations*, 2021.
- [62] D. Hendrycks, C. Burns, A. Chen, S. Ball, CUAD: an expert-annotated NLP dataset for legal contract review, in: *Thirty-Fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021, <https://openreview.net/forum?id=71lYgs3Bamw>.
- [63] D. Trautmann, A. Petrova, F. Schilder, Legal prompt engineering for multilingual legal judgement prediction, *arXiv preprint arXiv:2212.02199*, 2022, <https://doi.org/10.48550/arXiv.2212.02199>.
- [64] F. Yu, L. Quartey, F. Schilder, Legal prompting: teaching a language model to think like a lawyer, *arXiv preprint arXiv:2212.01326*, 2022, <https://doi.org/10.48550/arXiv.2212.01326>.
- [65] I. Chalkidis, Chatgpt may pass the bar exam soon, but has a long way to go for the lexglue benchmark, *arXiv preprint arXiv:2304.12202*, 2023, <https://doi.org/10.48550/arXiv.2304.12202>.
- [66] A. Blair-Stanek, N. Holzenberger, B. Van Durme, Can gpt-3 perform statutory reasoning? in: *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law: ICAIL2023*, ICAIL '23, Braga, Portugal, 2023, pp. 22–31, <https://doi.org/10.1145/3594536.3595163>.
- [67] L. Zhu, L. Yang, C. Li, S. Hu, L. Liu, B. Yin, Legilm: a fine-tuned legal language model for data compliance, *arXiv preprint arXiv:2409.13721*, 2024, <https://doi.org/10.48550/arXiv.2409.13721>.
- [68] Z. Wan, Y. Zhang, Y. Wang, F. Cheng, S. Kurohashi, Reformulating domain adaptation of large language models as adapt-retrieve-revise: a case study on Chinese legal domain, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, Bangkok, Thailand, 2024, pp. 5030–5041, <https://doi.org/10.18653/v1/2024.findings-acl.299>.
- [69] C. Nguyen, L.-M. Nguyen, Employing label models on chatgpt answers improves legal text entailment performance, *arXiv preprint arXiv:2401.17897*, 2024, <https://doi.org/10.48550/arXiv.2401.17897>.
- [70] V. Magesh, F. Surani, M. Dahl, M. Suzgun, C.D. Manning, D.E. Ho, Hallucination-free? assessing the reliability of leading ai legal research tools, *arXiv preprint arXiv:2405.20362*, 2024, <https://doi.org/10.48550/arXiv.2405.20362>.
- [71] Y. Hu, L. Gan, W. Xiao, K. Kuang, F. Wu, Fine-tuning large language models for improving factuality in legal question answering, in: O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. Eugenio di, S. Schockaert (Eds.), *Proceedings of the 31st International Conference on Computational Linguistics*, Abu Dhabi, UAE, 2025, pp. 4410–4427.
- [72] F. Yu, L. Quartey, F. Schilder, Exploring the effectiveness of prompt engineering for legal reasoning tasks, in: *Findings of the Association for Computational Linguistics: ACL 2023*, 2023, pp. 13582–13596, <https://doi.org/10.18653/v1/2023.findings-acl.858>.
- [73] A.H. Parizi, Y. Liu, P. Nokku, S. Gholamian, D. Emerson, A comparative study of prompting strategies for legal text classification, in: *Proceedings of the Natural Legal Language Processing Workshop 2023*, 2023, pp. 258–265, <https://doi.org/10.18653/v1/2023.nllp-1.25>.
- [74] B. Lester, R. Al-Rfou, N. Constant, The power of scale for parameter-efficient prompt tuning, in: M.-F. Moens, X. Huang, L. Specia, S.-W.-T. Yih (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, Online and Punta Cana, Dominican Republic, 2021, pp. 3045–3059, <https://doi.org/10.18653/v1/2021.emnlp-main.243>.
- [75] E.J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, et al., Lora: low-rank adaptation of large language models, *ICLR 1 (2)* (2022) 3.
- [76] L. Xu, H. Xie, S.-Z.J. Qin, X. Tao, F.L. Wang, Parameter-efficient fine-tuning methods for pretrained language models: a critical review and assessment, *arXiv preprint arXiv:2312.12148*, 2023.
- [77] M. Akter, E. Cano, E. Weber, D. Dobler, I. Habernal, A comprehensive survey on legal summarization: challenges and future directions, *arXiv preprint arXiv:2501.17830*, 2025, <https://doi.org/10.48550/arXiv.2501.17830>.
- [78] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q.V. Le, D. Zhou, et al., Chain-of-thought prompting elicits reasoning in large language models, *Adv. Neural Inf. Process. Syst.* 35 (2022) 24824–24837.
- [79] I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androustopoulos, D. Katz, N. Aletras, LexGLUE: a benchmark dataset for legal language understanding in English, in: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Dublin, Ireland, 2022, pp. 4310–4330, <https://doi.org/10.18653/v1/2022.acl-long.297>.
- [80] H. Li, Y. Chen, Y. Hu, Q. Ai, J. Chen, X. Yang, J. Yang, Y. Wu, Z. Liu, Y. Liu, Lexrag: benchmarking retrieval-augmented generation in multi-turn legal consultation conversation, *arXiv preprint arXiv:2502.20640*, 2025, <https://doi.org/10.48550/arXiv.2502.20640>.
- [81] F. Mo, A. Ghaddar, K. Mao, M. Rezagholizadeh, B. Chen, Q. Liu, J.-Y. Nie, CHIQC: contextual history enhancement for improving query rewriting in conversational

- search, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Miami, Florida, USA, 2024, pp. 2253–2268, <https://doi.org/10.18653/v1/2024.emnlp-main.135>
- [82] F. Mo, K. Mao, Y. Zhu, Y. Wu, K. Huang, J.-Y. Nie, ConvQGR: generative query reformulation for conversational search, in: A. Rogers, J. Boyd-Graber, N. Okazaki (Eds.), Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Toronto, Canada, 2023, pp. 4998–5012, <https://doi.org/10.18653/v1/2023.acl-long.274>
- [83] M. Li, H. Zhuang, K. Hui, Z. Qin, J. Lin, R. Jagerman, X. Wang, M. Bendersky, Can query expansion improve generalization of strong cross-encoder rankers? in: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2024, pp. 2321–2326, <https://doi.org/10.1145/3626772.3657979>
- [84] X. Ma, L. Wang, N. Yang, F. Wei, J. Lin, Fine-tuning Llama for multi-stage text retrieval, in: Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2024, pp. 2421–2425, <https://doi.org/10.1145/3626772.3657951>
- [85] Z. Liu, C. Li, S. Xiao, Y. Shao, D. Lian, Llama2vec: unsupervised adaptation of large language models for dense retrieval, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Bangkok, Thailand, 2024, pp. 3490–3500, <https://doi.org/10.18653/v1/2024.acl-long.191>
- [86] R. Xu, W. Shi, Y. Yu, Y. Zhuang, Y. Zhu, M.D. Wang, J.C. Ho, C. Zhang, C. Yang, BMRetriever: tuning large language models as better biomedical text retrievers, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Miami, Florida, USA, 2024, pp. 22234–22254, <https://doi.org/10.18653/v1/2024.emnlp-main.1241>
- [87] K. Mao, C. Deng, H. Chen, F. Mo, Z. Liu, T. Sakai, Z. Dou, ChatRetriever: adapting large language models for generalized and robust conversational dense retrieval, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Miami, Florida, USA, 2024, pp. 1227–1240, <https://doi.org/10.18653/v1/2024.emnlp-main.71>
- [88] H. Zhang, C. Chen, L. Mei, Q. Liu, J. Mao, Mamba retriever: utilizing mamba for effective and efficient dense retrieval, in: Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, 2024, pp. 4268–4272, <https://doi.org/10.1145/3627673.3679959>
- [89] R. Pradeep, S. Sharifmoghaddam, J. Lin, Rankzephyr: effective and robust zero-shot listwise reranking is a breeze!, arXiv preprint [arXiv:2312.02724](https://arxiv.org/abs/2312.02724), 2023, <https://doi.org/10.48550/arXiv.2312.02724>
- [90] X. Zhang, S. Hofstätter, P. Lewis, R. Tang, J. Lin, Rank-without-gpt: building gpt-independent listwise rerankers on open-source large language model, arXiv preprint [arXiv:2312.02969](https://arxiv.org/abs/2312.02969), 2023, <https://doi.org/10.48550/arXiv.2312.02969>
- [91] H. Déjean, S. Clinchant, T. Formal, A thorough comparison of cross-encoders and llms for reranking splade, arXiv preprint [arXiv:2403.10407](https://arxiv.org/abs/2403.10407), 2024, <https://doi.org/10.48550/arXiv.2403.10407>
- [92] T. Niu, S. Joty, Y. Liu, C. Xiong, Y. Zhou, S. Yavuz, Judgerank: leveraging large language models for reasoning-intensive reranking, arXiv preprint [arXiv:2411.00142](https://arxiv.org/abs/2411.00142), 2024, <https://doi.org/10.48550/arXiv.2411.00142>
- [93] S. Zhuang, X. Ma, B. Koopman, J. Lin, G. Zucco, Rank-r1: enhancing reasoning in llm-based document rerankers via reinforcement learning, arXiv preprint [arXiv:2503.06034](https://arxiv.org/abs/2503.06034), 2025, <https://doi.org/10.48550/arXiv.2503.06034>
- [94] Y. Tay, V. Tran, M. Dehghani, J. Ni, D. Bahri, H. Mehta, Z. Qin, K. Hui, Z. Zhao, J. Gupta, et al., Transformer memory as a differentiable search index, *Adv. Neural Inf. Process. Syst.* 35 (2022) 21831–21843.
- [95] M. Bevilacqua, G. Ottaviano, P. Lewis, S. Yih, S. Riedel, F. Petroni, Autoregressive search engines: generating substrings as document identifiers, *Adv. Neural Inf. Process. Syst.* 35 (2022) 31668–31683, <https://doi.org/10.5555/3600270.3602566>
- [96] Y. Wang, Y. Hou, H. Wang, Z. Miao, S. Wu, Q. Chen, Y. Xia, C. Chi, G. Zhao, Z. Liu, et al., A neural corpus indexer for document retrieval, *Adv. Neural Inf. Process. Syst.* 35 (2022) 25600–25614, <https://doi.org/10.5555/3600270.3602126>
- [97] H. Zeng, C. Luo, B. Jin, S.M. Sarwar, T. Wei, H. Zamani, Scalable and effective generative information retrieval, in: Proceedings of the ACM Web Conference 2024, WWW '24, Singapore, Singapore, 2024, pp. 1441–1452, <https://doi.org/10.1145/3589334.3645477>
- [98] Q. Tang, J. Chen, Z. Li, B. Yu, Y. Lu, H. Yu, H. Lin, F. Huang, B. He, X. Han, et al., Self-retrieval: end-to-end information retrieval with one large language model, *Adv. Neural Inf. Process. Syst.* 37 (2024) 63510–63533.
- [99] H. Li, Q. Ai, J. Chen, Q. Dong, Y. Wu, Y. Liu, C. Chen, Q. Tian, Sailer: structure-aware pre-trained language model for legal case retrieval, in: Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2023, pp. 1035–1044, <https://doi.org/10.1145/3539618.3591761>
- [100] Y. Zhou, H. Huang, Z. Wu, Boosting legal case retrieval by query content selection with large language models, in: Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region, SIGIR-AP '23, Beijing, China, 2023, pp. 176–184, <https://doi.org/10.1145/3624918.3625328>
- [101] C. Deng, K. Mao, Z. Dou, Learning interpretable legal case retrieval via knowledge-guided case reformulation, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, Miami, Florida, USA, 2024, pp. 1253–1265, <https://doi.org/10.18653/v1/2024.emnlp-main.73>
- [102] S. Ma, C. Chen, Q. Chu, J. Mao, Leveraging large language models for relevance judgments in legal case retrieval, arXiv preprint [arXiv:2403.18405](https://arxiv.org/abs/2403.18405), 2024, <https://doi.org/10.48550/arXiv.2403.18405>
- [103] H.-L. Nguyen, T.-M. Nguyen, D.-M. Nguyen, T.-H.-Y. Vuong, H.-T. Nguyen, X.-H. Phan, Exploiting llms' reasoning capability to infer implicit concepts in legal information retrieval, arXiv preprint [arXiv:2410.12154](https://arxiv.org/abs/2410.12154), 2024, <https://doi.org/10.48550/arXiv.2410.12154>
- [104] Y.-H. Wang, F.-D. Su, T.-Y. Yeh, Y.-C. Fan, A cross-lingual statutory article retrieval dataset for taiwan legal studies, arXiv preprint [arXiv:2410.11450](https://arxiv.org/abs/2410.11450), 2024.
- [105] Z. Chen, P. Ren, F. Sun, X. Wang, Y. Li, S. Zhao, T. Yang, SLARD: a Chinese superior legal article retrieval dataset, in: Proceedings of the 31st International Conference on Computational Linguistics, Association for Computational Linguistics, 2025, Abu Dhabi, UAE, pp. 740–754, <https://aclanthology.org/2025.coling-main.50/>
- [106] H.-L. Nguyen, D.-M. Nguyen, T.-M. Nguyen, H.-T. Nguyen, T.-H.-Y. Vuong, K. Satoh, Enhancing legal document retrieval: a multi-phase approach with large language models, arXiv preprint [arXiv:2403.18093](https://arxiv.org/abs/2403.18093), 2024, <https://doi.org/10.48550/arXiv.2403.18093>
- [107] C. Nguyen, P. Nguyen, L.-M. Nguyen, Retrieve-revise-refine: a novel framework for retrieval of concise entailing legal article set, *Inf. Process. Manag.* 62 (1) (2025) 103949.
- [108] Q.H. Trung, N.V.H. Phuc, L.T. Hoang, Q.H. Hieu, V.N. Duy le, Adaptive two-phase finetuning llms for japanese legal text retrieval, arXiv preprint [arXiv:2412.13205](https://arxiv.org/abs/2412.13205), 2024, <https://doi.org/10.48550/arXiv.2412.13205>
- [109] A. Deroy, K. Ghosh, S. Ghosh, How ready are pre-trained abstractive models and llms for legal case judgement summarization? in: Proceedings of the Third International Workshop on Artificial Intelligence and Intelligent Assistance for Legal Professionals in the Digital Workplace (LegalAIIA 2023), Held in Conjunction with ICAIL 2023, Braga, Portugal, 2023, pp. 8–19.
- [110] T.Y.S.S. Santosh, C. Jia, P. Goroncy, M. Grabmair, RELEXED: retrieval-enhanced legal summarization with exemplar diversity, in: L. Chiruzzo, A. Ritter, L. Wang (Eds.), Findings of the Association for Computational Linguistics: NAACL 2025, Albuquerque, New Mexico, 2025, pp. 427–434.
- [111] M. Sie, R. Beek, M. Bots, S. Brinkkemper, A. Gatt, Summarizing long regulatory documents with a multi-step pipeline, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, D. Preoticiu-Pietro, G. Spanakis (Eds.), Proceedings of the Natural Language Processing Workshop 2024, Miami, FL, USA, 2024, pp. 18–32, <https://doi.org/10.18653/v1/2024.nllp-1.2>
- [112] A. Deroy, K. Ghosh, S. Ghosh, Applicability of large language models and generative models for legal case judgement summarization, *Artif. Intell. Law* (2024) 1–44, <https://doi.org/10.1007/s10506-024-09411-z>
- [113] J. Gesnouin, Y. Tannier, C.G. Da Silva, H. Tapory, C. Brier, H. Simon, R. Rozenberg, H. Woehrel, M.E. Yakaabi, T. Binder, et al., Llamandement: large language models for summarization of french legislative proposals, arXiv preprint [arXiv:2401.16182](https://arxiv.org/abs/2401.16182), 2024, <https://doi.org/10.48550/arXiv.2401.16182>
- [114] T.Y.S.S. Santosh, C. Weiss, M. Grabmair, LexSumm and text5: benchmarking and modeling legal summarization tasks in English, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, D. Preoticiu-Pietro, G. Spanakis (Eds.), Proceedings of the Natural Language Processing Workshop 2024, Miami, FL, USA, 2024, pp. 381–403, <https://doi.org/10.18653/v1/2024.nllp-1.35>
- [115] I. Benedetto, L. Cagliero, M. Ferro, F. Tarasconi, C. Bernini, G. Giacalone, Leveraging large language models for abstractive summarization of Italian legal news, *Artif. Intell. Law* (2025) 1–21, <https://doi.org/10.1007/s10506-025-09431-3>
- [116] S. Liu, J. Cao, Y. Li, R. Yang, Z. Wen, Low-resource court judgment summarization for common law systems, *Inf. Process. Manag.* 61 (5) (2024) 103796, <https://doi.org/10.1016/j.ipm.2024.103796>
- [117] H. Mentzingen, N. António, F. Bacao, Effectiveness in retrieving legal precedents: exploring text summarization and cutting-edge language models toward a cost-efficient approach, *Artif. Intell. Law* (2025) 1–21, <https://doi.org/10.1007/s10506-025-09440-2>
- [118] O.S. Chlapanis, D. Galanis, I. Androutsopoulos, LAR-ECHR: a new legal argument reasoning task and dataset for cases of the european court of human rights, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, D. Preoticiu-Pietro, G. Spanakis (Eds.), Proceedings of the Natural Language Processing Workshop 2024, Miami, FL, USA, 2024, pp. 267–279, <https://doi.org/10.18653/v1/2024.nllp-1.22>
- [119] A. Gupta, D. Rice, B. O'Connor, δ -stance: a large-scale real world dataset of stances in legal argumentation, in: W. Che, J. Nabende, E. Shutova, M.T. Pilehvar (Eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria, 2025, pp. 31450–31467, <https://doi.org/10.18653/v1/2025.acl-long.1517>, <https://aclanthology.org/2025.acl-long.1517/>
- [120] D. Bernsohn, G. Semo, Y. Vazana, G. Hayat, B. Hagag, J. Niklaus, R. Saha, K. Trusovskiy, LegalLens: leveraging LLMs for legal violation identification in unstructured text, in: Y. Graham, M. Purver (Eds.), Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), St. Julian's, Malta, 2024, pp. 2129–2145.
- [121] J. Savelka, Unlocking practical applications in legal domain: evaluation of GPT for zero-shot semantic annotation of legal texts, in: Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law, ICAIL '23, Braga, Portugal, 2023, pp. 447–451, <https://doi.org/10.1145/3594536.3595161>
- [122] L. Martin, N. Whitehouse, S. Yiu, L. Catterson, R. Perera, Better call gpt, comparing large language models against lawyers, arXiv preprint [arXiv:2401.16212](https://arxiv.org/abs/2401.16212), 2024, <https://doi.org/10.48550/arXiv.2401.16212>
- [123] N. Whitehouse, N. Lincoln, S. Yiu, L. Catterson, R. Perera, Better bill gpt: comparing large language models against legal invoice reviewers, arXiv preprint [arXiv:2504.02881](https://arxiv.org/abs/2504.02881), 2025, <https://doi.org/10.48550/arXiv.2504.02881>
- [124] C. Zheng, S. Wong, X. Su, Y. Tang, A. Nawaz, M. Kassem, Automating construction contract review using knowledge graph-enhanced large language models, *Autom. Constr.* 175 (2025) 106179, <https://doi.org/10.1016/j.autcon.2025.106179>

- [125] S. Wong, C. Zheng, X. Su, Y. Tang, Construction contract risk identification based on knowledge-augmented language models, *Comput. Ind.* 157 (2024) 104082, <https://doi.org/10.1016/j.compind.2024.104082>
- [126] A. Roegiest, R. Chitta, J. Donnelly, M. Lash, A. Vtyurina, F. Longtin, Questions about contracts: prompt templates for structured answer generation, in: D. Preotiuc-Pietro, C. Goanta, I. Chalkidis, L. Barrett, G. Spanakis, N. Aletras (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2023*, Singapore, 2023, pp. 62–72, <https://doi.org/10.18653/v1/2023.nllp-1.8>
- [127] A. Roegiest, R. Chitta, Answering questions in stages: prompt chaining for contract qa, *arXiv preprint arXiv:2410.12840*, 2024, <https://doi.org/10.48550/arXiv.2410.12840>
- [128] A. Singhal, C. Jain, P.R. Anish, A. Chakraborty, S. Ghaisas, Generating clarification questions for disambiguating contracts, in: N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, N. Xue (Eds.), *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italy, 2024, pp. 7611–7622.
- [129] M. Aspromonte, A. Ferraris, F. Galli, G. Contissa, LLMs to the rescue: explaining DSA statements of reason with platform's terms of services, in: *Proceedings of the Natural Legal Language Processing Workshop 2024*, Miami, FL, USA, 2024, pp. 205–215, <https://doi.org/10.18653/v1/2024.nllp-1.17>
- [130] C. Loeffler, A.M. Freile, T.R. Pizarro, Predicting potentially unfair clauses in Chilean terms of services with natural language processing, *arXiv preprint arXiv:2502.00865*, 2025, <https://doi.org/10.48550/arXiv.2502.00865>
- [131] D. Braun, F. Matthes, AGB-DE: a corpus for the automated legal assessment of clauses in German consumer contracts, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, 2024, pp. 10389–10405, <https://doi.org/10.18653/v1/2024.acl-long.559>
- [132] S. Vats, A. Zope, S. De, A. Sharma, U. Bhattacharya, S.K. Nigam, S. Guha, K. Rudra, K. Ghosh, LLMs – the good, the bad or the indispensable?: a use case on legal statute prediction and legal judgment prediction on Indian court cases, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023*, Singapore, 2023, pp. 12451–12474, <https://doi.org/10.18653/v1/2023.findings-emnlp.831>
- [133] S. Paul, R. Bhatt, P. Goyal, S. Ghosh, Legal statute identification: a case study using state-of-the-art datasets and methods, in: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 2231–2240.
- [134] X. Chi, W. Zhong, Y. Wu, W. Wang, K. Kuang, F. Wu, M. Xiong, Universal legal article prediction via tight collaboration between supervised classification model and llm, *arXiv preprint arXiv:2509.22119*, 2025.
- [135] Y. Le, C. He, M. Chen, Y. Wu, X. He, B. Zhou, Learning to predict charges for legal judgment via self-attentive capsule network, in: *Proceedings of 24th European Conference on Artificial Intelligence: ECAI 2020*, 2020, pp. 1802–1809, <https://doi.org/10.3233/FAIA200295>
- [136] C. He, T.-P. Tan, S. Xue, Y. Tan, Simulating judicial trial logic: dual residual cross-attention learning for predicting legal judgment in long documents, *Expert Syst. Appl.* 261 (2025) 125462, <https://doi.org/10.1016/j.eswa.2024.125462>
- [137] Y. Wu, S. Zhou, Y. Liu, W. Lu, X. Liu, Y. Zhang, C. Sun, F. Wu, K. Kuang, Precedent-enhanced legal judgment prediction with LLM and domain-model collaboration, in: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Singapore, 2023, pp. 12060–12075, <https://doi.org/10.18653/v1/2023.emnlp-main.740>
- [138] X. Peng, L. Chen, Athena: retrieval-augmented legal judgment prediction with large language models, *arXiv preprint arXiv:2410.11195*, 2024, <https://doi.org/10.48550/arXiv.2410.11195>
- [139] C. Jiang, X. Yang, Legal syllogism prompting: teaching large language models for legal judgment prediction, in: *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law, ICAIL '23*, Braga, Portugal, 2023, pp. 417–421, <https://doi.org/10.1145/3594536.3595170>
- [140] B. Wei, Y. Yu, L. Gan, F. Wu, An llms-based neuro-symbolic legal judgment prediction framework for civil cases, *Artif. Intell. Law* (2025) 1–35, <https://doi.org/10.1007/s10506-025-09433-1>
- [141] K. Zhang, H. Yang, X. Tang, W. Yu, J. Xu, Beyond guilt: legal judgment prediction with trichotomous reasoning, *arXiv preprint arXiv:2412.14588*, 2024, <https://doi.org/10.48550/arXiv.2412.14588>
- [142] W. Deng, J. Pei, K. Kong, Z. Chen, F. Wei, Y. Li, Z. Ren, Z. Chen, P. Ren, Syllogistic reasoning for legal judgment analysis, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Singapore, 2023, pp. 13997–14009, <https://doi.org/10.18653/v1/2023.emnlp-main.864>
- [143] H. Chen, L. Zhang, Y. Liu, Y. Yu, Rethinking the development of large language models from the causal perspective: a legal text prediction case study, in: *Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Fourteenth Symposium on Innovative Applications of Artificial Intelligence, AAAI'24/IAAI'24/AAAI'24*, 2024, <https://doi.org/10.1609/aaai.v38i19.30086>
- [144] M.B. Kmainasi, A.E. Shahroor, A. Al-Ghraibah, Can large language models predict the outcome of judicial decisions? *arXiv preprint arXiv:2501.09768*, 2025, <https://doi.org/10.48550/arXiv.2501.09768>
- [145] N. Prasad, M. Boughanem, T. Dkaki, Exploring large language models and hierarchical frameworks for classification of large unstructured legal documents, in: *European Conference on Information Retrieval*, 2024, pp. 221–237, https://doi.org/10.1007/978-3-031-56060-6_15
- [146] C. He, T.-P. Tan, S. Xue, Y. Tan, Explaining legal judgments: a multitask learning framework for enhancing factual consistency in rationale generation, *J. King Saud Univ.-Comput. Inf. Sci.* 35 (10) (2023) 101868, <https://doi.org/10.1016/j.jksuci.2023.101868>
- [147] I. Benedetto, A. Koudounas, L. Vaiani, E. Pastor, L. Cagliero, F. Tarasconi, E. Baralis, Boosting court judgment prediction and explanation using legal entities, *Artif. Intell. Law* (2024) 1–36, <https://doi.org/10.1007/s10506-024-09397-8>
- [148] S.K. Nigam, A. Sharma, D. Khanna, N. Shallum, K. Ghosh, A. Bhattacharya, Legal judgment reimaged: PredEx and the rise of intelligent AI interpretation in Indian courts, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, Bangkok, Thailand, 2024, pp. 4296–4315, <https://doi.org/10.18653/v1/2024.findings-acl.255>
- [149] Q. Xu, X. Wei, H. Yu, Q. Liu, H. Fei, Divide and conquer: legal concept-guided criminal court view generation, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024*, Miami, Florida, USA, 2024, pp. 3395–3410, <https://doi.org/10.18653/v1/2024.findings-emnlp.194>
- [150] L. Yue, Q. Liu, L. Zhao, L. Wang, W. Gao, Y. An, Event grounded criminal court view generation with cooperative (large) language models, in: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR '24*, Association for Computing Machinery, New York, NY, USA, 2024, pp. 2221–2230, <https://doi.org/10.1145/3626772.3657698>
- [151] C. Xiao, H. Zhong, Z. Guo, C. Tu, Z. Liu, M. Sun, Y. Feng, X. Han, Z. Hu, H. Wang, et al., Cail2018: a large-scale legal dataset for judgment prediction, *arXiv preprint arXiv:1807.02478*, 2018, <https://doi.org/10.48550/arXiv.1807.02478>
- [152] M. Medvedeva, P. McBride, Legal judgment prediction: if you are going to do IT, do it right, in: D. Preotiuc-Pietro, C. Goanta, I. Chalkidis, L. Barrett, G. Spanakis, N. Aletras (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2023*, Singapore, 2023, pp. 73–84, <https://doi.org/10.18653/v1/2023.nllp-1.9>
- [153] S.K. Nigam, A. Deroy, S. Maity, A. Bhattacharya, Rethinking legal judgement prediction in a realistic scenario in the era of large language models, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goantă, D. Preotiuc-Pietro, G. Spanakis (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2024*, Miami, FL, USA, 2024, pp. 61–80, <https://doi.org/10.18653/v1/2024.nllp-1.6>
- [154] E.L. Rissland, K.D. Ashley, R.P. Loui, AI and law: a fruitful synergy, *Artif. Intell.* 150 (1–2) (2003) 1–15.
- [155] K. Atkinson, T. Bench-Capon, D. Bollegala, Explanation in AI and law: past, present and future, *Artif. Intell.* 289 (2020) 103387.
- [156] X. Yang, Z. Wang, Q. Wang, K. Wei, K. Zhang, J. Shi, Large language models for automated q&a involving legal documents: a survey on algorithms, frameworks and applications, *Int. J. Web Inf. Syst.* 20 (4) (2024) 413–435.
- [157] W. Su, Y. Hu, A. Xie, Q. Ai, Q. Bing, N. Zheng, Y. Liu, W. Shen, Y. Liu, STARD: a Chinese statute retrieval dataset derived from real-life queries by non-professionals, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024*, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 10658–10671, <https://doi.org/10.18653/v1/2024.findings-emnlp.625>, <https://aclanthology.org/2024.findings-emnlp.625/>
- [158] L. Zheng, N. Guha, J. Arifov, S. Zhang, M. Skreta, C.D. Manning, P. Henderson, D.E. Ho, A reasoning-focused legal retrieval benchmark, in: *Proceedings of the 2025 Symposium on Computer Science and Law, CSLAW '25*, Munich, Germany, 2025, pp. 169–193, <https://doi.org/10.1145/3709025.3712219>
- [159] U. Ujwal, S.S.H. Surampudi, S. Mitra, T. Saha, “reasoning before responding”: towards legal long-form question answering with interpretability, in: *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, Boise, ID, USA, 2024, pp. 4922–4930, <https://doi.org/10.1145/3627673.3680082>
- [160] M. Tao, D. Zhao, Y. Feng, Chain-of-discussion: a multi-model framework for complex evidence-based question answering, in: O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. Eugenio di, S. Schockaert (Eds.), *Proceedings of the 31st International Conference on Computational Linguistics*, Abu Dhabi, UAE, 2025, pp. 11070–11085.
- [161] D. Trautmann, N. Ostapuk, Q. Grail, A. Pol, G. Bonifazi, S. Gao, M. Gajek, Measuring the groundedness of legal question-answering systems, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goantă, D. Preotiuc-Pietro, G. Spanakis (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2024*, Miami, FL, USA, 2024, pp. 176–186, <https://doi.org/10.18653/v1/2024.nllp-1.14>
- [162] R. Russo, D. Russo, G.M. Orlando, A. Romano, G. Riccio, V. La Gatta, M. Postiglione, V. Moscato, Europeanlawadvisor: an open source search engine for european laws, in: *2024 IEEE International Conference on Big Data (BigData)*, 2024, pp. 4751–4756, <https://doi.org/10.1109/BigData62323.2024.10826025>
- [163] F. Redelaar, R. Van Drie, S. Verberne, M. De Boer, Attributed question answering for preconditions in the Dutch law, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goantă, D. Preotiuc-Pietro, G. Spanakis (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2024*, Miami, FL, USA, 2024, pp. 154–165, <https://doi.org/10.18653/v1/2024.nllp-1.12>
- [164] N. Xie, Y. Bai, H. Gao, Z. Xue, F. Fang, Q. Zhao, Z. Li, L. Zhu, S. Ni, M. Yang, Delilaw: a Chinese legal counselling system based on a large language model, in: *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, 2024, pp. 5299–5303, <https://doi.org/10.1145/3627673.3679219>
- [165] R. Yao, Y. Wu, C. Wang, J. Xiong, F. Wang, X. Liu, Elevating legal LLM responses: harnessing trainable logical structures and semantic knowledge with legal reasoning, in: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Albuquerque, New Mexico, 2025, pp. 5630–5642.

- [166] K. Zhang, W. Yu, Z. Sun, J. Xu, Syler: a framework for explicit syllogistic legal reasoning in large language models, arXiv preprint arXiv:2504.04042, 2025, <https://doi.org/10.48550/arXiv.2504.04042>
- [167] S. Chowdhury, S. Joshi, L. Dey, Cross examine: an ensemble-based approach to leverage large language models for legal text analytics, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, D. Preotiu-Pietro, G. Spanakis (Eds.), Proceedings of the Natural Language Processing Workshop 2024, Miami, FL, USA, 2024, pp. 194–204, <https://doi.org/10.18653/v1/2024.nllp-1.16>
- [168] K.-Y. Lam, V.C.W. Cheng, Z.K. Yeong, Applying large language models for enhancing contract drafting, in: LegalAIIA@ ICAIL, 2023, pp. 70–80.
- [169] A. Chouhan, M. Gertz, LexDrafter: terminology drafting for legislative documents using retrieval augmented generation, in: N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, N. Xue (Eds.), Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), Torino, Italy, 2024, pp. 10448–10458.
- [170] M. Kant, S. Nabi, M. Kant, R. Scharrer, M. Ma, M. Nabi, Towards robust legal reasoning: harnessing logical llms in law, arXiv preprint arXiv:2502.17638, 2025, <https://doi.org/10.48550/arXiv.2502.17638>
- [171] A. Trozze, T. Davies, B. Kleinberg, Large language models in cryptocurrency securities cases: can a GPT model meaningfully assist lawyers? Artif. Intell. Law (2024) 1–47.
- [172] R. Hemrajani, Evaluating the role of large language models in legal practice in india, arXiv preprint arXiv:2508.09713, 2025.
- [173] H. Li, J. Ye, Y. Hu, J. Chen, Q. Ai, Y. Wu, J. Chen, Y. Chen, C. Luo, Q. Zhou, et al., Casegen: a benchmark for multi-stage legal case documents generation, arXiv preprint arXiv:2502.17943, 2025, <https://doi.org/10.48550/arXiv.2502.17943>
- [174] C.-H. Lin, P.-J. Cheng, Assisting drafting of Chinese legal documents using fine-tuned pre-trained large language models, Rev. Socionetwork Strateg. (2025) 1–28.
- [175] T.Y.S.S. Santosh, M. Aly, O. Ichim, M. Grabmair, LexGenie: automated generation of structured reports for european court of human rights case law, in: G. Rehm, Y. Li (Eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track), Association for Computational Linguistics, Vienna, Austria, 2025, pp. 672–683, <https://doi.org/10.18653/v1/2025.acl-industry.47>, <https://aclanthology.org/2025.acl-industry.47/>.
- [176] P. Kulkarni, Y. Liu, H.-M. Fu, S. Yang, I. Gunasekara, M. Peloquin, N. Spitzer-Williams, X. Zhou, X. Liu, Z. Ji, et al., Auto-drafting police reports from noisy asr outputs: a trust-centered llm approach, arXiv preprint arXiv:2502.07677, 2025, <https://doi.org/10.48550/arXiv.2502.07677>
- [177] H. Westermann, Dallma: semi-structured legal reasoning and drafting with large language models, in: 2nd Workshop on Generative AI and Law, Co-Located with the International Conference on Machine Learning (ICML) (Vienna, Austria, 2024), 2024.
- [178] S.K. Nigam, B.D. Patnaik, A.V. Thomas, N. Shallum, K. Ghosh, A. Bhattacharya, Structured legal document generation in India: a model-agnostic wrapper approach with vidhikdastaavej, arXiv preprint arXiv:2504.03486, 2025.
- [179] N. Miao, H. Zhou, L. Mou, R. Yan, L. Li, Cgmh: constrained sentence generation by metropolis-hastings sampling, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, 2019, pp. 6834–6842.
- [180] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y.J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, ACM Comput. Surv. 55 (12) (2023) 1–38.
- [181] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-T. Yih, T. Rocktäschel, et al., Retrieval-augmented generation for knowledge-intensive NLP tasks, Adv. Neural Inf. Process. Syst. 33 (2020) 9459–9474.
- [182] R. Puduppully, L. Dong, M. Lapata, Data-to-text generation with content selection and planning, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, 2019, pp. 6908–6915.
- [183] J. Shi, Q. Guo, Y. Liao, S. Liang, Legalgpt: legal chain of thought for the legal large language model multi-agent framework, in: International Conference on Intelligent Computing: ICIC 2024, Tianjin, China, 2024, pp. 25–37, https://doi.org/10.1007/978-981-97-5678-0_3
- [184] J. Sun, C. Dai, Z. Luo, Y. Chang, Y. Li, Lawluo: a chinese law firm co-run by llm agents, arXiv preprint arXiv:2407.16252, 2024, <https://doi.org/10.48550/arXiv.2407.16252>
- [185] S. Yue, T. Huang, Z. Jia, S. Wang, S. Liu, Y. Song, X. Huang, Z. Wei, Multi-agent simulator drives language models for legal intensive interaction, in: L. Chiruzzo, A. Ritter, L. Wang (Eds.), Findings of the Association for Computational Linguistics: NAACL 2025, Albuquerque, New Mexico, 2025, pp. 6537–6570.
- [186] Z. He, P. Cao, C. Wang, Z. Jin, Y. Chen, J. Xu, H. Li, K. Liu, J. Zhao, AgentsCourt: building judicial decision-making agents with court debate simulation and legal knowledge augmentation, in: Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, Florida, USA, 2024, pp. 9399–9416, <https://doi.org/10.18653/v1/2024.findings-emnlp.549>
- [187] C. Jiang, X. Yang, Agents on the bench: Large language model based multi agent framework for trustworthy digital justice, arXiv preprint arXiv:2412.18697, 2024, <https://doi.org/10.48550/arXiv.2412.18697>
- [188] X. Chen, M. Mao, S. Li, H. Shangguan, Debate-feedback: a multi-agent framework for efficient legal judgment prediction, in: Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers), Albuquerque, New Mexico, 2025, pp. 462–470.
- [189] S. Jung, J. Jung, Courtroom-LLM: a legal-inspired multi-LLM framework for resolving ambiguous text classifications, in: O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. Eugenio di, S. Schockaert (Eds.), Proceedings of the 31st International Conference on Computational Linguistics, Abu Dhabi, UAE, 2025, pp. 7367–7385.
- [190] W. Yuan, J. Cao, Z. Jiang, Y. Kang, J. Lin, K. Song, T. Lin, P. Yan, C. Sun, X. Liu, Can large language models grasp legal theories? Enhance legal reasoning with insights from multi-agent collaboration, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, Florida, USA, 2024, pp. 7577–7597, <https://doi.org/10.18653/v1/2024.findings-emnlp.445>
- [191] W. Watson, N. Cho, N. Srishankar, Z. Zeng, L. Cecchi, D. Scott, S. Siddagangappa, R. Kaur, T. Balch, M. Veloso, LAW: legal agentic workflows for custody and fund services contracts, in: Proceedings of the 31st International Conference on Computational Linguistics: Industry Track, Abu Dhabi, UAE, 2025, pp. 583–594.
- [192] M. Bommarito II, D.M. Katz, Gpt takes the bar exam, arXiv preprint arXiv:2212.14402, 2022, <https://doi.org/10.48550/arXiv.2212.14402>
- [193] J.H. Choi, K.E. Hickman, A.B. Monahan, D. Schwarcz, Chatgpt goes to law school, J. Legal Educ. 71 (2021) 387.
- [194] S. Hargreaves, 'words are flowing out like endless rain into a paper cup': Chatgpt & law school assessments, Leg. Educ. Rev. 33 (2023) 69.
- [195] E. Martínez, Re-evaluating gpt-4's bar exam performance, Artif. Intell. Law (2024) 1–24, <https://doi.org/10.1007/s10506-024-09396-9>
- [196] R. Fernandes, A. Biedenkapp, F. Hutter, N. Awad, A llama walks into the bar: efficient supervised fine-tuning for legal reasoning in the multi-state bar exam, arXiv preprint arXiv:2504.04945, 2025, <https://doi.org/10.48550/arXiv.2504.04945>
- [197] K. Luo, Q. Huang, C. Jiang, Y. Feng, Automating legal concept interpretation with llms: retrieval, generation, and evaluation, arXiv preprint arXiv:2501.01743, 2025, <https://doi.org/10.48550/arXiv.2501.01743>
- [198] H. Jiang, X. Zhang, R. Mahari, D. Kessler, E. Ma, T. August, I. Li, A. Pentland, Y. Kim, D. Roy, J. Kabbara, Leveraging large language models for learning complex legal concepts through storytelling, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Bangkok, Thailand, 2024, pp. 7194–7219, <https://doi.org/10.18653/v1/2024.acl-long.388>
- [199] J. Nelson, The other 'llm': large language models and the future of legal education, Eur. J. Leg. Educ. 5 (1) (2024) 127–155.
- [200] L. Peoples, Artificial intelligence and legal analysis: Implications for legal education and the profession, arXiv preprint arXiv:2502.03487, 2025, <https://doi.org/10.48550/arXiv.2502.03487>
- [201] M. Ajevski, K. Barker, A. Gilbert, L. Hardie, F. Ryan, Chatgpt and the future of legal education and practice, Law Teach. 57 (3) (2023) 352–364, <https://doi.org/10.1080/03069400.2023.2207426>
- [202] T.P. Oltz, Chatgpt, professor of law, U. Ill. J.L. Tech. Pol'y (2023) 207.
- [203] J.J. Nay, Law informs code: a legal informatics approach to aligning artificial intelligence with humans, Nw. J. Tech. Intell. Prop. 20 (2022) 309.
- [204] H. Westermann, J. Savelka, K. Benyekhlef, Lmediator: Gpt-4 assisted online dispute resolution, arXiv preprint arXiv:2307.16732, 2023, <https://doi.org/10.48550/arXiv.2307.16732>
- [205] J. Breton, M.M. Billami, M. Chevalier, H.T. Nguyen, K. Satoh, C. Trojahn, M.M. Zin, Leveraging llms for legal terms extraction with limited annotated data, Artif. Intell. Law (2025) 1–27, <https://doi.org/10.1007/s10506-025-09448-8>
- [206] J. Kieffaber, K. Gandall, K. McLaren, We built judge. AI. And you should buy IT, and you should buy IT. (January 28, 2025), 2025.
- [207] Y. Kano, M.-Y. Kim, M. Yoshioka, Y. Lu, J. Rabelo, N. Kiyota, R. Goebel, K. Satoh, Coliee-2018: evaluation of the competition on legal information extraction and entailment, in: New Frontiers in Artificial Intelligence: JSAI-ISA 2018 Workshops, JURISIN, AI-Biz, SKL, LENLS, IDAA, Yokohama, Japan, November 12–14, 2018, Revised Selected Papers, 2019, pp. 177–192, https://doi.org/10.1007/978-3-030-31605-1_14
- [208] J. Rabelo, M.-Y. Kim, R. Goebel, M. Yoshioka, Y. Kano, K. Satoh, A summary of the coliee 2019 competition, in: JSAI International Symposium on Artificial Intelligence, 2019, pp. 34–49, https://doi.org/10.1007/978-3-030-58790-1_3
- [209] J. Rabelo, M.-Y. Kim, R. Goebel, M. Yoshioka, Y. Kano, K. Satoh, Coliee 2020: methods for legal document retrieval and entailment, in: New Frontiers in Artificial Intelligence: JSAI-ISA 2020 Workshops, JURISIN, LENLS 2020 Workshops, Virtual Event, November 15–17, 2020, Revised Selected Papers 12, 2021, pp. 196–210, https://doi.org/10.1007/978-3-030-79942-7_13
- [210] J. Rabelo, R. Goebel, M.-Y. Kim, Y. Kano, M. Yoshioka, K. Satoh, Overview and discussion of the competition on legal information extraction/entailment (coliee) 2021, Rev. Socionetwork Strateg. 16 (1) (2022) 111–133, <https://doi.org/10.1007/s12626-022-00105-z>
- [211] M.-Y. Kim, J. Rabelo, R. Goebel, M. Yoshioka, Y. Kano, K. Satoh, Coliee 2022 summary: methods for legal document retrieval and entailment, in: JSAI International Symposium on Artificial Intelligence, 2022, pp. 51–67, https://doi.org/10.1007/978-3-031-29168-5_4
- [212] R. Goebel, Y. Kano, M.-Y. Kim, J. Rabelo, K. Satoh, M. Yoshioka, Overview and discussion of the competition on legal information, extraction/entailment (coliee) 2023, Rev. Socionetwork Strateg. 18 (1) (2024) 27–47, <https://doi.org/10.1007/s12626-023-00152-0>
- [213] R. Goebel, Y. Kano, M.-Y. Kim, J. Rabelo, K. Satoh, M. Yoshioka, Overview of benchmark datasets and methods for the legal information extraction/entailment competition (coliee) 2024, in: JSAI International Symposium on Artificial Intelligence, 2024, pp. 109–124, https://doi.org/10.1007/978-981-97-3076-6_8
- [214] Y. Ma, Y. Shao, Y. Wu, Y. Liu, R. Zhang, M. Zhang, S. Ma, Lecard: a legal case retrieval dataset for Chinese law system, in: Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021, pp. 2342–2348, <https://doi.org/10.1145/3404835.3463250>

- [215] H. Li, Y. Shao, Y. Wu, Q. Ai, Y. Ma, Y. Liu, Leca2v2: a large-scale Chinese legal case retrieval dataset, in: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024, pp. 2251–2260, <https://doi.org/10.1145/3626772.3657887>
- [216] M. Wrzalik, D. Krehel, GerDaLIR: a German dataset for legal information retrieval, in: N. Aletras, I. Androustopoulos, L. Barrett, C. Goanta, D. Preotiu-Pietro (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2021*, Punta Cana, Dominican Republic, 2021, pp. 123–128, <https://doi.org/10.18653/v1/2021.nllp-1.13>
- [217] A.B. Hou, O. Weller, G. Qin, E. Yang, D. Lawrie, N. Holzenberger, A. Blair-Stanek, B. Van Durme, CLERC: a dataset for U. S. Legal case retrieval and retrieval-augmented analysis generation, in: L. Chiruzzo, A. Ritter, L. Wang (Eds.), *Findings of the Association for Computational Linguistics: NAACL 2025*, Albuquerque, New Mexico, 2025, pp. 7898–7913.
- [218] T.Y.S.S. Santosh, R. Haddad, M. Grabmair, ECtHR-PCR: a dataset for precedent understanding and prior case retrieval in the european court of human rights, in: N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, N. Xue (Eds.), *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italy, 2024, pp. 5473–5483.
- [219] A. Mandal, K. Ghosh, A. Bhattacharya, A. Pal, S. Ghosh, Overview of the fire 2017 ired track: information retrieval from legal documents, in: *FIRE (Working Notes)*, 2017, pp. 63–68.
- [220] C. Xiao, H. Zhong, Z. Guo, C. Tu, Z. Liu, M. Sun, T. Zhang, X. Han, Z. Hu, H. Wang, et al., Cail2019-scm: a dataset of similar case matching in legal domain, *arXiv preprint arXiv:1911.08962*, 2019.
- [221] A. Louis, G. Spanakis, A statutory article retrieval dataset in French, in: S. Muresan, P. Nakov, A. Villavicencio (Eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Association for Computational Linguistics, Dublin, Ireland, 2022, pp. 6789–6803, <https://doi.org/10.18653/v1/2022.acl-long.468>, <https://aclanthology.org/2022.acl-long.468/>.
- [222] I. Habernal, D. Faber, N. Recchia, S. Brethauer, I. Gurevych, I. Spiecker genannt Döhmann, C. Burchard, Mining legal arguments in court decisions, *Artif. Intell. Law* 32 (3) (2024) 1–38.
- [223] G. Grundler, A. Galassi, P. Santin, A. Fidelangeli, F. Galli, E. Palmieri, F. Lagioia, G. Sartor, P. Torroni, et al., Amelia-argument mining evaluation on legal documents in italian: a calamita challenge, in: *CLIC-It 2024: Tenth Italian Conference on Computational Linguistics*, 2024, pp. 1–10.
- [224] G. Zhang, P. Nulty, D. Lillis, A decade of legal argumentation mining: datasets and approaches, in: *International Conference on Applications of Natural Language to Information Systems*, Springer, 2022, pp. 240–252.
- [225] L. Yue, Q. Liu, H. Wu, Y. An, L. Wang, S. Yuan, D. Wu, Circumstances enhanced criminal court view generation, in: *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021, pp. 1855–1859.
- [226] Z. Hou, K. Zeng, Shizhi: a chinese lightweight large language model for court view generation, *arXiv preprint arXiv:2510.09297*, 2025.
- [227] Q. Li, Q. Zhang, Court opinion generation from case fact description with legal basis, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, 2021, pp. 14840–14848.
- [228] M. Malik, Z. Zhao, M. Fonseca, S. Rao, S.B. Cohen, Civilsum: a dataset for abstractive summarization of Indian court decisions, in: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '24, Washington DC, USA, 2024, pp. 2241–2250, <https://doi.org/10.1145/3626772.3657859>.
- [229] D. Aumiller, A. Chouhan, M. Gertz, EUR-lex-sum: a multi- and cross-lingual dataset for long-form summarization in the legal domain, in: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, Abu Dhabi, United Arab Emirates, 2022, pp. 7626–7639, <https://doi.org/10.18653/v1/2022.emnlp-main.519>
- [230] S. Klaus, R. Van Hecke, K. Djafari Naini, I.S. Altingovde, J. Bernabé-Moreno, E. Herrera-Viedma, Summarizing legal regulatory documents using transformers, in: *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '22, Madrid, Spain, 2022, pp. 2426–2430, <https://doi.org/10.1145/3477495.3531872>.
- [231] A. Kornilova, V. Eidelman, BillSum: a corpus for automatic summarization of US legislation, in: L. Wang, J.C.K. Cheung, G. Carenini, F. Liu (Eds.), *Proceedings of the 2nd Workshop on New Frontiers in Summarization*, Hong Kong, China, 2019, pp. 48–56, <https://doi.org/10.18653/v1/D19-5406>
- [232] M. Elaraby, D. Litman, ArgLegalSumm: improving abstractive summarization of legal documents with argument mining, in: N. Calzolari, C.-R. Huang, H. Kim, J. Pustejovsky, L. Wanner, K.-S. Choi, P.-M. Ryu, H.-H. Chen, L. Donatelli, H. Ji, S. Kurohashi, P. Paggio, N. Xue, S. Kim, Y. Hamm, Z. He, T.K. Lee, E. Santus, F. Bond, S.-H. Na (Eds.), *Proceedings of the 29th International Conference on Computational Linguistics, Gyeongju, Republic of Korea*, 2022, pp. 6187–6194.
- [233] A. Shukla, P. Bhattacharya, S. Poddar, R. Mukherjee, K. Ghosh, P. Goyal, S. Ghosh, Legal case document summarization: extractive and abstractive methods and their evaluation, in: Y. He, H. Ji, S. Li, Y. Liu, C.-H. Chang (Eds.), *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Online only, 2022, pp. 1048–1064, <https://doi.org/10.18653/v1/2022.acl-main.77>
- [234] Z. Shen, K. Lo, L. Yu, N. Dahlberg, M. Schlanger, D. Downey, Multi-lexsum: real-world summaries of civil rights lawsuits at multiple granularities, in: *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, New Orleans, LA, USA, 2022, <https://doi.org/10.5555/3600270.3601226>.
- [235] I. Glaser, S. Moser, F. Matthes, Summarization of German court rulings, in: *Proceedings of the Natural Legal Language Processing Workshop 2021*, 2021, pp. 180–189, <https://doi.org/10.18653/v1/2021.nllp-1.19>
- [236] W. Gao, S. Yu, Y. Qin, C. Yang, R. Huang, Y. Chen, C. Lin, LsdK-legalsum: improving legal judgment summarization using logical structure and domain knowledge, *J. King Saud Univ. Comput. Inf. Sci.* 37 (1) (2025) 1–15, <https://doi.org/10.1007/s44443-025-00022-5>
- [237] M. Heddaya, K. MacMillan, H. Mei, C. Tan, A. Malani, CaseSumm: a large-scale dataset for long-context summarization from U.S. Supreme court opinions, in: L. Chiruzzo, A. Ritter, L. Wang (Eds.), *Findings of the Association for Computational Linguistics: NAACL 2025*, Albuquerque, New Mexico, 2025, pp. 1917–1942.
- [238] L. Rolshoven, V. Rasiah, S.B. Bose, M. Stürmer, J. Niklaus, Unlocking legal knowledge: a multilingual dataset for judicial summarization in switzerland, *arXiv preprint arXiv:2410.13456*, 2024, <https://doi.org/10.48550/arXiv.2410.13456>
- [239] L. Ragazzi, G. Moro, S. Guidi, G. Frisoni, Lawsuit: a large expert-written summarization dataset of Italian constitutional court verdicts, *Artif. Intell. Law* (2024) 1–37, <https://doi.org/10.1007/s10506-024-09414-w>
- [240] D. de Vargas Feijó, V.P. Moreira, Rulingbr: a summarization dataset for legal texts, in: *International Conference on Computational Processing of the Portuguese Language*, Springer, 2018, pp. 255–264.
- [241] L.C. Fernandes, G.Z.R. Dobins, R. Lotufo, J.A. Pereira, Publichearingbr: a brazilian portuguese dataset of public hearing transcripts for summarization of long documents, *arXiv preprint arXiv:2410.07495*, 2024.
- [242] I. Chalkidis, I. Androustopoulos, N. Aletras, Neural legal judgment prediction in English, in: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Florence, Italy, 2019, pp. 4317–4323, <https://doi.org/10.18653/v1/P19-1424>
- [243] H.J. Spaeth, L. Epstein, A.D. Martin, J.A. Segal, T.J. Ruger, S.C. Benesh, Supreme court database, version 2024 release 01, 2024, <http://supremecourtdatabase.org>.
- [244] J. Niklaus, I. Chalkidis, M. Stürmer, Swiss-judgment-prediction: a multilingual legal judgment prediction benchmark, in: N. Aletras, I. Androustopoulos, L. Barrett, C. Goanta, D. Preotiu-Pietro (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2021*, Punta Cana, Dominican Republic, 2021, <https://doi.org/10.18653/v1/2021.nllp-1.3>
- [245] V. Malik, R. Sanjay, S.K. Nigam, K. Ghosh, S.K. Guha, A. Bhattacharya, A. Modi, ILDC for CJPE: Indian legal documents corpus for court judgment prediction and explanation, in: C. Zong, F. Xia, W. Li, R. Navigli (Eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Online, 2021, pp. 4046–4062, <https://doi.org/10.18653/v1/2021.acl-long.313>
- [246] W. Hwang, D. Lee, K. Cho, H. Lee, M. Seo, A multi-task benchmark for Korean legal language understanding and judgement prediction, in: *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, New Orleans, LA, USA, 2022, <https://doi.org/10.5555/3600270.3602628>.
- [247] M. Sesodia, A. Petrova, J. Armour, T. Lukaszewicz, O.-M. Camburu, P.K. Dokania, P. Torr, C.S. de Witt, Annocaslaw: A richly-annotated dataset for benchmarking explainable legal judgment prediction, *arXiv preprint arXiv:2503.00128*, 2025, <https://doi.org/10.48550/arXiv.2503.00128>
- [248] O.-M. Şulea, M. Zampieri, M. Vela, J. van Genabith, Predicting the law area and decisions of French supreme court cases, in: R. Mitkov, G. Angelova (Eds.), *Proceedings of the International Conference Recent Advances in Natural Language Processing*, RANLP 2017, Varna, Bulgaria, 2017, pp. 716–722, <https://doi.org/10.26615/978-954-452-049-6.092>
- [249] I. Chalkidis, M. Fergadiotis, D. Tsarapatsanis, N. Aletras, I. Androustopoulos, P. Malakasiotis, Paragraph-level rationale extraction through regularization: a case study on european court of human rights cases, in: K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tur, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, Y. Zhou (Eds.), *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Online, 2021, pp. 226–241, <https://doi.org/10.18653/v1/2021.naacl-main.22>
- [250] S. Urrchs, J. Mitrovic, M. Granitzer, Design and implementation of German legal decision corpora, in: *ICAART (2)*, 2021, pp. 515–521.
- [251] S. Abbata, M. Hafez, A. Kazzaz, A. Alhothali, A. Alsolami, Aljp: An arabic legal judgment prediction in personal status cases using machine learning models, *arXiv preprint arXiv:2309.00238*, 2023, <https://doi.org/10.48550/arXiv.2309.00238>
- [252] A. Lage-Freitas, H. Allende-Cid, O. Santana, L. Oliveira-Lage, Predicting Brazilian court decisions, *PeerJ Comput. Sci.* 8 (2022) e904, <https://doi.org/10.7717/peerj-cs.904>
- [253] B. Fang, T. Cohn, T. Baldwin, L. Frermann, Super-SCOTUS: a multi-sourced dataset for the supreme court of the US, in: D. Preotiu-Pietro, C. Goanta, I. Chalkidis, L. Barrett, G. Spanakis, N. Aletras (Eds.), *Proceedings of the Natural Legal Language Processing Workshop 2023*, Singapore, 2023, pp. 202–214, <https://doi.org/10.18653/v1/2023.nllp-1.20>
- [254] W. Huang, Y. Feng, C. Li, H. Wu, J. Ge, V. Ng, CMDL: a large-scale Chinese multi-defendant legal judgment prediction dataset, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, Bangkok, Thailand, 2024, pp. 5895–5906, <https://doi.org/10.18653/v1/2024.findings-acl.351>
- [255] S. Paul, P. Goyal, S. Ghosh, Lesicin: a heterogeneous graph-based approach for automatic legal statute identification from Indian legal documents, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, 2022, pp. 11139–11146.
- [256] L. Zheng, N. Guha, B.R. Anderson, P. Henderson, D.E. Ho, When does pretraining help? Assessing self-supervised learning for law and the casehold dataset of 53, 000+ legal holdings, in: *Proceedings of the Eighteenth International Conference*

- on Artificial Intelligence and Law, ICAIL '21, São Paulo, Brazil, 2021, pp. 159–168, <https://doi.org/10.1145/3462757.3466088>.
- [257] A. Louis, G. van Dijk, G. Spanakis, Interpretable long-form legal question answering with retrieval-augmented large language models, in: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence and Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence and Fourteenth Symposium on Educational Advances in Artificial Intelligence, AAAI'24/IAAI'24/EAAI'24, 2024, <https://doi.org/10.1609/aaai.v38i20.30232>.
- [258] B. Liu, Z. Zhu, Q. Ai, Y. Liu, Y. Wu, Ledqa: a Chinese legal case document-based question answering dataset, in: Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24, Boise, ID, USA, 2024, pp. 5385–5389, <https://doi.org/10.1145/3627673.3679154>.
- [259] H. Zhong, C. Xiao, C. Tu, T. Zhang, Z. Liu, M. Sun, Jec-qa: a legal-domain question answering dataset, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 34, 2020, pp. 9701–9708, <https://doi.org/10.1609/aaai.v34i05.6519>.
- [260] Y. Wang, X. Shen, Z. Huang, L. Niu, S. Ou, Clegal-qa: a Chinese legal question answering with natural language generation methods, *Complex Intell. Syst.* 11 (1) (2025) 77, <https://doi.org/10.1007/s40747-024-01675-x>.
- [261] F. Sovrano, M. Palmirani, B. Distefano, S. Sapienza, F. Vitali, A dataset for evaluating legal question answering on private international law, in: Proceedings of the Eighteenth International Conference on Artificial Intelligence and Law, 2021, pp. 230–234, <https://doi.org/10.1145/3462757.3466094>.
- [262] J. Monteiro, P.-A. Noël, E. Marcotte, S. Rajeswar, V. Zantedeschi, D. Vázquez, N. Chapados, C. Pal, P. Taslakian, Repliqa: a question-answering dataset for benchmarking llms on unseen reference content, in: A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, C. Zhang (Eds.), *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 24242–24276.
- [263] F. Faisal, U. Yousaf, Legal-uqa: a low-resource urdu-english dataset for legal question answering, *arXiv preprint arXiv:2410.13013*, 2024, <https://doi.org/10.48550/arXiv.2410.13013>.
- [264] A. Chen, F. Yao, X. Zhao, Y. Zhang, C. Sun, Y. Liu, W. Shen, Equals: a real-world dataset for legal question answering via reading Chinese laws, in: Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law, 2023, pp. 71–80, <https://doi.org/10.1145/3594536.3595159>.
- [265] A. Askari, Z. Yang, Z. Ren, S. Verberne, Answer retrieval in legal community question answering, in: European Conference on Information Retrieval, 2024, pp. 477–485, https://doi.org/10.1007/978-3-031-56063-7_40.
- [266] Anonymous, ALLQA: evaluating AI-driven legal question answering systems for the Indian legal system, in: Submitted to ACL Rolling Review - April 2024, 2024, under review, <https://openreview.net/forum?id=JnAHWgKeSt>.
- [267] M. Büttner, I. Habernal, Answering legal questions from laymen in German civil law system, in: Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), 2024, pp. 2015–2027.
- [268] C.-G. Crăciun, R.-A. Smădu, D.-C. Cercel, M.-C. Cercel, Graf: Graph retrieval augmented by facts for legal question answering, *arXiv preprint arXiv:2412.04119*, 2024, <https://doi.org/10.48550/arXiv.2412.04119>.
- [269] D. Tuggenier, P. von Däniken, T. Peetz, M. Cieliebak, LEDGAR: a large-scale multi-label corpus for text classification of legal provisions in contracts, in: N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, S. Piperidis (Eds.), *Proceedings of the Twelfth Language Resources and Evaluation Conference*, Marseille, France, 2020, pp. 1235–1241.
- [270] S. Wang, A. Scardigli, L. Tang, W. Chen, D. Levkin, A. Chen, S. Ball, T. Woodside, O. Zhang, D. Hendrycks, MAUD: an expert-annotated legal NLP dataset for merger agreement understanding, in: H. Bouamor, J. Pino, K. Bali (Eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, Singapore, 2023, pp. 16369–16382, <https://doi.org/10.18653/v1/2023.emnlp-main.1019>, <https://aclanthology.org/2023.emnlp-main.1019/>.
- [271] W. Su, B. Yue, Q. Ai, Y. Hu, J. Li, C. Wang, K. Zhang, Y. Wu, Y. Liu, Judge: benchmarking judgment document generation for chinese legal system, *arXiv preprint arXiv:2503.14258*, 2025, <https://doi.org/10.48550/arXiv.2503.14258>.
- [272] S. Zhou, Y. Wu, H. Chen, X. Hu, K. Kuang, A. Jatowt, C. Zheng, F. Wu, ClaimGen-CN: a large-scale Chinese dataset for legal claim generation, in: C. Christodoulopoulos, T. Chakraborty, C. Rose, V. Peng (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2025*, Association for Computational Linguistics, Suzhou, China, 2025, pp. 12296–12323, <https://aclanthology.org/2025.findings-emnlp.658/>.
- [273] J. Niklaus, V. Matoshi, M. Stürmer, I. Chalkidis, D. Ho, MultiLegalPile: a 689gb multilingual legal corpus, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, 2024, pp. 15077–15094, <https://doi.org/10.18653/v1/2024.acl-long.805>.
- [274] I. Chalkidis, E. Fergadiotis, P. Malakasiotis, I. Androutopoulos, Large-scale multi-label text classification on EU legislation, in: A. Korhonen, D. Traum, L. Márquez (Eds.), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics, Florence, Italy, 2019, pp. 6314–6322, <https://doi.org/10.18653/v1/P19-1636>, <https://aclanthology.org/P19-1636/>.
- [275] E. Leitner, G. Rehm, J. Moreno-Schneider, A dataset of German legal documents for named entity recognition, in: N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, S. Piperidis (Eds.), *Proceedings of the Twelfth Language Resources and Evaluation Conference*, European Language Resources Association, Marseille, France, 2020, pp. 4478–4485, <https://aclanthology.org/2020.lrec-1.551/>.
- [276] P.H. Luz de Araujo, T.E. de Campos, R.R.R. de Oliveira, M. Stauffer, S. Couto, P. Bermejo, Lener-br: a dataset for named entity recognition in Brazilian legal text, in: *International Conference on Computational Processing of the Portuguese Language*, Springer, 2018, pp. 313–323.
- [277] P.H. Luz de Araujo, T.E. de Campos, F. Ataiades Braz, N. Correia da Silva, VICTOR: a dataset for Brazilian legal documents classification, in: N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, S. Piperidis (Eds.), *Proceedings of the Twelfth Language Resources and Evaluation Conference*, European Language Resources Association, Marseille, France, 2020, pp. 1449–1458, <https://aclanthology.org/2020.lrec-1.181/>.
- [278] P. Henderson, M. Krass, L. Zheng, N. Guha, C.D. Manning, D. Jurafsky, D. Ho, Pile of law: learning responsible data filtering from the law and a 256gb open-source legal dataset, *Adv. Neural Inf. Process. Syst.* 35 (2022) 29217–29234.
- [279] A. Östling, H. Sargeant, H. Xie, L. Bull, A. Terenin, L. Jonsson, M. Magnusson, F. Steffel, The Cambridge law corpus: a dataset for legal AI research, *Adv. Neural Inf. Process. Syst.* 36 (2023) 41355–41385.
- [280] F. Yao, C. Xiao, X. Wang, Z. Liu, L. Hou, C. Tu, J. Li, Y. Liu, W. Shen, M. Sun, LEVEN: a large-scale Chinese legal event detection dataset, in: S. Muresan, P. Nakov, A. Villavicencio (Eds.), *Findings of the Association for Computational Linguistics: ACL 2022*, Association for Computational Linguistics, Dublin, Ireland, 2022, pp. 183–201, <https://doi.org/10.18653/v1/2022.findings-acl.17>, <https://aclanthology.org/2022.findings-acl.17/>.
- [281] C. Ji, Y. Zhang, A. Jatowt, H. Wu, CDD: a large scale dataset for legal intelligence research, in: M. Wang, I. Zitouni (Eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*, Association for Computational Linguistics, Singapore, 2023, pp. 66–73, <https://doi.org/10.18653/v1/2023.emnlp-industry.7>, <https://aclanthology.org/2023.emnlp-industry.7/>.
- [282] A. Spangher, Z. Xue, T.-L. Wu, M. Hansen, J. May, Legaldiscourse: interpreting when laws apply and to whom, in: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 8528–8551.
- [283] M.-Y. Kim, R. Goebel, S. Ken, Collee-2015: evaluation of legal question answering, in: *Ninth International Workshop on Juris-Informatics (JURISIN 2015)*, 2015.
- [284] M.-Y. Kim, R. Goebel, Y. Kano, K. Satoh, Collee-2016: evaluation of the competition on legal information extraction and entailment, in: *International Workshop on Juris-Informatics (JURISIN 2016)*, 2016.
- [285] Y. Kano, M.-Y. Kim, R. Goebel, K. Satoh, Overview of collee 2017, in: *COLIEE@ ICAIL*, 2017, pp. 1–8.
- [286] R. Goebel, Y. Kano, M.-Y. Kim, J. Rabelo, K. Satoh, M. Yoshioka, Summary of the competition on legal information, extraction/entailment (collee) 2023, in: *Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law*, 2023, pp. 472–480, <https://doi.org/10.1145/3594536.3595176>.
- [287] Y. Dai, D. Feng, J. Huang, H. Jia, Q. Xie, Y. Zhang, W. Han, W. Tian, H. Wang, LaiW: a Chinese legal large language models benchmark, in: O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. Eugenio di, S. Schockaert (Eds.), *Proceedings of the 31st International Conference on Computational Linguistics*, Abu Dhabi, UAE, 2025, pp. 10738–10766.
- [288] Z. Fei, X. Shen, D. Zhu, F. Zhou, Z. Han, A. Huang, S. Zhang, K. Chen, Z. Yin, Z. Shen, J. Ge, V. Ng, LawBench: benchmarking legal knowledge of large language models, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Miami, Florida, USA, 2024, pp. 7933–7962, <https://doi.org/10.18653/v1/2024.emnlp-main.452>.
- [289] H. Li, Y. Chen, Q. Ai, Y. Wu, R. Zhang, Y. Liu, Lexeval: a comprehensive Chinese legal benchmark for evaluating large language models, in: A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, C. Zhang (Eds.), *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 25061–25094.
- [290] N. Guha, J. Nyarko, D.E. Ho, C. Ré, A. Chilton, A. Narayana, A. Chohlas-Wood, A. Peters, B. Waldon, D.N. Rockmore, D. Zambrano, D. Talisman, E. Hoque, F. Surani, F. Fagan, G. Sarfaty, G.M. Dickinson, H. Porat, J. Hegland, J. Wu, J. Nudell, J. Niklaus, J. Nay, J.H. Choi, K. Tobia, M. Hagan, M. Ma, M. Livermore, N. Rasumov-Rahe, N. Holzenberger, N. Kolt, P. Henderson, S. Rehaag, S. Goel, S. Gao, S. Williams, S. Gandhi, T. Zur, V. Iyer, Z. Li, Legalbench: a collaboratively built benchmark for measuring legal reasoning in large language models, in: *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS '23, New Orleans, LA, USA, 2023, <https://doi.org/10.5555/3666122.3668037>.
- [291] Y. Kim, Y. Choi, E. Choi, J. Choi, H.J. Park, W. Hwang, Developing a pragmatic benchmark for assessing Korean legal language understanding in large language models, in: Y. Al-Onaizan, M. Bansal, Y.-N. Chen (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024*, Miami, Florida, USA, 2024, pp. 5573–5595, <https://doi.org/10.18653/v1/2024.findings-emnlp.319>.
- [292] A. Joshi, S. Paul, A. Sharma, P. Goyal, S. Ghosh, A. Modi, IL-TUR: benchmark for Indian legal text understanding and reasoning, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, 2024, pp. 11460–11499, <https://doi.org/10.18653/v1/2024.acl-long.618>.
- [293] V. Rasiah, R. Stern, V. Matoshi, M. Stürmer, I. Chalkidis, D.E. Ho, J. Niklaus, One law, many languages: benchmarking multilingual legal reasoning for judicial support, in: *ICLR 2024 Workshop on Data-Centric Machine Learning Research (DMLR): Harnessing Momentum for Science*, 2024, <https://openreview.net/forum>.
- [294] B. Canaverde, T.P. Pires, L.M. Ribeiro, A.F.T. Martins, Legalbench. pt: a benchmark for portuguese law, *arXiv preprint arXiv:2502.16357*, 2025, <https://doi.org/10.48550/arXiv.2502.16357>.

- [295] N. Pipitone, G.H. Alami, Legalbench-rag: a benchmark for retrieval-augmented generation in the legal domain, arXiv preprint [arXiv:2408.10343](https://arxiv.org/abs/2408.10343), 2024, <https://doi.org/10.48550/arXiv.2408.10343>
- [296] K. Zhang, W. Yu, S. Dai, J. Xu, Citalaw: enhancing llm with citations in legal domain, arXiv preprint [arXiv:2412.14556](https://arxiv.org/abs/2412.14556), 2024, <https://doi.org/10.48550/arXiv.2412.14556>
- [297] J. Niklaus, J. Merane, L. Nenadic, S. Ahmadi, Y. Gao, C.A.H. Chevalley, C. Humbel, C. Gösen, L. Tanzi, T. Lüthi, et al., Swiltra-bench: The swiss legal translation benchmark, arXiv preprint [arXiv:2503.01372](https://arxiv.org/abs/2503.01372), 2025, <https://doi.org/10.48550/arXiv.2503.01372>
- [298] R. Zhang, H. Li, Y. Wu, Q. Ai, Y. Liu, M. Zhang, S. Ma, Evaluation ethics of llms in legal domain, arXiv preprint [arXiv:2403.11152](https://arxiv.org/abs/2403.11152), 2024, <https://doi.org/10.48550/arXiv.2403.11152>
- [299] F.J. Bex, AI, law and beyond. A transdisciplinary ecosystem for the future of AI & law, *Artif. Intell. Law* 33 (1) (2025) 253–270.
- [300] Y. Yin, I. Habernal, Privacy-preserving models for legal natural language processing, in: N. Aletras, I. Chalkidis, L. Barrett, C. Goanță, D. Preotiuc-Pietro (Eds.), Proceedings of the Natural Legal Language Processing Workshop 2022, Abu Dhabi, United Arab Emirates (Hybrid), 2022, pp. 172–183, <https://doi.org/10.18653/v1/2022.nllp-1.14>
- [301] I. Chalkidis, T. Pasini, S. Zhang, L. Tomada, S. Schwemer, A. Søgaard, FairLex: a multilingual benchmark for evaluating fairness in legal text processing, in: S. Muresan, P. Nakov, A. Villavicencio (Eds.), Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Dublin, Ireland, 2022, pp. 4389–4406, <https://doi.org/10.18653/v1/2022.acl-long.301>
- [302] S. De Capitani Di Vimercati, S. Foresti, G. Livraga, P. Samarati, Data privacy: definitions and techniques, *Int. J. Uncertain. Fuzziness Knowl.-Based Syst.* 20 (6) (2012) 793–817, <https://doi.org/10.1142/S0218488512400247>
- [303] W. Luo, Q. Zhang, T. Lu, X. Liu, Y. Zhao, Z. Xiang, C. Xiao, Doxing via the lens: revealing privacy leakage in image geolocation for agentic multi-modal large reasoning model, arXiv preprint [arXiv:2504.19373](https://arxiv.org/abs/2504.19373), 2025, <https://doi.org/10.48550/arXiv.2504.19373>
- [304] J. Jiao, S. Afroogh, Y. Xu, C. Phillips, Navigating llm ethics: Advancements, challenges, and future directions, arXiv preprint [arXiv:2406.18841](https://arxiv.org/abs/2406.18841), 2024, <https://doi.org/10.48550/arXiv.2406.18841>
- [305] Z. Xu, Human judges in the era of artificial intelligence: challenges and opportunities, *Appl. Artif. Intell.* 36 (1) (2022) 2013652, <https://doi.org/10.1080/08839514.2021.2013652>
- [306] A.W. Flores, K. Bechtel, C.T. Lowenkamp, False positives, false negatives, and false analyses: a rejoinder to machine bias: there's software used across the country to predict future criminals. And it's biased against blacks, *Fed. Probation* 80 (2016) 38.
- [307] N.A. Smuha, Regulation 2024/1689 of the EUR. Parl. & Council of June 13, 2024 (EU artificial intelligence act), *Int. Leg. Mater.* (2024) 1–148.
- [308] C. Draper, N. Gillibrand, The potential for jurisdictional challenges to AI or LLM training datasets, in: *AI4AJ@ ICAIL*, 2023.
- [309] H.B. Dixon Jr., A.H. Goddard, M.R. Grossman, X. Rodriguez, S.U. Schlegel, S.A. Thumma, Navigating AI in the judiciary: new guidelines for judges and their chambers, *Sedona Conf. J.* 26 (2025) 1, <https://thesedonaconference.org/sites/default/files/publications/Navigating>.
- [310] C. Novelli, M. Taddeo, L. Floridi, Accountability in artificial intelligence: what it is and how it works, *Ai Soc.* 39 (4) (2024) 1871–1882, <https://doi.org/10.1007/s00146-023-01635-y>